

Introduction to Stochastic Analysis
MATH5975

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Contents

1	Background Material	5
1.1	Probability Space	5
1.2	Random Variables and the Borel σ -algebra	9
1.2.1	Independence	11
1.3	Expectation and Conditional Expectation	12
1.3.1	Lebesgue-Stieltjes Integral	12
1.3.2	Conditional Expectation	14
2	Filtrations and Stochastic Processes	19
2.1	Martingales, Submartingales and Supermartingales	21
2.2	Stopping Times and Martingales	23
2.2.1	Discrete Time	23
2.2.2	Continuous Time	28
2.2.3	The Usual Conditions	29
3	The Brownian Motion	31
3.1	Total and Quadratic Variations	34
3.2	Reflection Principle	39
4	Itô Stochastic Integral	43
4.1	Itô Integral for Elementary Processes	43
4.1.1	Itô Integral for Processes from $\mathcal{L}_{\mathbb{P}}^2(W)$	45
4.1.2	Itô Integral for processes from $\mathcal{L}(W)$	46
4.2	Itô Processes and Itô's Lemma	47
4.2.1	One-Dimensional Case	48
4.2.2	Multidimensional Case	48
4.3	Computational Examples	50

5 Semimartingales	53
5.1 Continuous Local Martingales	53
5.1.1 Continuous Semimartingales	56
5.2 Lévy Characterisation	60
5.2.1 One-Dimensional Case	60
5.2.2 Multidimensional Case	63
5.3 Martingale Representation Property	63
6 Stochastic Differential Equations	65
6.1 Linear Stochastic Differential Equation	66
6.2 Itô's Existence and Uniqueness Theorem	68
6.2.1 Stochastic Exponential	69
7 Radon-Nikodým Density and Girsanov Theorem	71
7.1 Abstract Bayes Formula	72
7.2 Girsanov's Theorem	73
7.3 Applications	78
7.3.1 Martingale Measures	78
7.3.2 Distribution of the First Passage Time	79
7.3.3 Brownian motion with drift and its running supremum.	82
8 Feynman-Kac Formula	87
8.1 One-Dimensional Case	87
8.1.1 Multidimensional Case	89
9 Exercise	91
10 Appendix	97
10.1 Modes of convergences	97
10.2 Convergence Theorems	98

Chapter 1

Background Material

1.1 Probability Space

The goal of this chapter is to revisit results from the theory of probability. The standard probability model has three key components, which are the state space Ω , a σ -algebra $\mathcal{A}$ on Ω and a probability measure $\mathbb{P}$ defined on the σ -algebra $\mathcal{A}$.

Definition 1.1.1. The set of all possible outcomes of an experiment is called the *state space* and is denoted by Ω .

Example 1.1.1. Some example of the state space Ω .

- Tossing a two sided coin: One can take $\Omega = \{T, H\}$ where T stands for tails and H stands for heads.
- The life time of a light bulb: One can take $\Omega = [0, \infty)$.
- All possible path of a financial stock: One can take $\Omega = C[0, T)$, that is continuous functions on the time interval $[0, T)$.

To model the flow of information, we introduce a σ -algebra. To this end, we first introduce some standard notations and terminologies.

- An element of the state space Ω is usually denoted by ω .
- If A is a subset of Ω , then the complement of A is $A^c = \{\omega \in \Omega : \omega \notin A\}$.
- The subset ‘ A or B ’ is $A \cup B$.
- The subset ‘ A and B ’ is $A \cap B$.
- The subsets A and B are said to be disjoint if $A \cap B = \emptyset$.
- The empty set $\emptyset$.
- The family of all possible subset of Ω , that is the power set is denoted by 2^Ω .

Lemma 1.1.1 (De Morgan's law). *Given a family of sets $(A_i)_{i \geq 1}$, we have*

$$\left(\bigcup_{i=1}^{\infty} A_i \right)^c = \left(\bigcap_{i=1}^{\infty} A_i^c \right)$$

Definition 1.1.2. A family of subsets $\mathcal{A} \subset 2^\Omega$ is called a *algebra* if

1. $\Omega \in \mathcal{A}$
2. $A \in \mathcal{A}$ implies $A^c \in \mathcal{A}$.
3. $\mathcal{A}$ is closed under finite unions (or finite intersections). More specifically, if $A_1, \dots, A_n$ belongs to $\mathcal{A}$ then $\bigcup_{i=1}^n A_i$ (or $\bigcap_{i=1}^n A_i$) also belong to $\mathcal{A}$.

Definition 1.1.3. A family of subsets $\mathcal{A} \subset 2^\Omega$ is called a σ -algebra or σ -field if

1. $\Omega \in \mathcal{A}$
2. $A \in \mathcal{A}$ implies $A^c \in \mathcal{A}$.
3. $\mathcal{A}$ is closed under countable unions (or countable intersections). More specifically, if $(A_i)_{i \geq 1}$ belongs to $\mathcal{A}$ then $\bigcup_{i=1}^{\infty} A_i$ (or $\bigcap_{i=1}^{\infty} A_i$) also belong to $\mathcal{A}$.

It is clear that an σ -algebra also an algebra, but reverse is usually not true.

Definition 1.1.4. Given a state space Ω and an σ -algebra $\mathcal{A} \subseteq 2^\Omega$, the duple $(\Omega, \mathcal{A})$ is called a measurable space.

Definition 1.1.5. A subset $A \subseteq \Omega$ is said to be measurable with respect to $\mathcal{A}$, if $A \in \mathcal{A}$. One can simply say A is $\mathcal{A}$ -measurable or call A an *event*.

Intuitively, the σ -algebra represents the family of all possible sets of outcomes which can be observed and measured. It is a model of available information.

Definition 1.1.6. Given a measurable space $(\Omega, \mathcal{A})$, a probability measure $\mathbb{P}$ is a set function from the σ -algebra $\mathcal{A}$ to $[0, 1]$ which satisfies the following conditions.

1. $\mathbb{P}(\Omega) = 1$
2. (Countably additive) For every countable sequence of pairwise disjoint events $(A_n)_{n \geq 1}$, i.e, for $n \neq m$, $A_n \cap A_m = \emptyset$, then

$$\mathbb{P} \left(\bigcup_{n=1}^{\infty} A_n \right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n)$$

Example 1.1.2. In the coin toss model, we have $\Omega = \{T, H\}$ and $\mathcal{A} = 2^\Omega = \{\Omega, \emptyset, \{T\}, \{H\}\}$, if we want to model a fair coin then $\mathbb{P} : 2^\Omega \rightarrow [0, 1]$ is uniquely determined by setting $\mathbb{P}(\{T\}) := \frac{1}{2}$.

Definition 1.1.7. A set function $\mathbb{Q}$ is additive if for every finite sequence of pairwise disjoint events $(A_n)_{n=1,\dots,k}$, i.e, for $n \neq m \leq k$, $A_n \cap A_m = \emptyset$, then

$$\mathbb{Q}\left(\bigcup_{n=1}^k A_n\right) = \sum_{n=1}^k \mathbb{Q}(A_n)$$

Theorem 1.1.1. If $\mathbb{P}$ is a probability measure then

1. $\mathbb{P}(\emptyset) = 0$.
2. $\mathbb{P}$ is additive.
3. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$

Proof. For any finite sequence of pairwise disjoint events $(A_i)_{i=1,\dots,n}$ we extend the sequence to a countable sequence by setting $A_i = \emptyset$ for $i > n$. The countable sequence $(A_i)_{i \geq 1}$ is again pairwise disjoint, since the empty set is disjoint from itself.

1. Using the countable additive property of $\mathbb{P}$ and the fact that for any subset A , $A \cup \emptyset = A$, we must have

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^n A_i\right) &= \mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) \\ &= \sum_{i=1}^{\infty} \mathbb{P}(A_i) = \sum_{i=1}^n \mathbb{P}(A_i) + \sum_{i=n+1}^{\infty} \mathbb{P}(\emptyset) \end{aligned}$$

By definition of a probability measure, $\mathbb{P}(\bigcup_{i=1}^n A_i)$ takes value between $[0, 1]$, this implies that on the right hand side $\sum_{i=n+1}^{\infty} \mathbb{P}(\emptyset) < \infty$, which shows that $\mathbb{P}(\emptyset) = 0$.

2. From 1. we know $\mathbb{P}(\emptyset) = 0$, which clearly implies that $\mathbb{P}$ is additive.
3. Given any $A \in \mathcal{A}$, we note that $\Omega = A \cup A^c$. Then by additivity of $\mathbb{P}$ and the fact that $\mathbb{P}(\Omega) = 1$, we have $1 = \mathbb{P}(\Omega) = \mathbb{P}(A \cup A^c) = \mathbb{P}(A) + \mathbb{P}(A^c)$. $\square$

Given a set A and an increase sequence of sets $(A_i)_{i \geq 1}$, that is $A_1 \subseteq A_2 \subseteq A_3 \dots$, we write $A_i \uparrow A$ if $\bigcup_{i=1}^{\infty} A_i = A$. Similarly, given a set A and an decrease sequence of sets $(A_i)_{i \geq 1}$, that is $A_1 \supseteq A_2 \supseteq A_3 \dots$, we write $A_i \downarrow A$ if $\bigcap_{i=1}^{\infty} A_i = A$.

Theorem 1.1.2. Let $\mathcal{A}$ be a σ -algebra and suppose $\mathbb{P} : \mathcal{A} \rightarrow [0, 1]$ is a set function such that $\mathbb{P}(\Omega) = 1$ and $\mathbb{P}$ is finitely additive. Then the following are equivalent.

1. $\mathbb{P}$ is countably additive.
2. If $A_n \in \mathcal{A}$ and $A_n \downarrow \emptyset$ then $\mathbb{P}(A_n) \downarrow 0$,
3. If $A_n \in \mathcal{A}$ and $A_n \downarrow A$ then $\mathbb{P}(A_n) \downarrow \mathbb{P}(A)$,
4. If $A_n \in \mathcal{A}$ and $A_n \uparrow \Omega$ then $\mathbb{P}(A_n) \uparrow 1$,
5. If $A_n \in \mathcal{A}$ and $A_n \uparrow A$ then $\mathbb{P}(A_n) \uparrow \mathbb{P}(A)$,

Proof. By De Morgan's law

$$\left(\bigcup_{i=1}^{\infty} A_i \right)^c = \left(\bigcap_{i=1}^{\infty} A_i^c \right)$$

which implies that $A_i \uparrow A$ is equivalent to $A_i^c \downarrow A^c$. Similarly we have

$$\left(\bigcap_{i=1}^{\infty} A_i \right) = \left(\bigcup_{i=1}^{\infty} A_i^c \right)^c$$

which implies that $A_i \downarrow A$ is equivalent to $A_i^c \uparrow A^c$. This show the equivalence between 2. and 4. and also the equivalence between 3. and 5..

Next, by taking $A = \Omega$, it is clear that 5. implies 4.. To this end, to see that 4. implies 5. Let $(A_i)_{i \geq 1}$ be an increasing family of sets such that $A_i \uparrow A$. Then we let $B_i = A_i \cup A^c$ and we observe that $B_i \uparrow \Omega$. Then by 4. we have

$$1 = \lim_{i \uparrow \infty} \mathbb{P}(B_i) = \lim_{i \uparrow \infty} \mathbb{P}(A_i \cup A^c) = \lim_{i \uparrow \infty} \mathbb{P}(A_i) + \mathbb{P}(A^c)$$

This implies that $\mathbb{P}(A) = 1 - \mathbb{P}(A^c) = \lim_{i \uparrow \infty} \mathbb{P}(A_i)$.

Next we show that 1. and 5. are equivalent. Firstly, suppose 5. holds and we show that $\mathbb{P}$ is countably additive. Let $(A_i)_{i \geq 1}$ be a family of pairwise disjoint sets and let

$$B_n = \bigcup_{i=1}^n A_i \quad \text{and} \quad B = \bigcup_{i=1}^{\infty} A_i.$$

where we note that $B_n \uparrow B$. By finite additivity of $\mathbb{P}$

$$\mathbb{P}(B_n) = \sum_{i=1}^n \mathbb{P}(A_i) \implies \lim_{n \uparrow \infty} \mathbb{P}(B_n) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

On the other hand, from 5. we have

$$\lim_{n \uparrow \infty} \mathbb{P}(B_n) = \mathbb{P}(B) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right)$$

Assume that 1. holds. i.e $\mathbb{P}$ is finitely additive, we show that 5. is true. Let $(A_i)_{i \geq 1}$ be increasing sequence of sets such that $A_i \uparrow A$, then one can construct a sequence of pairwise disjoint sets B_i by setting

$$B_1 := A_1 \quad \text{and} \quad B_n := A_n \setminus A_{n-1} = A_n \cap A_{n-1}^c$$

We note that $(B_i)_{i \geq 1}$ is a family of pairwise disjoint sets such that $\bigcup_{i=1}^n B_i = A_n$ and $\bigcup_{i=1}^{\infty} B_i = A$. By 1. i.e countable additivity of $\mathbb{P}$, we have

$$\mathbb{P}(A) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} B_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(B_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbb{P}(B_i).$$

Then by finite additivity of $\mathbb{P}$, we have

$$\sum_{i=1}^n \mathbb{P}(B_i) = \mathbb{P}\left(\bigcup_{i=1}^n B_i\right) = \mathbb{P}(A_n)$$

and 5. holds. □

Definition 1.1.8. Given a family of sets $\mathcal{C}$, we denote by $\sigma(\mathcal{C})$ the smallest σ -algebra which contains $\mathcal{C}$ and say that $\sigma(\mathcal{C})$ is the σ -algebra generated by $\mathcal{C}$.

1.2 Random Variables and the Borel σ -algebra

In the following, we assume that one is given a probability space $(\Omega, \mathcal{A}, \mathbb{P})$.

Definition 1.2.1. Let Ω be a topological space then the *Borel σ -algebra* on Ω denoted by $\mathcal{B}(\Omega)$ is the σ -algebra generated by the open sets of Ω .

Theorem 1.2.1. Let $\Omega = \mathbb{R}$, then the Borel σ -algebra $\mathcal{B}(\mathbb{R})$ is generated by all intervals of the form $(-\infty, x]$, $x \in \mathbb{Q}$.

Proof. (Sketch) Let $\mathcal{C}$ denote the family of all open intervals. We use the fact that every open set in $\mathbb{R}$ is the countable union of open intervals (a fact from real analysis) then we can claim that $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R})$.

To this end, let $\mathcal{D}$ denote all intervals of the form $(-\infty, a]$ where $a \in \mathbb{Q}$. We show that $\mathcal{C} \subset \sigma(\mathcal{D})$ and it is sufficient to note that for $a < b$,

$$(a, b) = \bigcup_{n=1}^{\infty} \left(\left(-\infty, b - \frac{1}{n} \right] \setminus \left(-\infty, a \right] \right).$$

□

We now introduce the indicator function of a set $H \in \mathcal{A}$ given

$$\mathbf{1}_H(\omega) = \begin{cases} 1 & \omega \in H \\ 0 & \omega \notin H \end{cases}$$

Definition 1.2.2. Let $(\Omega, \mathcal{A}, \mathbb{P})$ and $(\Psi, \mathcal{G})$ be two measurable spaces. A function $X : \Omega \rightarrow \Psi$ is said to be measurable with respect to $\mathcal{A}$ and $\mathcal{G}$ if $X^{-1}(G) = \{\omega : X(\omega) \in G\} \in \mathcal{A}$ for all $G \in \mathcal{G}$.

In the special case where $(\Psi, \mathcal{G}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, we call any function $X : \Omega \rightarrow \mathbb{R}$ a (real valued) random variable if X is a measurable function with respect to $\mathcal{A}$ and $\mathcal{B}(\mathbb{R})$ or we simply say $\mathcal{A}$ -measurable. A random variable is a real valued function from the sample space Ω . In addition, suppose Ω is a topological space endowed with its Borel σ -algebra $\mathcal{B}(\Omega)$ then as a consequence of the above definition, any continuous function $X : \Omega \rightarrow \mathbb{R}$ is a random variable.

Theorem 1.2.2. *Let $\mathcal{C}$ be a class of subsets of Ψ such that $\sigma(\mathcal{C}) = \mathcal{G}$. For a function $X : \Omega \rightarrow \Psi$ to be measurable with respect to $\mathcal{A}$ and $\mathcal{G}$, it is necessary and sufficient that $X^{-1}(\mathcal{C}) = \{X^{-1}(C) : C \in \mathcal{C}\} \subseteq \mathcal{A}$.*

We note from the above that, to check if a function is measurable, it is sufficient check the inverse image of the generating family of sets belongs to $\mathcal{A}$. In particular, since $\mathcal{B}(\mathbb{R})$ is generated by intervals of the form $(-\infty, a]$, we have the following Corollary.

Corollary 1.2.1. *Let $(\Omega, \mathcal{A})$ be a measurable space and $X : \Omega \rightarrow \mathbb{R}$. X is a random variable if and only if for all $a \in \mathbb{Q}$, the set $\{\omega : X(\omega) \leq a\} = X^{-1}((-\infty, a])$ belongs to the σ -algebra $\mathcal{A}$.*

One can place different σ -algebra on a space to model information.

Example 1.2.1. Let us consider the toss of a die. In this setting $\Omega = \{1, 2, 3, 4, 5, 6\}$ and we suppose that one only observes the sets $\{3\}$ and $\{1, 2\}$. That is we set $\mathcal{A} = \{\{1, 2\}, \{3\}\}$ and

$$\mathcal{A} := \sigma(\mathcal{A}) = \{\{1, 2\}, \{3\}, \{1, 2, 3\}, \{1, 2, 4, 5, 6\}, \{4, 5, 6\}, \{3, 4, 5, 6\}, \Omega, \emptyset\}$$

note that here $\mathcal{A} \subset 2^\Omega$, i.e. we observe only a subset of all possible outcomes.

Exercise: Consider the function $X : \Omega \mapsto \mathbb{R}$ defined by $X(\omega) = \omega$, e.g, $X(6) = 6$. Is X a random variable? In other words, is X measurable with respect to the σ -algebras $\mathcal{B}(\mathbb{R})$ and $\mathcal{A}$?

To check if X is random variable, we need only to check that $X^{-1}((-\infty, a]) \in \mathcal{A}$ for all $a \in \mathbb{Q}$. Let us take $a = 6$ and it is clear that $X^{-1}((-\infty, 6]) = \{\omega : X(\omega) \leq 6\} = \{1, 2, 3, 4, 5, 6\} = \Omega$. However, when $a = 5$, we have $X^{-1}((-\infty, 5]) = \{\omega : X(\omega) \leq 5\} = \{1, 2, 3, 4, 5\} \notin \mathcal{A}$. This shows that X is not a random variable.

Definition 1.2.3. Given $X : (\Omega, \mathcal{A}) \rightarrow (\Psi, \mathcal{G})$, we denote by $\sigma(X)$ the smallest σ -algebra on Ω such that X is a random variable.

Example 1.2.2. Construction of a continuous random variables with a given distribution: Let $\Omega = ([0, 1], \mathcal{B}((0, 1)), \lambda)$ where λ is the "Lebesgue measure", that is $\lambda([a, b]) = b - a$ for $a < b \in [0, 1]$. On this space, consider the identity function, that is $X(\omega) = \omega$. First we check that X is a random variable. For any $a \in \mathbb{Q}$

$$X^{-1}((-\infty, a]) = \{\omega : X(\omega) \leq a\} = \{\omega \in [0, 1] : \omega \leq a\} = [0, a]$$

which belongs to $\mathcal{B}((0, 1))$. Secondly, we compute the CDF of X , that is

$$\mathbb{P}(X \leq a) = \lambda(\{\omega \in [0, 1] : \omega \leq a\}) = \lambda([0, a]) = a.$$

Therefore we have constructed a uniformly random variable X on $[0, 1]$.

In general, one can construct a random variable with a given probability density function by changing the measure. That is instead of the Lebesgue measure, we consider $F(dx) = f(x)dx$. That is $F([a, b]) = \int_a^b f(x)dx$ then X is a random variable with CDF given by F .

Definition 1.2.4. Two random variables X and Y are said to be equal almost surely (a.s) if the set $A = \{\omega : X(\omega) \neq Y(\omega)\}$ is of probability zero.

Example 1.2.3. Consider the probability space $([0, 1], \mathcal{B}([0, 1]), \lambda)$ and

$$X(\omega) = \omega^2$$

$$Y(\omega) = 1 \times \mathbb{1}_{\{\frac{1}{2}\}}(\omega) + \omega^2 \mathbb{1}_{[0, \frac{1}{2}) \cup (\frac{1}{2}, 1]}(\omega)$$

That is the function Y is the same as X except at the point $\omega = \frac{1}{2}$. i.e $\{\omega : X(\omega) \neq Y(\omega)\} = \{\frac{1}{2}\}$. However the probability of the point $\{\frac{1}{2}\}$ is zero. That is because

$$\lambda(\{\frac{1}{2}\}) = \lim_{n \downarrow 0} \lambda([\frac{1}{2} - \frac{1}{n}, \frac{1}{2} + \frac{1}{n}]) = \lim_{n \rightarrow \infty} \frac{2}{n} = 0.$$

This implies that $X = Y$ a.s., but $X \neq Y$.

1.2.1 Independence

Definition 1.2.5.

1. A family of sub σ -algebra $(A_i)_{i \in I}$ where I is some index set is said to be independent if for every finite subset J of I and all $A_i \in \mathcal{A}_i$,

$$\mathbb{P}\left(\bigcap_{i \in J} A_i\right) = \prod_{i \in J} \mathbb{P}(A_i)$$

2. Random variables $(X_i)_{i \in I}$ taking values in $(\Psi_i, \mathcal{G}_i)$ are independent if the σ -algebras generated by X_i , i.e $X_i^{-1}(\mathcal{G}_i)$ are independent.

Theorem 1.2.3. *In order for X and Y taking values in $(\Psi_X, \mathcal{G}_X)$ and $(\Psi_Y, \mathcal{G}_Y)$ to be independent, it is necessary and sufficient to have any one of the following:*

1. $\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$ for all $A \in \mathcal{G}_X$ and $B \in \mathcal{G}_Y$.
2. $\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$ for all $A \in \mathcal{H}_X$ and $B \in \mathcal{H}_Y$, where $\mathcal{H}_X$ and $\mathcal{H}_Y$ are classes of sets stable under finite intersection which generates $\mathcal{G}_X$ and $\mathcal{G}_Y$ respectively. i.e $\sigma(\mathcal{H}_X) = \mathcal{G}_X$ and $\sigma(\mathcal{H}_Y) = \mathcal{G}_Y$.
3. $f(X)$ and $g(Y)$ are independent for each pair of measurable function (f, g) .
4. $\mathbb{E}(f(X)g(Y)) = \mathbb{E}(f(X))\mathbb{E}(g(Y))$ for each pair of measurable function (f, g) .

From 2. of theorem 1.2.3 we see that, for any real valued random variables X and Y , to check X and Y are independent, it is sufficient to check that for all $a, b \in \mathbb{Q}$,

$$\mathbb{P}(X \leq a, Y \leq b) = \mathbb{P}(X \leq a)\mathbb{P}(Y \leq b).$$

Definition 1.2.6. The characteristic functions of a r.v X is given by

$$\Phi_X(t) = \mathbb{E}[e^{itX}]$$

where $i = \sqrt{-1}$ and $e^{it} = \cos(t) + i \sin(t)$ is bounded.

1.3 Expectation and Conditional Expectation

In this section, we introduce the notion of the Lebesgue-Stieltjes integral and conditional expectation.

1.3.1 Lebesgue-Stieltjes Integral

Definition 1.3.1. A random variable X is called *simple* if it takes on only finite number of values and hence can be written in the form

$$X = \sum_{i=1}^n x_i \mathbb{1}_{A_i}$$

where $x_i \in \mathbb{R}$ and $A_i \in \mathcal{A}$.

The expectation of a simple random variable X is defined by

$$\mathbb{E}(X) = \sum_{i=1}^n x_i \mathbb{P}(A_i)$$

and we often write $\int_{\Omega} X(\omega) \mathbb{P}(d\omega)$. The expected value of a simple random variables is a linear function on the space of simple random variables. To see this, let X and Y be two simple random variables given by

$$X = \sum_{i=1}^n x_i \mathbb{1}_{D_i^X} \quad \text{and} \quad Y = \sum_{i=1}^m y_i \mathbb{1}_{D_i^Y}$$

then it is possible to rewrite the random variable X and Y using a common family of partition given by $\{D_i^X \cap D_j^Y : i = 1, \dots, n, j = 1, \dots, m\}$. We denote this common partition by $\{A_i : i = 1, \dots, M\}$ and we can suppose X and Y is of the form

$$X = \sum_{i=1}^M x_i \mathbb{1}_{A_i} \quad \text{and} \quad Y = \sum_{i=1}^M y_i \mathbb{1}_{A_i}.$$

Let β be a real number, then it is clear that $\beta X = \sum_{i=1}^M \beta x_i \mathbb{1}_{D_i^X}$ and

$$\mathbb{E}[\beta X] = \sum_{i=1}^M \beta x_i \mathbb{P}(D_i^X) = \beta \sum_{i=1}^M x_i \mathbb{P}(D_i^X) = \beta \mathbb{E}[X].$$

On the other hand, we also have

$$X + Y = \sum_{i=1}^M (x_i + y_i) \mathbb{P}(A_i) = \sum_{i=1}^M x_i \mathbb{P}(A_i) + \sum_{i=1}^M y_i \mathbb{P}(A_i) = \mathbb{E}(X) + \mathbb{E}(Y).$$

These computations shows that the expectation $\mathbb{E}$ is a linear function on the space of simple random variables. Further more we see that if $X \leq Y$ then $\mathbb{E}(X) \leq \mathbb{E}(Y)$.

Definition 1.3.2. For any positive random variable X , we define

$$\mathbb{E}(X) := \sup \{ \mathbb{E}(Y) : Y \text{ is a positive simple random variable such that } Y \leq X \}$$

If X is a positive random variable then $\mathbb{E}(X)$ is allowed to take value infinity.

From a constructive point of view, we recall a well known result from analysis or measure theory. That is for every positive random variable X there exists a sequence $(X_n)_{n \geq 1}$ of positive bounded simple random variables such that for $n \leq m$, $X_n \leq X_m \leq X$ and $X_n \rightarrow X$ almost surely as $n \rightarrow \infty$. An example of such sequence is given by

$$X_n(\omega) = \begin{cases} k2^{-n} & \text{if } k2^{-n} \leq X(\omega) < (k+1)2^{-n} \text{ and } 0 < k \leq n2^n - 1 \\ n & \text{if } X(\omega) \geq n \end{cases}$$

Here $\mathbb{E}(X)$ is given by $\lim_{n \rightarrow \infty} \mathbb{E}(X_n)$, since the sequence is monotonically increasing then there always exists a point-wise limit.

To define the expectation for general random variable X , we consider it's positive and negative part, that is $X^+ = \max(X, 0)$ and $X^- = -\min(X, 0)$. From the representation $X = X^+ - X^-$ and the fact that both X^+ and X^- are positive random variables their expectations are well defined (see definition 1.3.2). Therefore, for a general random variable X we can define

$$\mathbb{E}(X) := \mathbb{E}(X^+) - \mathbb{E}(X^-)$$

as long as $\mathbb{E}(X^+) < \infty$ or $\mathbb{E}(X^-) < \infty$.

Definition 1.3.3. A random variable is said to be integrable if $\mathbb{E}(X) < \infty$ and the space of integrable random variables is denoted by $\mathcal{L}^1(\Omega, \mathbb{P})$.

We note that almost sure equality defines a equivalence relationship, that is $X \sim Y$ if $X = Y$ almost surely. We can then consider the equivalence class $[X] = \{X \in \mathcal{L}^1 : X \sim Y\}$ and denote by L^1 to be the space $\mathcal{L}^1$ modulo the equivalent class of almost surely equal random variables. That is

$$L^1(\Omega, \mathbb{P}) = \{[X] : X \in \mathcal{L}^1(\Omega, \mathbb{P})\}$$

Theorem 1.3.1. Let X and Y be random variables then

1. if $0 \leq X \leq Y$ and $Y \in L^1$ then $X \in L^1$ and $\mathbb{E}(X) \leq \mathbb{E}(Y)$.
2. $X \in L^1$ if and only if $|X| \in L^1$.

Similarly, for $p \geq 1$, the space $\mathcal{L}^p$ is defined to be the space of random variables X such that $|X|^p \in \mathcal{L}^1$.

1.3.2 Conditional Expectation

We first introduce the conditional expectation for a σ -algebra which are finitely generated. A finite family of sets $\mathcal{D} = \{D_1, \dots, D_n\}$ is said to be a '*decomposition*' or '*partition*' of Ω if the family of sets $\mathcal{D} = \{D_1, \dots, D_n\}$ are pairwise disjoint, i.e. $D_i \cap D_j = \emptyset$ for $i \neq j$, and $\Omega = \cup_{i=1}^n D_i$.

Given a simple random variable X , that is $X = \sum_{i=1}^n x_i \mathbb{1}_{D_i^X}$ where $D_i^X = \{X = x_i\}$. We write $\mathcal{D}(X) = \{D_1^X, \dots, D_n^X\}$ to denote the decomposition of Ω associated with X .

Definition 1.3.4. For any decomposition $\mathcal{D} = \{D_1, \dots, D_n\}$ of Ω and any event $A \in \mathcal{A}$ the conditional probability of A with respect to $\mathcal{D}$ is

$$\mathbb{P}(A|\mathcal{D}) = \sum_{i=1}^n \mathbb{P}(A|D_i) \mathbb{1}_{D_i}$$

where $\mathbb{P}(A|D_i) = \mathbb{P}(A \cap D_i) \mathbb{P}(D_i)^{-1}$, i.e. the usual conditional probability. Moreover if X is a simple random variable, i.e. $X = \sum_{i=1}^m x_i \mathbb{1}_{D_i^X}$, then its conditional expectation given $\mathcal{D}$ is

$$\mathbb{E}[X|\mathcal{D}] = \sum_{i=1}^m x_i \mathbb{P}(D_i^X|\mathcal{D}) = \sum_{i=1}^m \sum_{j=1}^n x_i \mathbb{P}(D_i^X|D_j) \mathbb{1}_{D_j}.$$

It is important to note that both $\mathbb{P}(A|\mathcal{D})$ and $\mathbb{E}(X|\mathcal{D})$ are again simple random variable. One can similarly define the conditional expectation with respect to a simple random variables. Suppose Y is another simple random variable, that is $Y = \sum_{i=1}^n y_i \mathbb{1}_{D_i^Y}$, then we set

$$\begin{aligned} \mathbb{E}[X|Y] &:= \mathbb{E}[X|\mathcal{D}(Y)] = \sum_{i=1}^m x_i \mathbb{P}(D_i^X|\mathcal{D}(Y)) \\ &= \sum_{i=1}^m \sum_{j=1}^n x_i \mathbb{P}(D_i^X|D_j^Y) \mathbb{1}_{D_j^Y} = \sum_{j=1}^n c_j \mathbb{1}_{D_j^Y} \end{aligned}$$

where we set $c_j = \sum_{i=1}^m x_i \mathbb{P}(D_i^X|D_j^Y)$.

The two important observations which we make is that (i) the quantity $\mathbb{E}(X|\mathcal{D}(Y))$ is again a simple random variable and it is $\sigma(Y)$ -measurable. (ii) The random variable $\mathbb{E}[X|\mathcal{D}(Y)]$ is equal to $g(Y)$ for some measurable function g . These properties are also true for general conditional expectations.

Theorem 1.3.2. Let X and Y be two simple random variables then

1. If $\mathcal{D}(X) \subseteq \sigma(\mathcal{D})$ then $\mathbb{E}(X|\mathcal{D}) = X$.
2. (Linearity) For $a, b \in \mathbb{R}$, $\mathbb{E}[aX + bY|\mathcal{D}] = a\mathbb{E}[X|\mathcal{D}] + b\mathbb{E}[Y|\mathcal{D}]$.
3. (Tower Property) If $\sigma(\mathcal{D}_1) \subseteq \sigma(\mathcal{D}_2)$ then

$$\mathbb{E}[\mathbb{E}[X|\mathcal{D}_1]|\mathcal{D}_2] = \mathbb{E}[\mathbb{E}[X|\mathcal{D}_2]|\mathcal{D}_1] = \mathbb{E}[X|\mathcal{D}_1].$$

4. If X is independent of $\mathcal{D}$ in the sense that $A \in \mathcal{D}(X)$ and $B \in \mathcal{D}$ then $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ then

$$\mathbb{E}[X|\mathcal{D}] = \mathbb{E}[X]$$

5. if X is $\mathcal{D}$ -measurable and Y is independent of $\mathcal{D}$ then for any function $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ we have

$$\mathbb{E}[h(X, Y)|\mathcal{D}] = H(X);$$

where the function $H : \mathbb{R} \rightarrow \mathbb{R}$ is given by the formula $H(x) = \mathbb{E}[h(x, Y)]$.

Proof. Left for exercise. □

Remark 1.3.1. The formula $\mathbb{E}[h(X, Y)|\mathcal{D}] = H(X)$ is often applied to the situation, where X represents the present, Y is the future and $\mathcal{D}$ is the information generated by past events.

Definition 1.3.5. (Conditional Expectation) Let X be a integrable random variable on $(\Omega, \mathcal{A}, \mathbb{P})$. Given a arbitrary σ -field $\mathcal{G} \subset \mathcal{A}$ then the conditional expectation of X with respect to $\mathcal{G}$, denoted by $\mathbb{E}[X|\mathcal{G}]$, is the unique integrable random variable satisfying the following conditions

1. $\mathbb{E}[X|\mathcal{G}]$ is a $\mathcal{G}$ -measurable random variable.
2. For any $A \in \mathcal{G}$, we have

$$\mathbb{E}[\mathbb{1}_A X] = \mathbb{E}[\mathbb{1}_A \mathbb{E}[X|\mathcal{G}]]$$

or in integral form $\int_A \mathbb{E}[X|\mathcal{G}](\omega) d\mathbb{P}(\omega) = \int_A X(\omega) d\mathbb{P}(\omega)$.

Exercise: Let $X = c$, that is X takes only a constant value then show that $\sigma(X)$ is the trivial σ -algebra $\mathcal{A}_0 = \{\Omega, \emptyset\}$. Also show that $\mathbb{E}[\cdot | \mathcal{A}_0] = \mathbb{E}[\cdot]$.

Exercise: Given a partition $\mathcal{D}$ of Ω , Check that $\mathbb{P}(A|\sigma(\mathcal{D})) = \mathbb{P}(A|\mathcal{D})$ for all $A \in \mathcal{A}$.

Hint: From the uniqueness of conditional expectation, it is sufficient to check for all $B \in \sigma(\mathcal{D})$,

$$\mathbb{E}[\mathbb{E}[\mathbb{1}_A|\mathcal{D}]\mathbb{1}_B] = \mathbb{E}[\mathbb{1}_A\mathbb{1}_B].$$

Exercise: For a simple random variable X , show that the σ -algebra $\sigma(X) = \sigma(\mathcal{D}(X))$.

Hint: Recall that $\sigma(X)$ is the smallest σ -algebra which makes X measurable (see definition 1.2.3).

Exercise: Let Ω be the outcomes of a fair six faced die and we consider the family $\mathcal{D} := \{\{2\}, \{4\}, \{6\}, \{1, 3, 5\}\}$. Suppose your pay-off is y if you have rolled y , e.g. you receive 2 dollars if you rolled a 2, or $P(x) = x$ for $x \in \{1, 2, 3, 4, 5, 6\}$.

1. Compute $\mathbb{E}[P|\mathcal{D}]$ and show that it is $\mathcal{D}$ measurable.
2. From the result of part 1., compute $\mathbb{E}[P]$.

3. Show that $3\mathbb{1}_{\{3\}}$ is not a $\mathcal{D}$ measurable random variable
4. Give another example of an $\mathcal{D}$ measurable random variable.

To proof the existence and uniqueness of the conditional expectation, we require the following result from measure theory.

Theorem 1.3.3 (Radon-Nikodym Theorem). *Let $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures on $(\Omega, \mathcal{A})$ such that for all $A \in \mathcal{A}$, $\mathbb{P}(A) = 0$ implies that $\mathbb{Q}(A) = 0$, in other words $\mathbb{Q}$ is absolutely continuous with respect to $\mathbb{P}$ (we write $\mathbb{Q} \ll \mathbb{P}$), then there exist a unique positive $\mathbb{P}$ -integrable random variable η (the Radon-Nikodym density) such that for all $A \in \mathcal{A}$, $\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}}(\mathbb{1}_A \eta)$.*

Example 1.3.1. Consider the space $\Omega = \{T, H\}$, that is the coin flip example, and the filtration $\mathbb{F} = (\mathcal{F}_n)_{n=0,1}$ where $\mathcal{F}_0 = \{\emptyset, \Omega\}$ and $\mathcal{F}_1 = \{\emptyset, \Omega, \{T\}, \{H\}\}$. Consider a probability $\mathbb{P}$ defined on $\mathcal{F}_1$, given by

$$\mathbb{P}(\{H\}) = \mathbb{P}(\{T\}) = \frac{1}{2}$$

and another probability measure $\tilde{\mathbb{P}}$, defined by

$$\mathbb{Q}(\{H\}) = \frac{1}{3} \quad \text{and} \quad \mathbb{Q}(\{T\}) = \frac{2}{3}$$

The probability measure $\mathbb{P}$ is equivalent to $\tilde{\mathbb{P}}$ on $\mathcal{F}_1$, since the null set under $\mathbb{P}$ (which is only the empty set here) is also a null set under $\tilde{\mathbb{P}}$ and vice versa. One can compute the Radon-Nikodým density of $\tilde{\mathbb{P}}$ with respect to $\mathbb{P}$. We note that

$$\begin{aligned} \mathbb{Q}(\{T\}) &= \mathbb{E}_{\mathbb{P}}[\eta \mathbb{1}_{\{T\}}] = \int_{\{T\}} \eta(\omega) d\mathbb{P}(\omega) = \eta(T) \mathbb{P}(\{T\}) \\ \mathbb{Q}(\{H\}) &= \mathbb{E}_{\mathbb{P}}[\eta \mathbb{1}_{\{H\}}] = \int_{\{H\}} \eta(\omega) d\mathbb{P}(\omega) = \eta(H) \mathbb{P}(\{H\}). \end{aligned}$$

The Radon-Nikodým density is then given by the $\mathcal{F}_1$ measurable random variable $\eta(\omega) = \frac{\mathbb{Q}(\omega)}{\mathbb{P}(\omega)}$ for $\omega \in \{T, H\}$.

Example 1.3.2. Consider $\Omega = [0, \infty)$ and $\mathcal{A} = \mathcal{B}([0, \infty))$. One can define two equivalent probability measure on $(\Omega, \mathcal{A})$ by setting, for every $A \in \mathcal{A}$,

$$\begin{aligned} \mathbb{P}(A) &= \int_0^\infty \mathbb{1}_A(x) \mathbb{1}_{[0,1]} dx = \int_0^\infty \mathbb{1}_A(x) \mathbb{P}(dx) \\ \mathbb{Q}(A) &= \int_0^\infty \mathbb{1}_A(x) e^{-x} dx = \int_0^\infty \mathbb{1}_A(x) \mathbb{Q}(dx) \end{aligned}$$

The probability measure $\mathbb{P}$ is the uniform measure on $[0, 1]$ and $\mathbb{Q}$ is the exponential measure. It is not difficult to see that $\mathbb{P}$ is absolutely continuous with respect to $\mathbb{Q}$ but not the reverse. To find the Radon-Nikodym density of $\mathbb{Q}$ with respect to $\mathbb{P}$ we note that for any $A \in \mathcal{A}$,

$$\mathbb{P}(A) = \int \mathbb{1}_A(x) \mathbb{1}_{[0,1]}(x) dx = \int \mathbb{1}_A(x) \frac{\mathbb{1}_{[0,1]}(x)}{e^{-x}} e^{-x} dx = \int \mathbb{1}_A(x) \frac{\mathbb{1}_{[0,1]}(x)}{e^{-x}} \mathbb{Q}(dx)$$

Therefore the Radon-Nikodym density of $\mathbb{P}$ wrt $\mathbb{Q}$ is given by $\eta(x) = \frac{\mathbb{1}_{[0,1]}(x)}{e^{-x}}$.

Theorem 1.3.4. *Let $\mathcal{G}$ be a sub- σ -algebra of $\mathcal{A}$. Then*

1. *there exists a conditional expectation $\mathbb{E}[X|\mathcal{G}]$ for any $X \in L^1$, and*
2. *any two conditional expectations of $X \in L^1$ respective to $\mathcal{G}$ are equal $\mathbb{P}$ -a.s.*

Proof. Existence: Suppose $X \in L^1(\Omega, \mathbb{P})$ then $X^\pm \in L^1$. Without loss of generality, we suppose that $X \geq 0$. Define a new probability measure $\mathbb{Q}$ on $\mathcal{A}$ by setting for any $A \in \mathcal{A}$,

$$\mathbb{Q}(A) := \frac{\mathbb{E}[X\mathbf{1}_A]}{\mathbb{E}[X]}$$

The probability $\mathbb{Q}$ is absolutely continuous with respect to $\mathbb{P}$ on $\mathcal{A}$ (more specifically, for $A \in \mathcal{A}$, $\mathbb{P}(A) = 0$ implies $\mathbb{Q}(A) = 0$). This implies that $\mathbb{Q}$ is absolutely continuous with respect to $\mathbb{P}$ on $\mathcal{G} \subset \mathcal{A}$. Therefore from the Radon-Nikodym Theorem or theorem 1.3.3, there exists a positive $\mathcal{G}$ -measurable Radon-Nikodym derivative $\zeta = d\mathbb{Q}/d\mathbb{P}|_{\mathcal{G}} \in L^1(\Omega, \mathbb{P})$ such that

$$\mathbb{E}[X\mathbf{1}_A] =: \mathbb{E}[X]\mathbb{Q}(A) = \mathbb{E}[X]\mathbb{E}[\zeta\mathbf{1}_A]$$

then $\mathbb{E}[X]\zeta$ is a version of the conditional expectation $\mathbb{E}[X|\mathcal{G}]$.

Uniqueness: Suppose Y and Y' are both $\mathcal{G}$ -conditional expectation of X , i.e $\mathbb{E}[X|\mathcal{G}]$. Let $G = \{\omega : Y(\omega) > Y'(\omega)\}$ and we assume that $\mathbb{P}(G) > 0$. To this end, we note that

$$G := \{Y - Y' > 0\} = \bigcup_{n=1}^{\infty} \{Y - Y' > n^{-1}\}$$

$$G_n := \bigcup_{j=1}^n \{Y - Y' > j^{-1}\} = \{Y - Y' > n^{-1}\}$$

By theorem 1.1.2, we have $G_n \uparrow G$ implies $\mathbb{P}(G_n) \uparrow \mathbb{P}(G)$ and from which we can conclude that there exists $m > 0$ such that $\mathbb{P}(G_m) > 0$.

Since Y and Y' are both conditional expectations given $\mathcal{G}$, we have by 2. of definition 1.3.5 that for every $A \in \mathcal{G}$, in particular G , we have $\mathbb{E}[\mathbf{1}_G(Y - Y')] = 0$. On the other hand

$$\mathbb{E}[\mathbf{1}_G(Y - Y')] \geq \mathbb{E}[\mathbf{1}_{G_m}(Y - Y')] \geq \frac{1}{m}\mathbb{E}[G_m] > 0$$

This is a contradiction and hence $\mathbb{P}(G) = 0$. □

Theorem 1.3.5. *Let X and Y be two integrable random variables on $(\Omega, \mathcal{A}, \mathbb{P})$, and $\mathcal{G}$ and $\mathcal{H}$ are two sub σ -algebras of $\mathcal{A}$. Then*

1. *If X is $\mathcal{G}$ measurable or equivalently $\sigma(X) \subseteq \mathcal{G}$ then $\mathbb{E}[X|\mathcal{G}] = X$.*
2. *(Linearity) For $a, b \in \mathbb{R}$, $\mathbb{E}[aX + bY|\mathcal{G}] = a\mathbb{E}[X|\mathcal{G}] + b\mathbb{E}[Y|\mathcal{G}]$.*

3. (Tower Property) If $\mathcal{H} \subset \mathcal{G}$ then

$$\mathbb{E}[\mathbb{E}[X|\mathcal{G}]|\mathcal{H}] = \mathbb{E}[\mathbb{E}[X|\mathcal{H}]|\mathcal{G}] = \mathbb{E}[X|\mathcal{H}].$$

in particular, if $\mathbb{E}[X|\mathcal{G}]$ is $\mathcal{H}$ -measurable then $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X|\mathcal{H}]$ a.s.

4. If X is independent of $\mathcal{G}$ in the sense that $\sigma(X)$ is independent of $\mathcal{G}$, then

$$\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$$

5. if X is $\mathcal{G}$ -measurable and Y is independent of $\mathcal{G}$ then for any function $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ we have

$$\mathbb{E}[h(X, Y)|\mathcal{G}] = H(X);$$

where the function $H : \mathbb{R} \rightarrow \mathbb{R}$ is given by the formula $H(x) = \mathbb{E}[h(x, Y)]$.

To give some intuition on the computation done in 5. of theorem 1.3.5 above, we note that since X is $\mathcal{G}$ -measurable, then X is "known with respect to the information of $\mathcal{G}$ ", using 1. of theorem 1.3.5 and one can "view it as a constant and take it out of the conditional expectation"

$$\mathbb{E}[h(X, Y)|\mathcal{G}] = \mathbb{E}[h(x, Y)|\mathcal{G}] \Big|_{x=X}$$

then by using the fact that Y is independent of $\mathcal{G}$, we can then use 4. of theorem 1.3.5 to obtain

$$\mathbb{E}[h(x, Y)|\mathcal{G}] \Big|_{x=X} = \mathbb{E}[h(x, Y)] \Big|_{x=X} = H(x) \Big|_{x=X}$$

and finally the claimed result by substituting X back into H .

Theorem 1.3.6 (Doob's measurability theorem). *Let X be a mapping from Ω to a measurable $(\Psi, \mathcal{G})$ (e.g the space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$) and Y be a real function on Ω . In order for Y to be $\sigma(X)$ -measurable, it is necessary and sufficient that there exist an $\mathcal{G}$ -measurable real function h on Ψ such that $Y = h(X)$.*

The Doob's measurability theorem tells us that $\mathbb{E}[Y|\sigma(X)]$, where Y is some integrable random variable, must be a function of X .

Exercise: Let Y and X be two square integrable random variables. Show that $\mathbb{E}[Y|\sigma(X)]$ is the unique $\sigma(X)$ -measurable random variable which minimises the quantity $\mathbb{E}[(Y - Z)^2]$ where Z ranges over all square integrable $\sigma(X)$ -measurable random variables.

Lemma 1.3.1. (Jensen's inequality) *Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function and X a r.v such that $\mathbb{E}[|g(X)|] < \infty$.*

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)]$$

(Conditional Jensen's inequality) *Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function and for any σ -algebra $\mathcal{G} \subset \mathcal{A}$*

$$g(\mathbb{E}[X|\mathcal{G}]) \leq \mathbb{E}[g(X)|\mathcal{G}]$$

Chapter 2

Filtrations and Stochastic Processes

Let us consider a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ equipped with a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]}$. A *filtration* is an increasing family of σ -fields, that is, we have that $\mathcal{F}_u \subseteq \mathcal{F}_t$ for any $0 \leq u \leq t \leq T$. For concreteness, we assume that T is a finite strictly positive real number and we set $\mathcal{A} = \mathcal{F}_T$. Hence in what follows we shall write $(\Omega, \mathbb{F}, \mathbb{P})$ instead of $(\Omega, \mathcal{A}, \mathbb{F}, \mathbb{P})$.

Definition 2.0.1. A real-valued stochastic process $(X_t)_{t \in \mathcal{I}}$ is a collection of $\mathcal{A}$ -measurable random variables X_t where t belongs to some index set $\mathcal{I}$.

Alternatively, one can think of a real-valued stochastic process X as map from $\Omega \times \mathcal{I}$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ which is measurable with respect to $\mathcal{B}(\mathbb{R})$ and the product σ -algebra $\sigma(\mathcal{A} \otimes \mathcal{B}(\mathcal{I}))$. For discrete time process, we let $\mathcal{I} = \{0, 1, 2, \dots, N\}$ for some $N \geq 1$ and write X_n for $n \in \mathcal{I}$.

Definition 2.0.2. A real-valued stochastic process $X = (X_t)_{t \in [0, T]}$ defined on $(\Omega, \mathbb{F}, \mathbb{P})$ is $\mathbb{F}$ -*adapted* if for any $t \in [0, T]$ the random variable X_t is $\mathcal{F}_t$ -measurable, that is, for any $x \in \mathbb{Q}$ the event $\{X_t \leq x\} \in \mathcal{F}_t$.

Similarly, an $\mathbb{R}^k$ -valued stochastic process $X = (X^1, \dots, X^k)$ defined on $(\Omega, \mathbb{F}, \mathbb{P})$ is said to be $\mathbb{F}$ -*adapted* if for any $t \in [0, T]$ the $\mathbb{R}^k$ -valued random variable $X_t = (X_t^1, \dots, X_t^k)$ is $\mathcal{F}_t$ -measurable, i.e., the event

$$\{X_t^1 \leq x_1, X_t^2 \leq x_2, \dots, X_t^k \leq x_k\}$$

belongs to the σ -algebra $\mathcal{F}_t$ for any choice of real numbers $x_1, \dots, x_k \in \mathbb{R}$.

Often we shall say that a process is adapted, rather than $\mathbb{F}$ -adapted, if the filtration $\mathbb{F}$ is and that there is no danger of confusion. Let us mention that any stochastic process X is adapted to its *natural filtration* (i.e., the filtration *generated* by X), which is defined as $\mathcal{F}_t^X = \sigma(X_u, u \leq t)$.

In the sequel all continuous time stochastic processes are assumed to be a *càdlàg* (continue à droite et limite à gauche) process, that is, almost all sample paths are

right-continuous functions with finite left-hand limits. We say the a stochastic process is *continuous* if almost all sample paths are continuous functions.

Definition 2.0.3. We say that the processes X and Y defined on a common probability space are *indistinguishable* if the event $\{X_t = Y_t, \forall t \in [0, T]\}$ has probability 1.

Definition 2.0.4. We say that the processes X and Y are modifications of each other if for all $t \geq 0$, $\mathbb{P}(X_t = Y_t) = 1$.

If X and Y are indistinguishable then X and Y are modifications of each other. However the converse is not true. In discrete time these two concepts coincide.

Exercise: Two discrete time processes X and Y are indistinguishable if and only if X and Y are modifications of each other.

Lemma 2.0.1. *If X and Y two càdlàg processes which are modification of each other, then X and Y are indistinguishable*

Example 2.0.1. Consider the process $X_t : \omega \rightarrow t^2$ for $t \in [0, T]$. The process X_t just a deterministic function, that is no matter which outcome ω you take, the sample path or trajectory of X is just $X_t = t^2$. It is continuous and differentiable for every ω . The natural filtration $\mathbb{F}^X$ generated by X is simply the trivial filtration given by $\mathcal{F}_t^X = \{\Omega, \emptyset\}$ for all $t \in [0, T]$.

Example 2.0.2. We give now an example where X and Y are modifications of each other, but not distinguishable. Consider the space $([0, 1], \mathcal{B}([0, 1]), \mathbb{P})$, where the probability $\mathbb{P}$ is the Lebesgue measure, i.e $\mathbb{P}([a, b]) = b - a$.

Let $(Y_t)_{t \in [0, 1]}$ denote a constant process given by $Y_t = 0$ and $(X_t)_{t \in [0, 1]}$ given by

$$X_t(\omega) = \begin{cases} 1 & \omega = t \\ 0 & \omega \neq t. \end{cases}$$

Note that here for a fixed ω , the trajectories $X_t(\omega)$ and $Y_t(\omega)$ differs only at the point $t = \omega$. To see that X and Y are modifications of each other, for every $t \in [0, 1]$, we have

$$\{\omega : X_t(\omega) = Y_t(\omega)\} = \{\omega : \omega \neq t\} = \Omega \setminus \{\omega : \omega = t\}.$$

This shows that $\mathbb{P}(X_t = Y_t) = 1 - \mathbb{P}(\{t\}) = 1$, since the probability (length) of a point is zero. Here the process X is not a right continuous process (càdlàg), therefore one cannot conclude that X and Y are indistinguishable from lemma 2.0.1. In fact, we see that

$$\{\omega : X_t(\omega) = Y_t(\omega), \forall t \in [0, 1]\} = \bigcap_{t \in [0, 1]} \{\omega : X_t(\omega) = Y_t(\omega)\} = \emptyset$$

since the complement is given by

$$\bigcup_{t \in [0, 1]} \{\omega : X_t(\omega) \neq Y_t(\omega)\} = \bigcup_{t \in [0, 1]} \{\omega : \omega = t\} = \Omega.$$

which means $(X_t)_{t \in [0, 1]}$ and $(Y_t)_{t \in [0, 1]}$ cannot be indistinguishable.

Example 2.0.3. Let $\Omega = \{T, H\}$ and define the random function

$$X_t(\omega) = \begin{cases} \sin(t), & \omega = T \\ \cos(t), & \omega = H \end{cases}$$

Therefore there is a 50% chance that we draw a sine curve and 50% chance that we draw a cosine curve.

Example 2.0.4. Consider the flip of a coin. That is $\Omega = \{T, H\}$, $\mathbb{P}(H) = p$ and $\mathbb{P}(T) = 1 - p$. Consider the random variable $Z(T) = 2$ and $Z(H) = 1$ and consider the stochastic process $(X_t)_{t \geq 0}$ defined by

$$X_t(\omega) = Z(\omega)t = \begin{cases} 2t & \omega = T \\ t & \omega = H. \end{cases}$$

Here one flips a coin and depending on the outcome of the coin, the process is a straight line with a different slope. One can also check that the natural filtration generated by X is given by $\mathcal{F}_t^X = \{\{T\}, \{H\}, \Omega, \emptyset\}$.

2.1 Martingales, Submartingales and Supermartingales

Generally speaking, a martingale is an $\mathbb{F}$ -adapted stochastic process which has the property that the conditional expected value of its future value at time t based on the information up to time u , i.e $\mathcal{F}_u$, coincides with its value at time u . The concept is related to the idea of a *fair game*, in that the expected future payoff should match the current stake.

Definition 2.1.1. A real-valued, $\mathbb{F}$ -adapted process $M = (M_t)_{t \in [0, T]}$, defined on a filtered probability space $(\Omega, \mathcal{A}, \mathbb{F}, \mathbb{P})$, is called an $\mathbb{F}$ -*martingale* with respect to the filtration $\mathbb{F}$ if the following conditions are satisfied:

1. M is integrable, that is, $\mathbb{E}[|M_t|] < \infty$ for $t \in [0, T]$,
2. the following *martingale equality* holds, for any $0 \leq s \leq t \leq T$,

$$\mathbb{E}(M_t | \mathcal{F}_s) = M_s. \tag{2.1}$$

If condition 2. in the above definition is replaced by the inequality

$$\mathbb{E}(M_t | \mathcal{F}_s) \leq M_s$$

then M is said to be an $\mathbb{F}$ -*supermartingale*. Finally, M is called an $\mathbb{F}$ -*submartingale* if for any $0 \leq s \leq t \leq T$

$$\mathbb{E}(M_t | \mathcal{F}_s) \geq M_s.$$

The expected value of a martingale is constant, $\mathbb{E}[M_t] = \mathbb{E}[M_0]$ for $t \in [0, T]$. For a supermartingale, the expected value is a non-increasing function, that is, $\mathbb{E}[M_t] \leq \mathbb{E}[M_0]$ for any $t \in [0, T]$. Finally, for a submartingale we have that $\mathbb{E}[M_t] \geq \mathbb{E}[M_0]$ for any $t \in [0, T]$.

Example 2.1.1. (Martingales, Submartingales, Supermartingales) Consider the two period stock model. That is $S_0 = s_0$, $S_1 = s_0 Z_1$ and $S_2 = s_0 Z_1 Z_2$, where Z_1, Z_2 are i.i.d random variables such that

$$Z_i = \begin{cases} u & \text{probability } p \\ d & \text{probability } (1 - p) \end{cases}$$

where $u \geq 1$, $d \leq 1$ is assumed to satisfy the condition $up + d(1 - p) = 1$. Then S is a martingale in its natural filtration $\mathbb{F}^S$. It is clear that S_i are bounded by $s_0 u^2$ and therefore integrable.

To check the martingale condition, we consider $\mathbb{E}[S_2 | \mathcal{F}_1^S]$ and we notice that s_0 is a constant and Z_1 is measurable with respect to $\sigma(s_0, Z_1)$ and Z_1 and Z_2 are independent. Therefore by using the properties 1, 2 and 4 of the conditional expectation given in theorem 1.3.5 we have

$$\begin{aligned} \mathbb{E}[S_2 | \mathcal{F}_1^S] &= \mathbb{E}[S_2 | \sigma(s_0, S_1)] = \mathbb{E}[s_0 Z_1 Z_2 | \sigma(s_0, Z_1)] \\ &= s_0 Z_1 \mathbb{E}[Z_2 | \sigma(s_0, Z_1)] \\ &= s_0 Z_1 \mathbb{E}[Z_2] \\ &= s_0 Z_1 = S_1 \end{aligned}$$

where in the third last equality above we have used the fact that

$$\mathbb{E}[Z_2] = up + d(1 - p) = 1.$$

Similarly $\mathbb{E}[S_1 | \mathcal{F}_0^S] = \mathbb{E}[s_0 Z_1 | \sigma(s_0)] = s_0 \mathbb{E}[Z_1]$.

Remark 2.1.1. If we have chosen the parameters such that $up + d(1 - p) < 1$ then $S = (S_i)_{i=0,1,2}$ is a supmartingale with respect to its natural filtration. In addition, if S is a martingale then the process $(S_t - K)^+ = \max(S_t - K, 0)$ is a submartingale, since the function $f(x) = (x - K)^+$ is convex and we can apply conditional Jensen's inequality given in lemma 1.3.1.

Example 2.1.2. Suppose $x \in \mathbb{R}$ and let

$$S_n = x + X_1 + \dots + X_n \quad 1 \leq n \leq N;$$

where $X_1, X_2, \dots$ are independent and identically distributed and

$$\mathbb{P}(X_k = 1) = p = 1 - \mathbb{P}(X_k = -1), \quad k \geq 1 :$$

It is easy to see that $\mathbb{E}[X_k] = 2p - 1$ and for $n \geq 0$,

$$S_{n+1} = S_n + X_{n+1}$$

Let $\mathbb{F} = (\mathcal{F}_n)_{n=0,1,\dots,N}$ where $\mathcal{F}_n = \sigma(X_1 \dots X_n)$. It follows that $\mathbb{E}(S_{n+1} | \mathcal{F}_n) = \mathbb{E}(S_n | \mathcal{F}_n) + \mathbb{E}(X_{n+1} | \mathcal{F}_n)$. Since S_n is $\mathcal{F}_n$ -measurable and X_{n+1} is independent of $\mathcal{F}_n$ we obtain

$$\mathbb{E}(S_{n+1} | \mathcal{F}_n) = S_n + \mathbb{E}(X_{n+1}) = S_n + 2p - 1$$

Therefore (S_n) is a $\mathbb{F}$ -martingale if $p = 1$, a $\mathbb{F}$ -submartingale if $p > 1/2$ and a $\mathbb{F}$ -supermartingale if $p < 1/2$.

Assume now that $p = 1/2$ and let

$$M_n = S_n^2 - n$$

The process (M_n) is adapted to the $\mathbb{F}$ and clearly $\mathbb{E}[|M_n|] < \infty$. To prove the martingale condition, we note that

$$\begin{aligned} \mathbb{E}[M_{n+1}|\mathcal{F}_n] &= \mathbb{E}[(S_{n+1}^2 - (n+1))|\mathcal{F}_n] \\ &= \mathbb{E}[(S_n + X_{n+1})^2 - (n+1)|\mathcal{F}_n] \\ &= S_n^2 + 2S_n\mathbb{E}[X_{n+1}|\mathcal{F}_n] + \mathbb{E}[X_{n+1}^2|\mathcal{F}_n] - (n+1) \\ &= M_n \end{aligned}$$

The computation allows from the fact that X_{n+1} is independent of σ -algebra $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$.

2.2 Stopping Times and Martingales

2.2.1 Discrete Time

We present below some definition and results in discrete time, analogous results also hold in continuous time.

Definition 2.2.1. A random variable $\tau : (\Omega, \mathbb{F}, \mathbb{P}) \rightarrow \{0, 1, \dots, N\}$ is called a *stopping time* with respect to the filtration $\mathbb{F}$ (or briefly, an $\mathbb{F}$ -*stopping time*) if, for every $n \in \{0, 1, \dots, N\}$, the event $\{\tau \leq n\}$ belongs to the σ -algebra $\mathcal{F}_n$.

Example 2.2.1. Let us consider independent repeated trials where there are two possible outcomes, i.e. success or failure. More explicitly, we define for $n \geq 1$

$$X_n = \begin{cases} 1, & \text{if the } n\text{-th trial succeeds} \\ 0, & \text{if the } n\text{-th trial fails,} \end{cases}$$

Let $(\mathcal{F}_n)_{n=1, \dots, N}$ be the natural filtration of (X_n) . Denote for $n > 1$

$$T_1 = \inf\{n \geq 1 : X_n = 1\} \quad \text{and} \quad T_{n+1} = \inf\{n > T_n : X_n = 1\}.$$

Then T_n is the waiting time for n -th success. Since (X_n) is i.i.d., one can prove that $T_1, T_2 - T_1, \dots, T_n - T_{n-1}$ are i.i.d and T_1 follows geometric distributed.

Exercise: Show that T_i for $i = 1, \dots, N$, in example 2.2.1, are stopping times with respect to the natural filtration generated by the process $(X_n)_{n=1, \dots, N}$.

Exercise: Argue that $\sup\{n \leq N : X_n = 1\}$, in example 2.2.1, is not a stopping time.

Lemma 2.2.1. *Let $X = (X_n)_{n=0,1,\dots,N}$ be a $\mathbb{F}$ -adapted process and τ be a $\mathbb{F}$ -stopping time. Then the stopped process X^τ is also $\mathbb{F}$ -adapted*

Proof. Let $X = (X_n)_{n=0,1,2,\dots,N}$ be a $\mathbb{F} = (\mathcal{F}_n)_{n=1,\dots,N}$ adapted process and τ be a $\mathbb{F}$ -stopping time taking values in $\{0, 1, 2, \dots, N\}$. We show that $X_n^\tau \in \mathcal{F}_n$ for all $n = 1, \dots, N$. Let $c \in \mathbb{Q}$ and we consider the set

$$\begin{aligned} \{X_{n \wedge \tau} \leq c\} &= \bigcup_{k=1}^N \{X_{n \wedge k} \leq c\} \cap \{\tau = k\} \\ &= \bigcup_{k=1}^{n-1} \{X_k \leq c\} \cap \{\tau = k\} \cup \bigcup_{k=n}^N \{X_n \leq c\} \cap \{\tau = k\} \\ &= A \cup B \end{aligned}$$

where $A = \bigcup_{k=1}^{n-1} \{X_k \leq c\} \cap \{\tau = k\}$ and $B = \bigcup_{k=n}^N \{X_n \leq c\} \cap \{\tau = k\}$.

We show that both A and B belongs to $\mathcal{F}_n$. Recall that since τ is a $\mathbb{F}$ -stopping time then $\{\tau \leq k\} \in \mathcal{F}_k$ and $\{\tau \leq k-1\} \in \mathcal{F}_{k-1} \subset \mathcal{F}_k$ which implies that $\{\tau = k\} = \mathcal{F}_k$. This shows that the set A belongs to $\mathcal{F}_n$ since for $k \leq n-1$, $\{X_k \leq c\} \in \mathcal{F}_k \subset \mathcal{F}_n$ and $\{\tau = k\} \in \mathcal{F}_k \subset \mathcal{F}_n$. On the other hand, the set B can be written as the follows

$$\bigcup_{k=n}^N \{X_n \leq c\} \cap \{\tau = k\} = \{X_n \leq c\} \cap \bigcup_{k=n}^N \{\tau = k\} = \{X_n \leq c\} \cap \{\tau \geq n\}$$

which is the intersection of two $\mathcal{F}_n$ -measurable sets and is therefore in $\mathcal{F}_n$.

Lemma 2.2.2. *Let $M = (M_n)_{n=0,1,\dots,N}$ be a $\mathbb{F}$ martingale and τ be a $\mathbb{F}$ -stopping time. Then the stopped process M^τ is also a $\mathbb{F}$ -martingale.*

Proof. Let $M = (M_n)_{n=0,\dots,N}$ be a discrete martingale and τ be a stopping time. It is sufficient to proof for all n that for all $n = 1, \dots, N$,

$$\mathbb{E}(M_n^\tau | \mathcal{F}_{n-1}) = M_{n-1}^\tau.$$

To this end, by using a telescoping sum, we see that

$$\begin{aligned} \mathbb{E}(M_n^\tau | \mathcal{F}_{n-1}) &= \mathbb{E} \left(\sum_{k=1}^{\tau \wedge n} (M_k - M_{k-1}) + M_0 \middle| \mathcal{F}_{n-1} \right) \\ &= \mathbb{E} \left(\mathbb{1}_{\{\tau \geq n\}} \sum_{k=1}^n M_k - M_{k-1} \middle| \mathcal{F}_{n-1} \right) + \mathbb{E} \left(\mathbb{1}_{\{\tau < n\}} \sum_{k=1}^{\tau} M_k - M_{k-1} \middle| \mathcal{F}_{n-1} \right) + M_0. \end{aligned}$$

Next, we note that $\{\tau \geq n\} = \{\tau > n-1\}$ and $\{\tau < n\} = \{\tau \leq n-1\}$ which are both contain in $\mathcal{F}_{n-1}$. Then

$$\begin{aligned} \mathbb{E} \left(\mathbb{1}_{\{\tau \geq n\}} \sum_{k=1}^n M_k - M_{k-1} \middle| \mathcal{F}_{n-1} \right) &= \mathbb{1}_{\{\tau > n-1\}} \sum_{k=1}^n \mathbb{E}(M_k - M_{k-1} | \mathcal{F}_{n-1}) \\ &= \mathbb{1}_{\{\tau > n-1\}} \sum_{k=1}^{n-1} (M_k - M_{k-1}) \end{aligned}$$

where in the last equality, we have used the fact that M is a martingale. On the other hand

$$\begin{aligned}\mathbb{E}\left(\mathbb{1}_{\{\tau < n\}} \sum_{k=1}^{\tau} M_k - M_{k-1} \middle| \mathcal{F}_{n-1}\right) &= \sum_{j=1}^{n-1} \mathbb{E}\left(\mathbb{1}_{\{\tau=j\}} \sum_{k=1}^j M_k - M_{k-1} \middle| \mathcal{F}_{n-1}\right) \\ &= \mathbb{1}_{\{\tau \leq n-1\}} \sum_{k=1}^{\tau} (M_k - M_{k-1})\end{aligned}$$

and by combining the above computations, we obtain

$$\mathbb{E}(M_n^\tau | \mathcal{F}_{n-1}) = \sum_{k=1}^{\tau \wedge (n-1)} (M_k - M_{k-1}) + M_0 = M_{n-1}^\tau$$

and this concludes the proof. $\square$

Definition 2.2.2. Given a stopping time τ , let

$$\mathcal{F}_\tau = \{A \in \mathcal{F}_T : \forall t \in [0, T], A \cap \{\tau \leq t\} \in \mathcal{F}_t\}$$

The σ -algebra $\mathcal{F}_\tau$ is called the σ -algebra of events prior to τ .

We present below baby versions of the Doob optional sampling theorem and the Doob decomposition theorem for submartingales.

Theorem 2.2.1 (Doob's optional sampling theorem). *Let $\sigma \leq \tau$ be two stopping times taking values in $\{0, 1, \dots, N\}$. For any martingale, submartingale or supermartingale X . The random variable X_σ, X_τ are integrable and $\mathbb{E}[X_\tau | \mathcal{F}_\sigma] = X_\sigma$, $\mathbb{E}[X_\tau | \mathcal{F}_\sigma] \geq X_\sigma$ and $\mathbb{E}[X_\tau | \mathcal{F}_\sigma] \leq X_\sigma$ holds respectively.*

Proof. We first check that X_τ is integrable. To do that, we note $\mathbb{E}[|X_\tau|] \leq \sum_{n=0}^N \mathbb{E}[|X_n|] < \infty$. Let us first suppose $\tau - \sigma \leq 1$. For any $A \in \mathcal{F}_\sigma$,

$$\begin{aligned}\mathbb{E}[\mathbb{1}_A(X_\tau - X_\sigma)] &= \mathbb{E}[\mathbb{1}_A \mathbb{1}_{\{\sigma < \tau\}}(X_\tau - X_\sigma)] = \sum_{j=1}^N \mathbb{E}[\mathbb{1}_A \mathbb{1}_{\{\sigma < j\}}(X_j - X_{j-1})] \\ &= \sum_{j=1}^N \mathbb{E}[\mathbb{1}_{A \cap \{\sigma \leq j-1\}}(X_j - X_{j-1})]\end{aligned}$$

By definition of $\mathcal{F}_\sigma$, we have $A \cap \{\sigma \leq j-1\} \in \mathcal{F}_{j-1}$ and from the Tower property

$$\sum_{j=1}^N \mathbb{E}[\mathbb{1}_{A \cap \{\sigma \leq j-1\}}(X_j - X_{j-1})] = \sum_{j=1}^N \mathbb{E}[\mathbb{1}_{A \cap \{\sigma \leq j-1\}} \mathbb{E}[X_j - X_{j-1} | \mathcal{F}_{j-1}]] = 0$$

where in the last equality, we have used the martingale property of X . For the general case, we set $\sigma_j = \tau \wedge (\sigma + j)$ and we note that $\sigma = \sigma_0 \leq \sigma_1 \leq \sigma_2 \leq \dots \leq \sigma_n = \tau$ and $\sigma_j - \sigma_{j-1} \leq 1$ for $1 \leq j \leq n$. The result then follows from repeated applications of the tower property of conditional expectation. $\square$

Theorem 2.2.2 (Doob's decomposition theorem). *Let $X = (X_n)_{n=0,1,\dots,N}$ be a discrete time submartingale then it can be decomposed into*

$$X_n = M_n + A_n$$

where M_n is a martingale and A_n is an increasing process.

Proof. For every $k = 1, \dots, n$, we can write

$$X_k - X_{k-1} = X_k - \mathbb{E}[X_k | \mathcal{F}_{k-1}] + \mathbb{E}[X_k | \mathcal{F}_{k-1}] - X_{k-1}$$

Then by summing the above over all k and noticing that the left hand side above is a telescoping sum, we obtain

$$X_n - X_0 = \left\{ \sum_{k=1}^n X_k - \mathbb{E}[X_k | \mathcal{F}_{k-1}] \right\} + \left\{ \sum_{k=1}^n \mathbb{E}[X_k | \mathcal{F}_{k-1}] - X_{k-1} \right\}$$

where we set

$$M_n = \left\{ \sum_{k=1}^n X_k - \mathbb{E}[X_k | \mathcal{F}_{k-1}] \right\} \quad \text{and} \quad A_n = \left\{ \sum_{k=1}^n \mathbb{E}[X_k | \mathcal{F}_{k-1}] - X_{k-1} \right\}.$$

It is clear that A_n is increasing since $\mathbb{E}[X_k | \mathcal{F}_{k-1}] - X_{k-1} \geq 0$. To check that M_n is a martingale, we need only to check

$$\mathbb{E}[M_n | \mathcal{F}_{n-1}] = M_{n-1}$$

To do this, we can write

$$\mathbb{E}[M_n | \mathcal{F}_{n-1}] = \left\{ \sum_{k=1}^{n-1} X_k - \mathbb{E}[X_k | \mathcal{F}_{k-1}] \right\} + \mathbb{E}[X_n - \mathbb{E}[X_n | \mathcal{F}_{n-1}] | \mathcal{F}_{n-1}] = M_{n-1}$$

□

Example 2.2.2. Let us view example 2.1.2 as a game, where at step n you make a bet and receives the amount $X_n = -1$ if you lose and $X_n = 1$ if you win. You start the game with an initial capital $S_0 = x$ and the accumulated wealth at time n is given by $S_n = x + \sum_{i=1}^n X_i$. We suppose that the game can only be played for a maximum of N steps and the game ends if your accumulated wealth hits zero or a fixed level $c \geq x$.

Let $\tau = \inf\{n : S_n = 0 \text{ or } S_n = c\}$ and suppose that $\mathbb{E}[\tau] < \infty$, i.e. τ is a finite stopping time. Let us consider the stopping time $\tau \wedge N$. In the case $p = 1/2$, by Doob's optimal sampling theorem or the fact that S_n^τ is again a martingale, we have

$$\mathbb{E}[S_{\tau \wedge N}] = \mathbb{E}[S_{\tau \wedge N} | \mathcal{F}_0] = x$$

To simplify the left hand side, we see that

$$x = \mathbb{E}[S_{\tau \wedge N}] = \mathbb{E}[S_\tau \mathbf{1}_{\{S_\tau = c\}} \mathbf{1}_{\{\tau \leq N\}}] + \mathbb{E}[S_N \mathbf{1}_{\{\tau > N\}}]$$

which implies that

$$\frac{x - \mathbb{E}[S_N \mathbb{1}_{\{\tau > N\}}]}{c} = \mathbb{P}(\{S_\tau = c\} \cap \{\tau \leq N\}).$$

Assuming that one can play forever and let $N \rightarrow \infty$. Clearly we see that $\{S_\tau = c\} \cap \{\tau \leq N\} \subseteq \{S_\tau = c\} \cap \{\tau \leq N+1\}$ which implies that $\lim_{N \rightarrow \infty} \mathbb{P}(\{S_\tau = c\} \cap \{\tau \leq N\}) = \mathbb{P}(S_\tau = c)$. One can show that $\lim_{N \rightarrow \infty} \mathbb{E}[S_N \mathbb{1}_{\{\tau > N\}}] = 0$ (exercise?), which implies

$$\frac{x}{c} = \mathbb{P}(S_\tau = c).$$

That is, if the game is allowed to go on forever, the probability that you exit the game with a winning amount c is given by $\frac{x}{c}$ for $c \geq x$.

From the fact that $M_n := S_n^2 - n$ is a martingale, we deduce that the Doob decomposition of the submartingale S_n^2 is given by $M_n + n$.

Proposition 2.2.1. *The following properties are valid:*

1. *Let X be an $\mathcal{F}_N$ -measurable and integrable random variable. Then the process $M_n = \mathbb{E}(X|\mathcal{F}_n)$, $n \in \{0, 1, \dots, N\}$ is a $\mathbb{F}$ -martingale.*
2. *Assume that M is either a $\mathbb{F}$ -supermartingale or a $\mathbb{F}$ -submartingale. Then M is an $\mathbb{F}$ -martingale if and only if the expected value of M is constant, or equivalently, whenever $\mathbb{E}[M_0] = \mathbb{E}[M_n]$ for all $n \in \{0, 1, \dots, N\}$.*
3. *Let M be an $\mathbb{F}$ -martingale and let $h : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function. If the process $X_n = h(M_n)$, $n \in \{0, 1, \dots, N\}$ is integrable then it is an $\mathbb{F}$ -submartingale.*

Proof. We show only 2. and the rest are left for exercise. To show 2. we see that if M is a martingale then it is a submartingale and it is a supermartingale and $\mathbb{E}[M_T] = \mathbb{E}[M_0]$, therefore we prove only the converse.

Suppose that M is a submartingale, then $\mathbb{E}[M_N] \geq \mathbb{E}[M_0]$ and we consider the $\mathcal{F}_n$ -measurable set $A_{m,n} = \{\mathbb{E}[M_m|\mathcal{F}_n] > M_n\}$ and define for each $m \in \mathbb{N}^*$,

$$A_{m,n,k} = \{\mathbb{E}[M_m|\mathcal{F}_n] - M_n > k^{-1}\}.$$

We clearly have $A_{m,n,k} \subseteq A_{m,n,k+1}$ and $A_{m,n,k} \uparrow A_{m,n}$. Hence by theorem 1.1.2 we have $\mathbb{P}(A_{m,n,k}) \uparrow \mathbb{P}(A_{m,n})$.

To this end, we show that for all $m \geq n \geq 0$, we have $\mathbb{P}(A_{m,n}) = 0$. Since M is submartingale therefore we have for all $s \leq t$, $\mathbb{E}[M_t|\mathcal{F}_s] - M_s \geq 0$ and by using the tower property of the conditional expectation, we obtain

$$\mathbb{E}[M_m - M_n] = \mathbb{E}[\mathbb{E}[M_m|\mathcal{F}_n] - M_n] \geq \mathbb{E}[\mathbb{1}_{A_{m,n,k}}(\mathbb{E}[M_m|\mathcal{F}_n] - M_n)] > \frac{1}{k} \mathbb{P}(A_{m,n,k}).$$

Since the expectation $\mathbb{E}[M_n] = \mathbb{E}[M_m] = \mathbb{E}[M_0]$ is a constant, we must have $\mathbb{P}(A_{m,n,k}) = 0$ for all k and therefore $\mathbb{P}(A_{m,n}) = \lim_{k \rightarrow \infty} \mathbb{P}(A_{m,n,k}) = 0$. This implies that $\mathbb{E}[M_m|\mathcal{F}_n] = M_n$ a.s. $\square$

Lemma 2.2.3. *Let $M = (M_n)_{n=0,1,\dots,N}$ be an $\mathbb{F}$ -adapted and integrable stochastic process. Then M is an $\mathbb{F}$ -martingale if and only if we have $\mathbb{E}(M_\tau) = \mathbb{E}(M_0)$ for any stopping time τ with values in $\{0, 1, \dots, N\}$.*

Proof. It suffices to check that $M_n = \mathbb{E}(M_T | \mathcal{F}_n)$ for every $n = \{0, 1, \dots, N\}$. By assumption, we have that $\mathbb{E}(M_\tau) = \mathbb{E}(M_N)$ for any stopping time τ with values in $0, 1, \dots, N$ (this is true since $\tau = N$ is also a stopping time). Let us fix t and let us consider an event A belonging to $\mathcal{F}_t$. Then the random variable τ_A defined as follows

$$\tau_A = \begin{cases} n, & \text{if } \omega \in A, \\ N, & \text{if } \omega \notin A, \end{cases}$$

is a stopping time with values in $[0, T]$. The equality $\mathbb{E}(M_{\tau_A}) = \mathbb{E}(M_N)$ yields $\mathbb{E}(\mathbb{1}_A M_n) = \mathbb{E}(\mathbb{1}_A M_N)$. Since this equality holds for any event A in $\mathcal{F}_t$, by virtue of definition of conditional expectation in definition 1.3.5, we obtain the equality $M_n = \mathbb{E}(M_N | \mathcal{F}_n)$, which in turn implies that M is an $\mathbb{F}$ -martingale. $\square$

All results in discrete time has a continuous time counterpart, but the proof is much more technical and we omit them here. For exercise, one can try proving proposition 2.2.1 and lemma 2.2.3 in continuous time.

2.2.2 Continuous Time

Definition 2.2.3. A positive random variable τ is a $\mathbb{F}$ -stopping time if for every $t \geq 0$, the set $\{\tau \leq t\}$ belongs to $\mathcal{F}_t$.

Example 2.2.3. A stopping time is usually of the form of a hitting time. For example the first hitting time of a constant level, i.e. $\tau := \inf \{s : B_s \geq 1\}$, where B is a Brownian motion. To check that τ is a stopping time it is suffice to note that $\{\tau \leq t\} = \{1 \leq \sup_{s \leq t} B_s\}$ and use the fact that the running supremum process $B_t^* := \sup_{s \leq t} B_s$ is $\mathbb{F}$ -adapted.

Theorem 2.2.3 (Optional Sampling Theorem). *Let $\sigma \leq \tau$ be two bounded stopping times. For any càdlàg martingale, submartingale or supermartingale X , the random variables X_σ, X_τ are integrable and $\mathbb{E}[X_\tau | \mathcal{F}_\sigma] = X_\sigma$, $\mathbb{E}[X_\tau | \mathcal{F}_\sigma] \geq X_\sigma$ and $\mathbb{E}[X_\tau | \mathcal{F}_\sigma] \leq X_\sigma$ holds respectively.*

Proposition 2.2.2 (Doob-Meyer's decomposition). *Let X be a non-negative continuous submartingale. Then there exists a continuous increasing process A with $A_0 = 0$ such that the process $N = X - A$ is a continuous martingale. The processes N and A in the Doob-Meyer decomposition $X = N + A$ are unique up to indistinguishability of stochastic processes.*

In general, one needs to introduce processes of class (D) or class (DL) , i.e. the concept of uniformly integrability for stochastic processes. Non-negative condition in the above proposition can be removed and replaced by class (D) . The result is technical and we refer to the recommended textbooks.

2.2.3 The Usual Conditions

From now on, we assume that the filtration $\mathbb{F}$ satisfies the *usual conditions*:

1. $\mathbb{F}$ is $\mathbb{P}$ -complete, i.e., any event $A \in \mathcal{F}_T$ such that $\mathbb{P}(A) = 0$ belongs to the σ -algebra $\mathcal{F}_0$, and thus it belongs to the σ -algebra $\mathcal{F}_t$ for any $t \in [0, T]$,
2. $\mathbb{F}$ is right-continuous, that is $\mathcal{F}_t = \bigcap_{u>t} \mathcal{F}_u$ for any $t \in [0, T)$.

Remark 2.2.1. If a filtration $\mathbb{F}$ is not $\mathbb{P}$ -complete and/or right-continuous, we need first to modify it as follows. First, we set $\tilde{\mathcal{F}}_t = \sigma(\mathcal{F}_t \cup \mathcal{N})$, where $\mathcal{N}$ is the class of all $\mathbb{P}$ -null events from $\mathcal{F}_T$ (i.e., all events from $\mathcal{F}_T = \mathcal{A}$ of probability zero). Second, for any $t \in [0, T)$ we define $\hat{\mathcal{F}}_t = \tilde{\mathcal{F}}_{t+}$, where $\tilde{\mathcal{F}}_{t+} = \bigcap_{\epsilon>0} \tilde{\mathcal{F}}_{t+\epsilon}$. The filtration $\hat{\mathbb{F}}$ defined in that way is $\mathbb{P}$ -complete and right-continuous; it is referred to as the $\mathbb{P}$ -*augmentation* of $\mathbb{F}$.

Chapter 3

The Brownian Motion

We start by stating the definition of the *standard Brownian motion* (also known as the *Wiener process*).

Definition 3.0.1. A real-valued, $\mathbb{F}$ -adapted process $W = (W_t)_{t \in [0, T]}$, with $W_0 = 0$, defined on a filtered probability space $(\Omega, \mathbb{F}, \mathbb{P})$, is called a *one-dimensional standard Brownian motion* with respect to the filtration $\mathbb{F}$ if:

1. for any $0 \leq u \leq t \leq T$, the increment $W_t - W_u$ is a random variable independent of the σ -algebra $\mathcal{F}_u$,
2. for any $0 \leq u \leq t \leq T$, the probability distribution of $W_t - W_u$ is Gaussian with expected value 0 and variance $t - u$, that is, $W_t - W_u \stackrel{d}{=} N(0, t - u)$,
3. almost all sample paths of W are continuous functions, that is, the process W is sample-paths continuous.

The initial condition $W_0 = 0$ is not essential, and thus it may be replaced by the equality $W_0 = x$ for any $x \in \mathbb{R}$. In that case, we say that the Brownian motion W is started at x .

If the filtration $\mathbb{F}$ is not given in advance, by a *standard Brownian motion* we mean a process W satisfying definition 3.0.1 with $\mathbb{F}$ being the natural filtration of W . More precisely, we assume that $\mathbb{F} = \mathbb{F}^W$, meaning that $\mathbb{F}^W$ is (the $\mathbb{P}$ -augmentation of) the natural filtration $\mathcal{F}_t^W = \sigma(W_u, u \leq t)$ of the process W .

In that case, condition (1) in definition 3.0.1 can be replaced by the requirement that W is a *process of independent increments*, that is, for any $n \in \mathbb{N}$ and any sequence $0 < t_1 < \dots < t_n \leq T$ the random variables $W_{t_1} - W_{t_0}$, $W_{t_2} - W_{t_1}$, $\dots$, $W_{t_n} - W_{t_{n-1}}$ are independent.

We take here for granted the existence of a Brownian motion (see, for instance, Karatzas and Shreve (1998)) and we focus on the most important properties of this process.

Definition 3.0.2. A stochastic process X is said to satisfy the Markov property if for all Borel function f and $s \leq t$

$$\mathbb{E}[f(X_t)|\mathcal{F}_s] = \mathbb{E}[f(X_t)|\sigma(X_s)].$$

Roughly speaking, the Markov property says that *the future* X_t is independent from *all the past* $\mathcal{F}_s$ given *the present* X_s .

Lemma 3.0.1. *Given a Brownian motion W , it is*

1. *a continuous $\mathbb{F}$ -martingale.*
2. *A Brownian motion W has the Markov property with respect to $\mathbb{F}$, meaning that for any bounded Borel measurable function $g : \mathbb{R} \rightarrow \mathbb{R}$ and $0 \leq u \leq t \leq T$*

$$\mathbb{E}(g(W_t)|\mathcal{F}_u) = \mathbb{E}(g(W_t)|\sigma(W_u)). \quad (3.1)$$

Proof. To prove 1, we first note that it is clear that W is a continuous, $\mathbb{F}$ -adapted process and $\mathbb{E}(|W_t|) < \infty$ for any $t \in [0, T]$. Moreover the martingale property holds since for any $0 \leq u \leq t \leq T$,

$$\mathbb{E}(W_t|\mathcal{F}_u) = \mathbb{E}(W_t - W_u|\mathcal{F}_u) + \mathbb{E}(W_u|\mathcal{F}_u) = W_u. \quad (3.2)$$

The last equality in formula (3.2) holds since condition (ii) implies that $\mathbb{E}_{\mathbb{P}}(W_t - W_u|\mathcal{F}_u) = \mathbb{E}_{\mathbb{P}}(W_t - W_u) = 0$. In fact, W is a *square-integrable martingale* as manifestly $\mathbb{E}_{\mathbb{P}}(W_t^2) = t < \infty$.

To prove 2, we note that

$$\mathbb{E}(g(W_t)|\mathcal{F}_u) = \mathbb{E}(g(W_t - W_u + W_u)|\mathcal{F}_u) = \mathbb{E}(g(\eta + \psi)|\mathcal{F}_u),$$

where $\psi = W_u$ is $\mathcal{F}_u$ -measurable and $\eta = W_t - W_u$ is independent of $\mathcal{F}_u$. Consequently from property 5 of conditional expectation given in theorem 1.3.5 we see that $\mathbb{E}(g(W_t)|\mathcal{F}_u)$ is a function of W_u and therefore $\sigma(W_u)$ -measurable. Consequently, by taking again the conditional expectation with respect to $\sigma(W_u)$ we can conclude that the equality (3.1) holds. $\square$

Definition 3.0.3. A continuous time stochastic process is said to be a Gaussian process, if for any set of time indices $t_1 < t_2 < \dots < t_n$. The vector $(X_{t_1}, \dots, X_{t_n})$ follows a multivariate Gaussian distribution.

Proposition 3.0.1. *Let $W = (W_t)_{t \geq 0}$ be an $\mathbb{F}$ -Brownian motion. Then*

1. $\mathbb{E}[W_s W_t] = \min(s, t)$ for all $s, t \geq 0$ and in particular $\mathbb{E}[|W_t|^2] = t$.
2. *The process W is a Gaussian process.*

Proof. To proof 1., we see that since $W_0 = 0$, we can write $W_t = W_t - W_0$, it follows from the definition of Wiener process that for $s \leq t$

$$\begin{aligned}\text{Cov}(W_s, W_t) &= \mathbb{E}[W_s W_t] \\ &= \mathbb{E}[W_s (W_t - W_s)] + \mathbb{E}[W_s^2] \\ &= \mathbb{E}[W_s] \mathbb{E}[W_t - W_s] + \mathbb{E}[W_s^2] = s\end{aligned}$$

Similarly when $s \geq t$, $\text{Cov}(W_s, W_t) = t$. Therefore $\mathbb{E}[W_t W_s] = \min(s, t)$.

To proof 2., it is enough to show that for any $n \geq 1$ and any choice of $0 \leq t_1 < t_2 < \dots < t_n$ the random vector

$$X = \begin{pmatrix} W_{t_1} \\ W_{t_2} \\ \vdots \\ W_{t_n} \end{pmatrix}$$

is normally distributed. To this end, the random variables $W_{t_1}, W_{t_2} - W_{t_1}, \dots, W_{t_n} - W_{t_{n-1}}$ are independent and therefore the random vector

$$Y = \begin{pmatrix} W_{t_1} \\ W_{t_2} - W_{t_1} \\ \vdots \\ W_{t_n} - W_{t_{n-1}} \end{pmatrix}$$

is normally distributed. It is also easy to check that $X = AY$ with the matrix A given by the equation

$$A = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \dots & 1 \end{pmatrix}.$$

We can then conclude by using the fact that any linear transform of a Gaussian vector is again Gaussian. $\square$

Remark 3.0.1. The above is practically important, since if we want to compute the quantity $\mathbb{E}[f(W_{t_1}, \dots, W_{t_k})]$ for $t_1 < t_2 < \dots < t_k$, then we know that the density function of $(W_{t_1}, \dots, W_{t_k})$ is just multivariate Gaussian $\mathcal{N}(0, AVA^T)$, where $V = \text{Var}(Y)$ is the variance matrix of Y .

Example 3.0.1. Given $m \in \mathbb{R}$ and $\sigma > 0$, let $S_t = mt + \sigma W_t$. The process $(S_t)_{t \in [0, T]}$ is sometimes called Brownian motion with drift m and variance σ^2 . It is not hard to see that $\mathbb{E}[S_t] = mt$ and $\mathbb{E}[(S_t - mt)^2] = \sigma^2 t$.

We shall determine the joint density of the random variables (W_{t_1}, S_{t_2}) . The process S is a Gaussian process and therefore the random vector $(W_{t_1}, S_{t_2}) = (W_{t_1}, mt_2 + \sigma W_{t_2})$ is also jointly Gaussian (**exercise?**). Therefore, to find the distribution the random vector, it is sufficient to compute the mean vector and the variance matrix.

By the definition of the Brownian motion we obtain

$$\text{Cov}(W_{t_1}, mt_2 + \sigma W_{t_2}) = \sigma \mathbb{E}[W_{t_1} W_{t_2}] = \sigma \min(t_1, t_2).$$

Hence the covariance matrix V is

$$V = \begin{pmatrix} t_1 & \sigma \min(t_1, t_2) \\ \sigma \min(t_1, t_2) & \sigma^2 t_2 \end{pmatrix}$$

and the distribution of (W_{t_1}, S_{t_2}) is Gaussian, that is

$$(W_{t_1}, S_{t_2}) \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ mt_2 \end{pmatrix}, \begin{pmatrix} t_1 & \sigma \min(t_1, t_2) \\ \sigma \min(t_1, t_2) & \sigma^2 t_2 \end{pmatrix}\right).$$

Example 3.0.2. Let $x, m \in \mathbb{R}$ and $\sigma > 0$ and we set $Y_t = xe^{mt+\sigma W_t}$. The process Y is called an exponential Brownian motion with drift m and variance σ and is ubiquitous in mathematical finance. We first notice that the process Y is adapted to the filtration $\mathbb{F}$. Next,

$$\mathbb{E}[Y_t] = xe^{(m+\frac{1}{2}\sigma^2)t}.$$

we shall show now that the process Y is an $\mathbb{F}$ -martingale if and only if $m = -\frac{\sigma^2}{2}$. Indeed, for $s \leq t$ we obtain

$$\begin{aligned} \mathbb{E}[xe^{mt+\sigma W_t} | \mathcal{F}_s] &= \mathbb{E}[xe^{ms+m(t-s)+\sigma W_s} e^{\sigma(W_t-W_s)} | \mathcal{F}_s] \\ &= Y_s \mathbb{E}[e^{m(t-s)+\sigma(W_t-W_s)}] = Y_s e^{m(t-s)+\frac{1}{2}\sigma^2(t-s)} \end{aligned}$$

and the claim follows.

3.1 Total and Quadratic Variations

The trajectory of the Brownian motion can be oscillate wildly and in the following we introduce two measures of variations, that is the *total variation* and the *quadratic variation*.

Given a function f on $[0, T]$, the goal of this subsection is to understand how much a function can oscillate on a fixed interval $[0, T]$. To do this, for every $n \geq 1$ we divide $[0, T]$ into subintervals $0 = t_0^n < t_1^n < t_2^n < \dots < t_{m_n}^n = T$. In the following, a partition will be denoted by π_n and with every partition π_n of $[0, T]$ we associate its "size" defined by

$$|\pi_n| := \max_{k \leq m_n} (t_k^n - t_{k-1}^n),$$

where $t_0^n = 0$ and $t_{m_n}^n = T$, and we also set $n(\pi) = m_n$ to be the number of subintervals in the partition. For a given partition π_n we define the corresponding measure of the oscillation of a function f by

$$V_n(f) = \sum_{k=1}^{n(\pi)} |f(t_k^n) - f(t_{k-1}^n)|.$$

We claim that if $\sup_{\pi_n} V_n(f) < \infty$ then the limit

$$V(f) = \lim_{|\pi_n| \rightarrow 0} V_n(f)$$

exists and is called the *total variation* of f . A function f with the property $V(f) < \infty$ is called a *function of bounded variation* or a *function of finite variation*.

Remark 3.1.1. For any continuously differentiable function f on the interval $[0, T]$, we have

$$V_n(f) = \sum_{k=1}^{n(\pi)} \left| \int_{t_{k-1}^n}^{t_k^n} f'(y) dy \right| \leq \sum_{k=1}^{n(\pi)} \int_{t_{k-1}^n}^{t_k^n} |f'(y)| dy = \int_0^T |f'(y)| dy$$

and it follows that

$$V_n(f) \leq \int_0^T |f'(y)| dy < \infty.$$

where we have used the fact that any continuous function on a closed and bounded interval is bounded (recall that $f'(x)$ is assumed to be continuous). Note that the above upper bound is independent of the choice of the partition π_n . Therefore by taking the limit as $|\pi_n| \rightarrow 0$ we claim that any continuously differentiable function on $[0, T]$ is a function of bounded variation or finite variation.

For a given partition π_n , we shall define now the so-called *quadratic variation*. Similar to the previous section. Given a function f on $[0, T]$, we let

$$V_n^{(2)}(f) = \sum_{i=1}^{n(\pi)} \left| f(t_{k_i}^n) - f(t_{k_{i-1}}^n) \right|^2.$$

The quadratic variation of a function f (if it exists) is defined to be

$$V^{(2)}(f) = \lim_{|\pi_n| \rightarrow 0} V_n^{(2)}(f).$$

Example 3.1.1. Let f be a continuously differentiable function, then from the same arguments in remark 3.1.1, we obtain by the triangular inequality and the Schwartz inequality

$$\begin{aligned} V_n^{(2)}(f) &\leq \sum_{k=1}^{m_n} \left| \int_{t_{k-1}^n}^{t_k^n} |f'(y)| dy \right|^2 \leq \sum_{k=1}^{n(\pi)} (t_k^n - t_{k-1}^n) \int_{t_{k-1}^n}^{t_k^n} |f'(y)|^2 dy \\ &\leq \max_{i \leq m_n} (t_k^n - t_{k-1}^n) \sum_{k=1}^{n(\pi)} \int_{t_{k-1}^n}^{t_k^n} |f'(y)|^2 dy = |\pi_n| \int_0^T |f'(y)|^2 dy. \end{aligned}$$

Given that $\int_0^T |f'(y)|^2 dy < \infty$, which is true since here f is continuously differentiable on $[0, T]$ and by letting $|\pi_n| \rightarrow 0$, we have

$$\lim_{|\pi_n| \rightarrow 0} V_n^{(2)}(f) = 0.$$

This shows that the quadratic variation of any continuous differentiable function f is zero on $[0, T]$.

Exercise: Let f be a continuous function of finite variation on $[0, T]$, show that the quadratic variation of f is equal to zero. *Hint:* Any continuous function on a closed and bounded interval is uniformly continuous.

Definition 3.1.1. A stochastic process $\{X_t : t \geq 0\}$ is of *finite quadratic variation* if there exists a process $\langle X \rangle$ such that for every $t \geq 0$

$$\lim_{|\pi_n| \rightarrow 0} \mathbb{E}[(V_n^{(2)}(X) - \langle X \rangle_t)^2] = 0,$$

where π_n denotes a partition of the interval $[0, t]$.

A Brownian motion is an example of a stochastic process of *finite quadratic variation*, which is given by $\langle W \rangle_t = t$. We shall prove this fact below.

Proposition 3.1.1. For every $0 \leq u < t \leq T$, and any sequence $(\pi_n)_{n \in \mathbb{N}}$ of finite partitions $\pi_n = \{u = t_0^n < t_1^n < \dots < t_{m_n}^n = t\}$ of the interval $[u, t]$ satisfying $\lim_{n \rightarrow \infty} |\pi_n| = 0$, where $|\pi_n| := \max_{k=1, \dots, m_n} (t_k^n - t_{k-1}^n)$, we have

$$\lim_{n \rightarrow \infty} \sum_{k=1}^{n(\pi)} (W_{t_k^n} - W_{t_{k-1}^n})^2 = t - u, \quad (3.3)$$

where the convergence in (3.3) is in $L^2(\Omega, \mathcal{F}_T, \mathbb{P})$.

Proof. Let us fix $0 \leq u < t \leq T$ and let us denote $\Delta t_k^n = t_k^n - t_{k-1}^n$. Note that necessarily $\lim_{n \rightarrow \infty} n(\pi) = \infty$ and it suffices to check that $\lim_{n \rightarrow \infty} I_n = 0$, where we write

$$I_n = \lim_{n \rightarrow \infty} \mathbb{E} \left[\left| \sum_{k=1}^{n(\pi)} ((W_{t_k^n} - W_{t_{k-1}^n})^2 - \Delta t_k^n) \right|^2 \right].$$

Since the increments $W_{t_{j+1}^n} - W_{t_j^n}$ and $W_{t_{i+1}^n} - W_{t_i^n}$ are independent for $i \neq j$ and $\mathbb{E}_{\mathbb{P}}[(W_{t_k^n} - W_{t_{k-1}^n})^2] = \Delta t_k^n$, we obtain

$$\begin{aligned} I_n &= \sum_{k=1}^{n(\pi)} (\mathbb{E}(W_{t_k^n} - W_{t_{k-1}^n})^4 - 2\Delta t_k^n \mathbb{E}(W_{t_k^n} - W_{t_{k-1}^n})^2 + (\Delta t_k^n)^2) \\ &= 2 \sum_{k=1}^{n(\pi)} (\Delta t_k^n)^2 \leq 2(t - u) \max_{k=1, \dots, m_n} (t_k^n - t_{k-1}^n) = 2(t - u)|\pi_n| \end{aligned}$$

where in the second equality we used the fact that the assumption that the increment $W_{t_k^n} - W_{t_{k-1}^n}$ has the Gaussian distribution $N(0, \Delta t_k^n)$ yields

$$\mathbb{E}[(W_{t_k^n} - W_{t_{k-1}^n})^4] = 3(\Delta t_k^n)^2 = 3(t_k^n - t_{k-1}^n)^2.$$

It is now clear that the quantity I_n tends to zero as n tends to infinity, since $|\pi_n|$ tends to zero. $\square$

Remark 3.1.2. It is worth mentioning that if the series $\sum_{n=1}^{\infty} |\pi_n|$ is convergent then it can be deduced from the Borel-Cantelli lemma that the convergence in (3.3) is almost sure (that is, it holds with probability 1).

Let $\langle W \rangle_t$ stand for the quadratic variation of W on $[0, t]$ and from the above result we have that $\langle W \rangle_t = t$ for any $t \in [0, T]$.

Lemma 3.1.1. *The process $M = W^2 - \langle W \rangle$ is a continuous $\mathbb{F}$ -martingale.*

Proof. The process M is $\mathbb{F}$ -adapted and integrable since W^2 is $\mathbb{F}$ -adapted and integrable, while the quadratic variation $\langle W \rangle_t = t$ is clearly $\mathbb{F}$ -adapted and integrable. From the equality $\mathbb{E}(W_t W_u | \mathcal{F}_u) = W_u \mathbb{E}(W_t | \mathcal{F}_u) = W_u^2$ for $0 \leq u \leq t \leq T$, we deduce that

$$\mathbb{E}(W_t^2 - W_u^2 | \mathcal{F}_t) = \mathbb{E}((W_t - W_u)^2 | \mathcal{F}_t) = \mathbb{E}((W_t - W_u)^2) = t - u,$$

and thus, we have for $0 \leq u \leq t \leq T$, $\mathbb{E}(M_t | \mathcal{F}_u) = \mathbb{E}(W_t^2 - t | \mathcal{F}_u) = W_u^2 - u = M_u$. $\square$

In view of the above result, we note the quadratic variation $\langle W \rangle$ is also the unique increasing and continuous process such that when subtracted from the submartingale $X = W^2$ give us a $\mathbb{F}$ -martingale. This observation give raise to an alternative definition of the quadratic variation process $\langle X \rangle$ as the unique increasing process such that $X^2 - \langle X \rangle$ is martingale. The existence of $\langle X \rangle$ follows from the Doob-Meyer decomposition.

Let us recall that for any function $f : [0, T] \rightarrow \mathbb{R}$ the *total variation of f on the interval $[u, t]$* , denoted as $V_{[u, t]}(f)$, is defined as

$$V_{[u, t]}(f) = \sup_{\pi_{[u, t]}} \sum_{k=1}^{n(\pi)} |f(t_k^n) - f(t_{k-1}^n)|, \quad (3.4)$$

where $\pi_{[u, t]}$ is the class of all finite partitions $\pi_n = \{u = t_0^n < t_1^n < \dots < t_{n(\pi)}^n = t\}$ of the interval $[u, t]$. Note that $n(\pi)$ stands for the number of subintervals in a partition π .

Corollary 3.1.1. *Sample paths of a Brownian motion are almost surely functions of infinite variation on any interval.*

Proof. Observe that for any partition of $[u, t]$ we have

$$\begin{aligned} \sum_{k=1}^{n(\pi)} (f(t_k^n) - f(t_{k-1}^n))^2 &\leq \max_{m=1, \dots, n(\pi)} (f(t_m^n) - f(t_{m-1}^n)) \sum_{k=1}^{n(\pi)} |f(t_k) - f(t_{k-1})| \\ &\leq \max_{m=1, \dots, n(\pi)} (f(t_m^n) - f(t_{m-1}^n)) V_{[u, t]}(f). \end{aligned}$$

By using the fact that any continuous function is uniformly continuous on the compact interval $[u, t]$, it is not difficult to show that the last inequality would contradict proposition 3.1.1 if $\mathbb{P}(V_{[u, t]}(W) < +\infty) > 0$. We conclude that for any $t < u$ we have $V_{[u, t]}(W) = +\infty$ almost surely. $\square$

Remark 3.1.3. The above shows that the paths of Brownian motion is not continuously differentiable, since any function which is continuously differentiable on an interval $[u, t]$ must be of finite variation. In fact, the path of the Brownian is not differentiable with respect to time at any time point.

Lemma 3.1.2. *Let X be a stochastic process such that $\int_0^T \mathbb{E}[|X_s|] ds < \infty$, then*

$$\mathbb{E}\left[\int_0^T X_s ds\right] = \int_0^T \mathbb{E}[X_s] ds.$$

Moreover, if the process X is adapted to the filtration $\mathbb{F}$ then the process

$$Y_t = \int_0^t X_s ds, \quad t \leq T,$$

is adapted to $\mathbb{F}$. Additionally, if X is a Gaussian process then the process Y is also Gaussian.

Proof. We give only a sketch of the proof. By assumption the process X is integrable on $[0, T]$ and we can consider the approximating sequence for the process (Y_t) ,

$$Y_t^n = \sum_{0 < \frac{kT}{2^n} \leq t} X_{\frac{(k-1)T}{2^n}} \left(\frac{kT}{2^n} - \frac{(k-1)T}{2^n} \right)$$

It is not difficult to see that all the properties stated in the lemma hold if we replace Y with Y^n . By definition of the integral

$$\lim_{n \rightarrow \infty} Y_t^n = Y_t$$

for every $t > 0$ and it remains to prove only that the properties stated in the lemma are preserved in the limit.

The first result is an application of the Fubini-Tonelli theorem. The second result follows from the fact that sum and limits of measurable functions are again measurable, while the last result uses the fact that sum of Gaussian random variables are again Gaussian, and the limiting distribution can be obtained from computing the moment generating function and applying dominated convergence theorem. $\square$

Example 3.1.2. Consider the process

$$Y_t = \int_0^t W_s ds.$$

By Lemma 1.1 it is an $\mathbb{F}$ -adapted Gaussian process. We shall determine the distribution of the random variable Y_t for a fixed t . By first result of the Lemma we have

$\mathbb{E}[Y_t] = 0$. Now, for $s \leq t$, once more by Lemma 1

$$\begin{aligned}
\mathbb{E}[Y_s Y_t] &= \mathbb{E} \left[\left(\int_0^s W_u du \right) \left(\int_0^t W_v dv \right) \right] \\
&= \int_0^s \int_0^t \mathbb{E}[W_u W_v] dudv \\
&= \int_0^s \int_0^s \mathbb{E}[W_u W_v] dudv + \int_0^s \int_s^t \mathbb{E}[W_u W_v] dudv \\
&= \int_0^s \int_0^s \min(u, v) dudv + \int_0^s \int_s^t \mathbb{E}[W_v (W_u - W_s)] dudv + \int_0^s \int_s^t \mathbb{E}[W_v W_s] dudv \\
&= \frac{1}{3}s^3 + \int_0^s \int_s^t \min(v, s) dudv = \frac{1}{3}s^3 + \frac{1}{2}(t-s)s^2
\end{aligned}$$

because

$$\int_0^s \int_s^t \mathbb{E}[W_v (W_u - W_s)] dudv = 0.$$

Hence

$$\mathbb{E}[Y_s Y_t] = \frac{1}{3} \min(s^3, t^3) + \frac{1}{2}(t-s) \min(s^2, t^2).$$

In applications, if S is a stochastic process representing the stock price. Then the quantity

$$A_T := \frac{1}{T} \int_0^T S_u du$$

is the payoff of the so called Asian option. That is the payoff of the option is the time average of the stock price over the time period $[0, T]$.

3.2 Reflection Principle

In this section, we present some results related to the reflection principle for the Brownian motion and its application in computing the distribution of the hitting times and the running supremum of the Brownian motion. In the following, let $(B_t)_{t \geq 0}$ be a standard Brownian motion, for $b > 0$ we let $T_b = \inf\{s : B_s = b\}$.

The aim below is to compute the distribution of the stopping time T_b . We first write

$$\mathbb{P}(T_b < t) = \mathbb{P}(T_b < t, B_t > b) + \mathbb{P}(T_b < t, B_t \leq b)$$

and make the observation that $\{T_b < t\} \cap \{B_t > b\} = \{b \leq \sup_{s \leq t} B_s\} \cap \{B_t > b\}$. This follows from the fact that $\{B_t > b\} \subset \{b \leq \sup_{s \leq t} B_s\}$. Heuristically speaking, the reflection principle tells us that

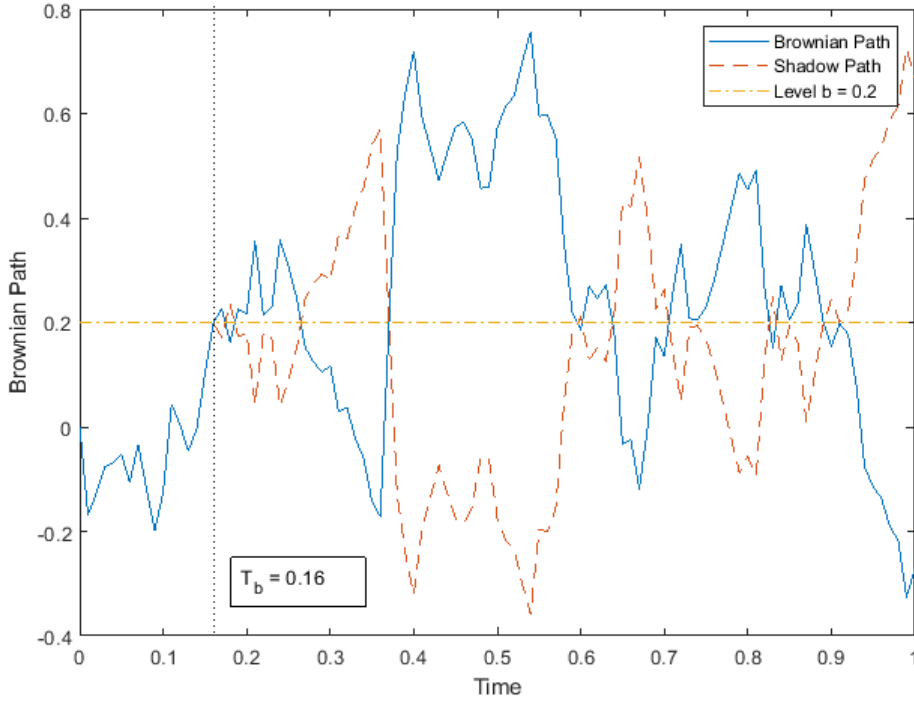
$$\mathbb{P}(T_b < t, B_t \leq b) = \mathbb{P}(T_b < t, B_t > b)$$

from which one can deduce that $\mathbb{P}(T_b < t) = 2\mathbb{P}(B_t > b)$.

We present below a heuristics explanation behind the equality

$$\mathbb{P}(T_b < t, B_t \leq b) = \mathbb{P}(T_b < t, B_t > b).$$

The mathematical proof requires the strong markov property of the Brownian motion which we do not cover here. We give below a heuristic argument.



The set $\{T_b < t\} \cap \{B_t \leq b\}$ is a collection of paths which before time t reached b at time T_b , and then starting from b the Brownian path travels to some point c less than b at time t . From the symmetry of Brownian motion, for every such path there should be a 'shadow path' which is reflected about the level b , that is there should be a 'shadow path', starting from b at time T_b , and travel to point $2b - c$ at time t . Therefore $\{T_b < t\} \cap \{B_t \leq b\}$ should have the same probability as $\{T_b < t\} \cap \{B_t > b\}$.

Formally, the process $-(W_t - W_{T_b}) + W_{T_b} = 2b - W_t$ for $t > T_b$ is a Brownian motion starting at T_b , which given the starting point W_{T_b} and time T_b is independent from the past. In other words, the process

$$\widetilde{W}_t = W_t \mathbb{1}_{\{T_b \geq t\}} + (2b - W_t) \mathbb{1}_{\{T_b < t\}}.$$

is another Brownian motion.

From the equality $\mathbb{P}(T_b < t) = 2\mathbb{P}(B_t > b)$, one can deduce the probability density function of the stopping time T_b and the following properties

Exercise.

- (i) Compute the probability density function of T_b .
- (ii) Show that $\mathbb{P}(T_b < \infty) = 1$ or $T_b < \infty$ almost surely.
- (iii) Show that $\mathbb{E}[T_b] = \infty$.
- (iv) Show that for any $t > 0$ the running supremum process $\sup_{s \leq t} B_s$ and the absolutely value process $|B_t|$ has the same distribution.

The exponential Brownian motion is a very important class of stochastic process. In particular, the Black-Scholes models assumes that the stock follows the process

$$X_t = X_0 \exp(\sigma W_t + \mu t), \quad t \geq 0.$$

The process X is a martingale if $\mu = -\frac{\sigma^2}{2}$ then X_t .

Lemma 3.2.1. *The moment generating function of T_b is $e^{-\sqrt{2\lambda}b}$ for $b, \lambda > 0$.*

Proof. From the previous example, we know that $X_t = X_0 \exp(\sigma W_t - \frac{\sigma^2}{2}t)$ is a martingale. Consider $S_0 = 1$ and S_t stopped at T_b . From the optional sampling theorem

$$\mathbb{E}[e^{\sigma W_{T_b \wedge t} - \frac{\sigma^2}{2}(t \wedge T_b)}] = S_0 = 1$$

By using the fact that $T_b < \infty$ a.s and letting $t \rightarrow \infty$, we have from the dominated convergence theorem,

$$\mathbb{E}[e^{\sigma b - \frac{\sigma^2}{2}T_b}] = 1.$$

Take $\sigma = \sqrt{2\lambda}$, we obtain $\mathbb{E}[e^{-\lambda T_b}] = e^{-\sqrt{2\lambda}b}$. □

More precisely, the hitting time T_b follow a Lévy distribution with parameters $(0, b^2)$

Lemma 3.2.2. *The joint distribution of $S_t = \sup_{s \leq t} W_s$ and W_t is given by*

$$\mathbb{P}(W_t \leq x, S_t > y) = \mathbb{P}(W_t > 2y - x) = \mathbb{P}(W_t < x - 2y)$$

for every $t \geq 0$, $y \geq 0$ and $x \leq y$.

Proof. The proof is based on the strong Markov property of the Brownian motion. We start by noticing that

$$\mathbb{E}(\mathbb{1}_{\{W_t \leq x\}} \mathbb{1}_{\{S_t > y\}}) = \mathbb{E}(\mathbb{1}_{\{W_t \leq x\}} \mathbb{1}_{\{t > T_y\}}) = \mathbb{E}(\mathbb{E}[\mathbb{1}_{\{W_t \leq x\}} | \mathcal{F}_{T_y}] \mathbb{1}_{\{t > T_y\}})$$

To compute the term $\mathbb{E}[\mathbb{1}_{\{W_t \leq x\}} | \mathcal{F}_{T_y}] = \mathbb{P}(W_t \leq x | \mathcal{F}_{T_y})$ we note that since $t > T_y$,

$$\begin{aligned} \mathbb{P}(W_t - W_{T_y} \leq x - W_{T_y} | \mathcal{F}_{T_y}) &= \mathbb{P}(W_t - W_{T_y} \leq x - y)_{y=W_{T_y}} \\ &= \mathbb{P}(W_t - y > y - x)_{y=W_{T_y}} \\ &= \mathbb{P}(W_t - W_{T_y} > W_{T_y} - x | \mathcal{F}_{T_y}) \\ &= \mathbb{P}(W_t > 2y - x | \mathcal{F}_{T_y}) \end{aligned}$$

where in the first equality, we have used the strong Markov property. This gives

$$\mathbb{P}(W_t \leq x, S_t > y) = \mathbb{P}(W_t > 2y - x, S_t > y) = \mathbb{P}(W_t > 2y - x)$$

and we have the claimed result. □

Corollary 3.2.1. *The joint density of a Brownian motion B_t and its supremum process M_t is given by*

$$\frac{2(2y-x)}{t} \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{(2y-x)^2}{2t}\right) \quad x \leq y, \quad y \geq 0$$

Proof. It is sufficient to differentiate the result in the previous lemma. □

Exercise: By computing the joint distribution of $(S_t - W_t, S_t)$ show that the distribution of the *drawdown* $S_t - W_t$ is the same as $|W_t|$.

Example 3.2.1. The joint distribution of Brownian motion and its running supremum can be useful in computing the price, for example, of the Up and Out Call Barrier option. That is the pay off is

$$(S_T - K)^+ \mathbb{1}_{\{M_T \leq b\}}$$

where S_t represents the stock and $M_t = \sup_{u \leq t} S_u$. The price of the *Up and Out Barrier option* is lower than the stand call option which has the pay-off $(S_T - K)^+ \geq (S_T - K)^+ \mathbb{1}_{\{M_T \leq b\}}$.

Chapter 4

Itô Stochastic Integral

Since almost all sample paths of a Brownian motion have infinite variation on every open interval, the classical Lebesgue-Stieltjes integration theory cannot be applied to define an integral of a stochastic process with respect to a Brownian motion. We shall now describe briefly the Itô integration theory, which is the basis of the Itô stochastic calculus. Let W be a standard d -dimensional Brownian motion defined on a filtered probability space $(\Omega, \mathbb{F}, \mathbb{P})$. For simplicity, the horizon date $T > 0$ will be fixed throughout and the notation $\cdot$ stands for the standard Euclidean dot product.

4.1 Itô Integral for Elementary Processes

Let $\mathcal{K}$ stand for the space of *elementary processes*, that is, processes of the form

$$\gamma(t) = \gamma_{-1} \mathbb{1}_0 + \sum_{j=0}^{m-1} \gamma_j \mathbb{1}_{(t_j, t_{j+1}]}(t), \forall t \in [0, T] \quad (4.1)$$

where $t_0 = 0 < t_1 < \dots < t_m = T$, the $\mathbb{R}^d$ -valued random variables γ_j , $j = 0, \dots, m-1$ are uniformly bounded and $\mathcal{F}_{t_j}$ -measurable and the random variable γ_{-1} is $\mathcal{F}_0$ -measurable. For any process $\gamma \in \mathcal{K}$, the *Itô stochastic integral* $\widehat{I}_T(\gamma)$ with respect to W over the time interval $[0, T]$ is defined by the formula

$$\widehat{I}_T(\gamma) = \int_0^T \gamma_u \cdot dW_u := \sum_{j=0}^{m-1} \gamma_j \cdot (W_{t_{j+1}} - W_{t_j}).$$

Similarly, the Itô stochastic integral of γ with respect to W over any interval $[0, t]$, where $t \leq T$, is defined by setting

$$\widehat{I}_t(\gamma) = \int_0^t \gamma_u \cdot dW_u := \widehat{I}_T(\gamma \mathbb{1}_{[0, t]}) = \sum_{j=0}^{m-1} \gamma_j \cdot (W_{t_{j+1} \wedge t} - W_{t_j \wedge t}), \quad (4.2)$$

where $x \wedge y = \min(x, y)$.

In the following, we check that for any process $\gamma \in \mathcal{K}$ the Itô integral $\widehat{I}_t(\gamma)$, $t \in [0, T]$, is a continuous $\mathbb{F}$ -martingale; in particular, the equality $\mathbb{E}(\widehat{I}_t(\gamma)|\mathcal{F}_u) = \widehat{I}_u(\gamma)$ holds for any $u \leq t \leq T$, or we can write

$$\mathbb{E}_{\mathbb{P}}\left(\int_0^t \gamma_s \cdot dW_s \middle| \mathcal{F}_u\right) = \int_0^u \gamma_s \cdot dW_s.$$

To check the martingale property, we note that

$$\begin{aligned} \mathbb{E}(\widehat{I}_t(\gamma)|\mathcal{F}_s) &= \mathbb{E}\left(\sum_{j=0}^{m-1} \gamma_j \cdot (W_{t_{j+1} \wedge t} - W_{t_j \wedge t}) \middle| \mathcal{F}_s\right) \\ &= \mathbb{E}\left(\mathbb{1}_{\{t_j > s\}} \sum_{j=0}^{m-1} \gamma_j \cdot (W_{t_{j+1} \wedge t} - W_{t_j \wedge t}) \middle| \mathcal{F}_s\right) + \mathbb{E}\left(\mathbb{1}_{\{t_j \leq s\}} \sum_{j=0}^{m-1} \gamma_j \cdot (W_{t_{j+1} \wedge t} - W_{t_j \wedge t}) \middle| \mathcal{F}_s\right) \end{aligned}$$

On the set $\{t_j \leq s\} \in \mathcal{F}_s$, we use the fact that a martingale stopped at a stopping time is again a martingale, see for example lemma 2.2.2, to obtain that

$$\sum_{j=0}^{m-1} \mathbb{1}_{\{t_j \leq s\}} \gamma_j \cdot \mathbb{E}[(W_{t_{j+1} \wedge t} - W_{t_j \wedge t}) | \mathcal{F}_s] = \sum_{j=0}^{m-1} \mathbb{1}_{\{t_j \leq s\}} \gamma_j \cdot (W_{t_{j+1} \wedge s} - W_{t_j \wedge s})$$

On the other hand, on the set $\{t_j > s\} \in \mathcal{F}_s$, by using the tower property, we have

$$\mathbb{E}\left[\sum_{j=0}^{m-1} \mathbb{1}_{\{t_j > s\}} \gamma_j \cdot (W_{t_{j+1} \wedge t} - W_{t_j \wedge t}) \middle| \mathcal{F}_s\right] = \mathbb{E}\left[\sum_{j=0}^{m-1} \mathbb{1}_{\{t_j > s\}} \gamma_j \cdot \mathbb{E}[(W_{t_{j+1} \wedge t} - W_{t_j \wedge t}) | \mathcal{F}_{t_j}] \middle| \mathcal{F}_s\right] = 0.$$

One can then conclude by noticing that

$$\sum_{j=0}^{m-1} \mathbb{1}_{\{t_j \leq s\}} \gamma_j \cdot (W_{t_{j+1} \wedge s} - W_{t_j \wedge s}) = \sum_{j=0}^{m-1} \gamma_j \cdot (W_{t_{j+1} \wedge s} - W_{t_j \wedge s}).$$

To this end, it is not hard to see that it is integrable and continuity in time is obvious.

In addition, the following properties are valid, for $t \in [0, T]$,

(i) (*linearity*) for any $a, b \in \mathbb{R}$ and any processes $\gamma, \eta \in \mathcal{K}$

$$\int_0^t (a\gamma_u + b\eta_u) \cdot dW_u = a \int_0^t \gamma_u \cdot dW_u + b \int_0^t \eta_u \cdot dW_u,$$

(ii) (*local property*) for any $\mathbb{F}$ -stopping time τ and any process $\gamma \in \mathcal{K}$ we have

$$\int_0^t \gamma_u \mathbb{1}_{[0, \tau]}(u) \cdot dW_u = \int_0^t \gamma_u^\tau \cdot dW_u^\tau = I_t^\tau(\gamma).$$

4.1.1 Itô Integral for Processes from $\mathcal{L}_{\mathbb{P}}^2(W)$

In the next step, the definition of the Itô integral will be extended from the class $\mathcal{K}$ to the class $\mathcal{L}_{\mathbb{P}}^2(W)$ of all $\mathbb{F}$ -progressively measurable processes γ defined on $(\Omega, \mathbb{F}, \mathbb{P})$ for which

$$\|\gamma\|_W^2 := \mathbb{E}\left[\int_0^T |\gamma_u|^2 du\right] < \infty, \quad (4.3)$$

where $|\cdot|$ stands for the Euclidean norm in $\mathbb{R}^d$.

Lemma 4.1.1. *The class $\mathcal{K}$ is a subset of $\mathcal{L}_{\mathbb{P}}^2(W)$ and for any $\gamma \in \mathcal{K}$*

$$\mathbb{E}\left[\left(\int_0^T \gamma_u \cdot dW_u\right)^2\right] = \|\hat{I}_T(\gamma)\|_{L^2}^2 = \|\gamma\|_W^2. \quad (4.4)$$

The space $\mathcal{L}_{\mathbb{P}}^2(W)$ equipped with the norm $\|\cdot\|_W$ is a complete normed linear space – that is, a Banach space. Moreover, the class $\mathcal{K}$ of elementary stochastic processes is a dense linear subspace of $\mathcal{L}_{\mathbb{P}}^2(W)$.

Proof. We proof only the Itô isometry and in $\mathbb{R}$, the multi-dimensional case can be studied using similar computations. To this end, since the expected value of the cross terms are zero, i.e, for $j \neq k$, $\mathbb{E}[(W_{t_{j+1}} - W_{t_j})(W_{t_{k+1}} - W_{t_k})] = 0$, we obtain

$$\begin{aligned} \mathbb{E}\left[\left(\sum_{i=0}^{m-1} \gamma_j(W_{t_{j+1}} - W_{t_j})\right)^2\right] &= \mathbb{E}\left[\left(\sum_{i=0}^{m-1} \gamma_j^2(W_{t_{j+1}} - W_{t_j})^2\right)\right] \\ &= \mathbb{E}\left[\sum_{i=0}^{m-1} \gamma_j^2 \mathbb{E}[(W_{t_{j+1}} - W_{t_j})^2 | \mathcal{F}_{t_j}]\right] \\ &= \mathbb{E}\left[\sum_{i=0}^{m-1} \gamma_j^2(t_{j+1} - t_j)\right] = \|\gamma\|_W^2 \end{aligned}$$

which shows that $\|\hat{I}_T(\gamma)\|_{L^2}^2 = \|\gamma\|_W^2$. This means that $\hat{I}_T : (\mathcal{K}, \|\cdot\|_W) \rightarrow L^2(\Omega, \mathcal{F}_T, \mathbb{P})$ is an isometry, i.e. a distance preserving transformation. $\square$

By virtue of lemma 4.1.1, the isometry $\hat{I}_T : (\mathcal{K}, \|\cdot\|_W) \rightarrow L^2(\Omega, \mathcal{F}_T, \mathbb{P})$ can be extended to the isometry $I_T : (\mathcal{L}_{\mathbb{P}}^2(W), \|\cdot\|_W) \rightarrow L^2(\Omega, \mathcal{F}_T, \mathbb{P})$. The extension is done using the fact that the class of simple process $\mathcal{K}$ is a dense subset in the Banach space $\mathcal{L}_{\mathbb{P}}^2(W)$ and the space $L^2(\Omega, \mathcal{F}_T, \mathbb{P})$ is also complete (since it a Banach space or more specifically a Hilbert space)¹. For the fact that $\mathcal{K}$ is dense in $\mathcal{L}_{\mathbb{P}}^2(W)$, we refer to Lemma 3.2.4 on page 132 of Karatzas and Shreve (1998)

Definition 4.1.1. For any process $\gamma \in \mathcal{L}_{\mathbb{P}}^2(W)$, the random variable $I_T(\gamma)$ is called the *Itô stochastic integral* of γ with respect to W over $[0, T]$, and it is denoted by $\int_0^T \gamma_u \cdot dW_u$.

¹A space is complete if every Cauchy sequence convergence to a limit in the space.

In other words, for any $\gamma \in \mathcal{L}_{\mathbb{P}}^2(W)$ we set $I_T(\gamma) = \lim_{n \rightarrow \infty} \widehat{I}_T(\gamma^n)$, where the limit is taken in $L^2(\Omega, \mathcal{F}_T, \mathbb{P})$, for any sequence of processes $(\gamma^n)_{n \in \mathbb{N}}$ such that $\gamma^n \in \mathcal{K}$ and

$$\lim_{n \rightarrow \infty} \mathbb{E}_{\mathbb{P}} \left(\int_0^T |\gamma_u - \gamma_u^n|^2 du \right) = \lim_{n \rightarrow \infty} \|\gamma - \gamma^n\|_W^2 = 0.$$

More generally, for every $\gamma \in \mathcal{L}_{\mathbb{P}}^2(W)$ and every $t \in [0, T]$, we set

$$I_t(\gamma) = \int_0^t \gamma_u \cdot dW_u := I_T(\gamma \mathbf{1}_{[0, t]}) \quad (4.5)$$

and thus we define the Itô stochastic integral $I(\gamma)$ as an $\mathbb{F}$ -adapted stochastic process. The next result summarizes its most important properties.

Proposition 4.1.1. *For any process $\gamma \in \mathcal{L}_{\mathbb{P}}^2(W)$, the Itô stochastic integral $I(\gamma)$ is a square-integrable continuous martingale on $(\Omega, \mathbb{F}, \mathbb{P})$. Moreover, the process*

$$(I_t(\gamma))^2 - \langle I(\gamma) \rangle_t, \forall t \in [0, T] \quad (4.6)$$

is a continuous martingale on $(\Omega, \mathbb{F}, \mathbb{P})$, where

$$\langle I(\gamma) \rangle_t := \int_0^t |\gamma_u|^2 du, \forall t \in [0, T]. \quad (4.7)$$

It is worth mentioning that the Itô integral is a linear map on $\mathcal{L}_{\mathbb{P}}^2(W)$ and it enjoys (i) the local property with respect to $\mathbb{F}$ -stopping times, that is, $I(\gamma \mathbf{1}_{[0, \tau]}) = I^\tau(\gamma)$ for any $\mathbb{F}$ -stopping time τ and any process $\gamma \in \mathcal{L}_{\mathbb{P}}^2(W)$ and (ii) the process $I(\gamma)$ is a martingale on the interval $[0, T]$.

4.1.2 Itô Integral for processes from $\mathcal{L}(W)$

It appears that the martingale property is too restrictive if we wish define the Itô integral for a large class of processes. Hence the following definition.

Definition 4.1.2. A process M is said to be a *local martingale* with respect to $\mathbb{F}$ if there exists an increasing sequence $(\tau_n)_{n \in \mathbb{N}}$ of stopping times such that τ_n tends to T almost surely, and for every n the process M^n , given by the formula

$$M_t^n = \begin{cases} M_{t \wedge \tau_n}(\omega), & \text{if } \tau_n(\omega) > 0, \\ 0, & \text{if } \tau_n(\omega) = 0, \end{cases}$$

is a uniformly integrable martingale. Any sequence $(\tau_n)_{n \in \mathbb{N}}$ with these properties is called the *reducing sequence* for a local martingale M .

By applying the optional stopping technique (also known as a *localization*), we shall extend the definition of Itô stochastic integral to the class $\mathcal{L}(W)$ of all progressively measurable processes γ for which

$$\mathbb{P} \left(\int_0^T |\gamma_u|^2 du < \infty \right) = 1. \quad (4.8)$$

Let a process γ belong to $\mathcal{L}(W)$. In order to define the Itô integral $I_T(\gamma)$, we first introduce an increasing sequence $(\tau_n)_{n \in \mathbb{N}}$ of $\mathbb{F}$ -stopping times by setting

$$\tau_n = \inf \left\{ t \in [0, T] : \int_0^t |\gamma_u|^2 du \geq n \right\},$$

where by convention $\inf \emptyset = T$. Since the process γ belongs to the class $\mathcal{L}(W)$, it is not difficult to verify that $\lim_{n \rightarrow \infty} \tau_n = T$.

Furthermore, for any $n \in \mathbb{N}$ the process γ^n , which is given by the formula $\gamma_t^n = \gamma_t \mathbb{1}_{[0, \tau_n]}(t)$, $t \in [0, T]$, manifestly belongs to the class $\mathcal{L}^2(W)$ and thus the Itô integral $I_T(\gamma^n)$ is well defined. Using the local property of the Itô integral, we obtain

$$I(\gamma^n) = I(\gamma \mathbb{1}_{[0, \tau_n]}) = (I(\gamma^n))^{\tau_n}.$$

We may thus define the Itô integral $I_T(\gamma)$ by setting

$$I_T(\gamma) = \lim_{n \rightarrow \infty} I_T(\gamma^n).$$

We need, of course, to check that the right-hand side in the last equality is well defined and does not depend on the choice of the *localizing* sequence of $\mathbb{F}$ -stopping times. The details of this construction are left to the reader (see e.g., Karatzas and Shreve (1998)).

We stress here that the Itô integral of a process from the class $\mathcal{L}(W)$ does not necessarily follow an $\mathbb{F}$ -martingale, in fact, they are $\mathbb{F}$ -local martingale.

Definition 4.1.3. A stochastic process X is said to be a *local martingale* with respect to the filtration $\mathbb{F}$, if there exists a sequence of $\mathbb{F}$ -stopping times $(T_n)_{n \geq 0}$ converging to infinity, i.e $T_n \rightarrow T$ (or ∞ if one is on a infinite time horizon), such that the stopped process X^{T_n} is a martingale with respect to the filtration $\mathbb{F}$.

4.2 Itô Processes and Itô's Lemma

Let W be a d -dimensional standard Brownian motion defined on a filtered probability space $(\Omega, \mathbb{F}, \mathbb{P})$. We shall now introduce a particular class of continuous semimartingales, referred to as *Itô processes*.

Definition 4.2.1. An $\mathbb{F}$ -adapted continuous process X is called an *Itô process* if it admits a representation

$$X_t = X_0 + \int_0^t \alpha_u du + \int_0^t \beta_u \cdot dW_u, \quad \forall t \in [0, T] \quad (4.9)$$

for some $\mathbb{F}$ -adapted processes α and β that are defined on $(\Omega, \mathbb{F}, \mathbb{P})$ and satisfy suitable integrability conditions.

It is customary to write the integral formula (4.9) using the differential notation as

$$dX_t = \alpha_t dt + \beta_t \cdot dW_t.$$

Remark 4.2.1. Note that from the property of Itô's integral, if $\mathbb{E}[\int_0^T |\beta_s|^2 ds] < \infty$ then the process is a $\int_0^t \beta_s \cdot dW_s$ is a continuous martingale.

In the following, we consider the following problem: What is the dynamics or trajectory interpretation of $g(X_t)$ when follow X is a Itô process and g is a sufficiently regular function?

4.2.1 One-Dimensional Case

Consider a real-valued function $g = g(x, t)$, where $x \in \mathbb{R}$ is the space variable and $t \in [0, T]$ is the time variable. It is evident that if X is a continuous semimartingale, and $g : \mathbb{R} \times [0, T] \rightarrow \mathbb{R}$ is a jointly continuous function, then the process $Y_t = g(X_t, t)$ is $\mathcal{F}_t$ -adapted and has almost all sample paths continuous.

The next result, which is a special case of Itô's lemma, states that Y is a semimartingale provided that the function g is sufficiently smooth.

Proposition 4.2.1. Suppose that $g : \mathbb{R} \times [0, T] \rightarrow \mathbb{R}$ is a function of class $C^{2,1}(\mathbb{R} \times [0, T], \mathbb{R})$. Then for any Itô process X , the process $Y_t = g(X_t, t)$, $t \in [0, T]$, is an Itô process. Moreover, its canonical decomposition is given by the Itô formula

$$dY_t = g_t(X_t, t) dt + g_x(X_t, t)\alpha_t dt + g_x(X_t, t)\beta_t dW_t + \frac{1}{2}g_{xx}(X_t, t)\beta_t^2 dt.$$

Example 4.2.1. Apply the Itô formula to $f(W_t) = (W_t)^2$. Then we have for all $t \in [0, T]$

$$\begin{aligned} W_t^2 &= W_0^2 + \int_0^t 2W_s dW_s + \frac{1}{2}2 \int_0^t d\langle W \rangle_s \\ &= \int_0^t 2W_s dW_s + t \end{aligned}$$

Note that this also gives the Doob-Meyer decomposition of the positive submartingale W^2 . Also recall that $W_t^2 - t = \int_0^t 2W_s dW_s$ which is a martingale.

Exercise: Compute the dynamics of $X_t = tW_t^2$ by applying the Itô's lemma. Is X a supermartingale, submartingale or neither?

4.2.2 Multidimensional Case

Recall that a process $W = (W^1, \dots, W^d)$ defined on a filtered probability space $(\Omega, \mathbb{F}, \mathbb{P})$ is called the d -dimensional standard Brownian motion if the components $W^1, \dots, W^d$ are independent one-dimensional standard Brownian motions with respect to $\mathbb{F}$.

In this paragraph, W is assumed to be a d -dimensional standard Brownian motion. Let γ belong to $\mathcal{L}_W(\mathbb{P})$, that is, γ is a $\mathbb{R}^d$ -valued $\mathbb{F}$ -progressively measurable process satisfying

$$\mathbb{P}\left(\int_0^T |\gamma_u|^2 du < \infty\right) = 1,$$

where $|\cdot|$ stands for the Euclidean norm in $\mathbb{R}^d$. Then the Itô stochastic integral of γ with respect to W is well defined and we have, for any $t \in [0, T]$,

$$I_t(\gamma) = \int_0^t \gamma_u \cdot dW_u = \sum_{i=1}^d \int_0^t \gamma_u^i dW_u^i.$$

Let $X = (X^1, \dots, X^k)$ be a k -dimensional Itô process given as

$$X_t^i = X_0^i + \int_0^t \alpha_u^i du + \int_0^t \beta_u^i \cdot dW_u, \quad (4.10)$$

where α^i are real-valued processes and $\beta^i = (\beta^{i1}, \dots, \beta^{id})$ are $\mathbb{R}^d$ -valued processes for $i = 1, \dots, k$. It is implicitly assumed in (4.10) that the processes $\alpha^i, \beta^i, i = 1, \dots, k$ are integrable in a suitable sense.

Let us introduce notation for the *cross-variation* (or the *quadratic covariation*) of two continuous semimartingales. If $X^i = X_0^i + M^i + A^i$ are in $\mathcal{S}^c(\mathbb{P})$ for $i = 1, \dots, k$ then we set $\langle X^i, X^j \rangle := \langle M^i, M^j \rangle$, where in turn $\langle M^i, M^j \rangle$ is given by the following *polarization equality*

$$\langle M^i, M^j \rangle = \frac{1}{2} (\langle M^i + M^j \rangle - \langle M^i \rangle - \langle M^j \rangle) = \frac{1}{4} (\langle M^i + M^j \rangle - \langle M^i - M^j \rangle).$$

It is easily seen that $\langle X^i, X^i \rangle = \langle X^i \rangle$. If X^i and X^j are the Itô processes given by (4.10) then it is easily seen that we have, for any $t \in [0, T]$,

$$\langle X^i, X^j \rangle_t = \int_0^t \beta_u^i \cdot \beta_u^j du = \int_0^t \sum_{l=1}^d \beta_u^{il} \beta_u^{jl} du$$

The following result extends Proposition 4.2.1.

Proposition 4.2.2. *Suppose that g is a function of class $C^2(\mathbb{R}^k, \mathbb{R})$. Then the following form of the Itô formula is valid*

$$dg(X_t) = \sum_{i=1}^k g_{x_i}(X_t) \alpha_t^i dt + \sum_{i=1}^k g_{x_i}(X_t) \beta_t^i \cdot dW_t + \frac{1}{2} \sum_{i,j=1}^k g_{x_i x_j}(X_t) \beta_t^i \cdot \beta_t^j dt.$$

Exercise: Let $R_t(d) := \sqrt{|W_t^1|^2 + |W_t^2|^2 + \dots + |W_t^d|^2}$ where $W = (W^1, W^2, \dots, W^d)$ is a standard d -dimensional Brownian motion for $d \geq 3$. The process R is called a Bessel process. Show that $1/R_t(3)$ for $1 \leq t \leq T$ is a local martingale but not a martingale. *Hint:* Compute directly $\mathbb{E}[1/R_t(d)]$ and show that it is not a constant.

Heuristically one can apply Itô formula to compute the dynamics of $1/R_t(3)$. However strictly speaking one cannot apply directly the Itô's formula, one note that the function $f(x) = \sqrt{x}$ is not differentiable at zero. But for $d \geq 3$, the 3D Brownian motion never return to zero after time zero.

4.3 Computational Examples

Given a filtered probability space $(\Omega, \mathbb{F}, \mathbb{P})$, we suppose that B is a $\mathbb{F}$ -Brownian Motion and γ_u is a progressively measure process such that

$$\mathbb{P} \left(\int_0^T \gamma_u^2 du < \infty \right) \quad a.s$$

then the stochastic integral $I(\gamma)$ satisfies the following properties

- The process $\int_0^t \gamma_u dW_u$ is a local martingale for $t \in [0, T]$.
- Given any constant a and $b \in \mathbb{R}$

$$\begin{aligned} \int_0^t (a + b\gamma_u) dW_u &= \int_0^t a dW_u + \int_0^t b\gamma_u dW_u \\ &= aW_t + b \int_0^t \gamma_u dW_u \end{aligned}$$

- The quadratic variation of $I(\gamma)_t = \int_0^t \gamma_u dW_u$ is given by

$$\langle I(\gamma) \rangle_t = \int_0^t \gamma_u^2 du.$$

- If in addition, we can show that

$$\mathbb{E} \left[\int_0^T \gamma_u^2 du \right] < \infty$$

then $I(\gamma)_t = \int_0^t \gamma_u dW_u$ is a true martingale, i.e. not just a local martingale.

- Given Itô processes $X_t = \int_0^t \alpha_s dB_s$ and $Y_t = \int_0^t \beta_s dB_s$ then to compute the quadratic covariation $\langle X, Y \rangle_t$, we can use the polarisation formula and obtain

$$\begin{aligned} \langle X, Y \rangle_t &= \frac{1}{2} (\langle X + Y \rangle_t - \langle X \rangle_t - \langle Y \rangle_t) \\ &= \frac{1}{2} \left(\left\langle \int_0^t (\alpha_s + \beta_s) dW_s, \int_0^t (\alpha_s + \beta_s) dW_s \right\rangle_t - \left\langle \int_0^t \alpha_s dW_s, \int_0^t \alpha_s dW_s \right\rangle_t - \left\langle \int_0^t \beta_s dW_s, \int_0^t \beta_s dW_s \right\rangle_t \right) \\ &= \frac{1}{2} \left(\int_0^t (\alpha_s + \beta_s)^2 ds - \int_0^t \alpha_s^2 ds - \int_0^t \beta_s^2 ds \right) \\ &= \int_0^t \alpha_s \beta_s ds \end{aligned}$$

We stress that the process $A_t = \int_0^t \alpha_s ds$ is always a process of finite variation. This is clear when α_s is positive and A_t is increasing, since the total variation on $[0, T]$ is just A_T . If α is not positive, we can write $\alpha = \alpha^+ - \alpha^-$ and $A_t = A_t^+ - A_t^-$ where

$$A_t^\pm = \int_0^t \alpha_s^\pm ds,$$

and it is not difficult to see that the total variation of A is bounded by $A_T^+ + A_T^-$. In general, by the *Jordan decomposition theorem* any process (function) of finite variation can be written as the difference of two increasing processes (function).

Lemma 4.3.1. (*Integration By Parts Formula*) *Given two continuous semimartingales X and Y then*

$$X_t Y_t = X_0 Y_0 + \int_0^t X_s dY_s + \int_0^t Y_s dX_s + \langle X, Y \rangle_t$$

Proof. Apply the multi-dimensional Itô formula to $f(x, y) = xy$ we have

$$\begin{aligned} d(XY)_t &= f_x(X_t, Y_t) dX_t + \frac{1}{2} f_{xx}(X_t, Y_t) d\langle X \rangle_t + f_y(X_t, Y_t) dY_t \\ &\quad + \frac{1}{2} f_{yy}(X_t, Y_t) d\langle Y \rangle_t + \frac{1}{2} f_{xy}(X_t, Y_t) d\langle X, Y \rangle_t + \frac{1}{2} f_{yx}(X_t, Y_t) d\langle Y, X \rangle_t. \end{aligned}$$

Then we use the fact that $f_x(X_t, Y_t) = Y_t$, $f_y(X_t, Y_t) = X_t$, $f_{xx} = f_{yy} = 0$ and $f_{xy} = f_{yx} = 1$ to obtain the formula. $\square$

Special case: In the case where both X_t and Y_t are of finite variation then the quadratic covariation process $\langle X, Y \rangle$ is equal to zero. We can recover the usual integration by parts formula

$$X_t Y_t = X_0 Y_0 + \int_0^t X_s dY_s + \int_0^t Y_s dX_s$$

To see why the quadratic covariation is zero, we use the polarization formula

$$\langle X, Y \rangle = \frac{1}{2} (\langle X + Y, X + Y \rangle - \langle X \rangle - \langle Y \rangle) = 0$$

since the quadratic variation of processes of finite variation is zero.

In fact, if only one of X or Y is a process of finite variation then the formula

$$X_t Y_t = X_0 Y_0 + \int_0^t X_s dY_s + \int_0^t Y_s dX_s$$

holds true.

Example 4.3.1. Let B be a Brownian motion, compute the dynamics of te^{B_t} . We can let $Y_t = t$ and $X_t = e^{B_t}$. Note that $Y_t = t$ is process of finite variation (it is increasing and the total variation is given by T), then by the integration by parts formula we have

$$d(te^{B_t}) = e^{B_t} dt + td(e^{B_t})$$

to compute $d(e^{B_t})$ we need to apply Itô formula to e^{B_t} with $f(x) = e^x$. This gives

$$e^{B_t} = e^0 + \int_0^t e^{B_s} dB_s + \int_0^t \frac{1}{2} e^{B_s} ds$$

or in differential notation $d(e^{B_t}) = e^{B_t} dB_t + \frac{1}{2}e^{B_t} dt$. This gives

$$d(te^{B_t}) = e^{B_t} dt + te^{B_t} dB_t + t \frac{1}{2} e^{B_t} dt$$

or in integral form

$$\begin{aligned} te^{B_t} &= 0 \times e^0 + \int_0^t e^{B_s} ds + \int_0^t se^{B_s} dB_s + \int_0^t s \frac{1}{2} e^{B_s} ds \\ &= \int_0^t e^{B_s} \left(\frac{s}{2} + 1 \right) ds + \int_0^t se^{B_s} dB_s \end{aligned}$$

The Itô formula gives an additive decomposition of the process te^{B_t} . We see that it is the sum of an increasing process $A_t = \int_0^t e^{B_s} \left(\frac{s}{2} + 1 \right) ds$ and a martingale $M_t = \int_0^t se^{B_s} dB_s$ on $[0, T]$.

From the semimartingale decomposition, we can claim that the process te^{B_t} is a positive submartingale, since the finite variation part A_t is increasing, and therefore the process is increasing on average. In fact one can check directly.

Let us first suppose that $\int_0^t se^{B_s} dB_s$ is a true martingale, we will show square integrability later. For $u \leq t$

$$\begin{aligned} \mathbb{E}[te^{B_t} | \mathcal{F}_u] &= \mathbb{E}\left[\int_0^t e^{B_s} \left(\frac{s}{2} + 1 \right) ds | \mathcal{F}_u\right] + \mathbb{E}\left[\int_0^t se^{B_s} dB_s | \mathcal{F}_u\right] \\ &= \mathbb{E}\left[\int_0^t e^{B_s} \left(\frac{s}{2} + 1 \right) ds | \mathcal{F}_u\right] + \int_0^u se^{B_s} dB_s \\ &= \mathbb{E}\left[\left(\int_0^t e^{B_s} \left(\frac{s}{2} + 1 \right) ds - \int_0^u e^{B_s} \left(\frac{s}{2} + 1 \right) ds\right) + \int_0^u e^{B_s} \left(\frac{s}{2} + 1 \right) ds | \mathcal{F}_u\right] + \int_0^u se^{B_s} dB_s \end{aligned}$$

Using the fact that $\int_0^u e^{B_s} ds$ is $\mathcal{F}_u$ measurable, we can write

$$\begin{aligned} \mathbb{E}[te^{B_t} | \mathcal{F}_u] &= \mathbb{E}\left[\int_u^t e^{B_s} \left(\frac{s}{2} + 1 \right) ds | \mathcal{F}_u\right] + \int_0^u e^{B_s} \left(\frac{s}{2} + 1 \right) ds + \int_0^u se^{B_s} dB_s \\ &= \mathbb{E}\left[\int_u^t e^{B_s} \left(\frac{s}{2} + 1 \right) ds | \mathcal{F}_u\right] + ue^{B_u} \\ &\geq ue^{B_u} \end{aligned}$$

where we used that the fact that $\mathbb{E}\left[\int_u^t e^{B_s} \left(\frac{s}{2} + 1 \right) ds | \mathcal{F}_u\right] \geq 0$. To show that $\int_0^t se^{B_s} dB_s$ is a true martingale, we just check that

$$\mathbb{E}\left[\int_0^t s^2 e^{2B_s} ds\right] < \infty.$$

By using the form of the moment generating function of a $\mathcal{N}(0, s)$ we obtain

$$\mathbb{E}\left[\int_0^T s^2 e^{2B_s} ds\right] = \int_0^T s^2 \mathbb{E}[e^{2B_s}] ds = \int_0^T s^2 e^{2s^2} ds \leq T^2 e^{2T^2} T < \infty$$

Example 4.3.2. Compute the quadratic variation of the martingale $N_t = W_t^2 - t$ by using the Itô formula. *Hint: Use the fact that the quadratic variation $\langle N \rangle$ is the unique process such that $N_t^2 - \langle N \rangle_t$ is a martingale.*

Chapter 5

Semimartingales

5.1 Continuous Local Martingales

In order to simplify the presentation, we assume that the σ -field $\mathcal{F}_0$ is trivial, that is, $\mathbb{P}(A) = 0$ or $\mathbb{P}(A) = 1$ for any $A \in \mathcal{F}_0$. This implies that the initial value M_0 is therefore a real number.

Definition 5.1.1. A stochastic process X is said to be a *local martingale* with respect to the filtration $\mathbb{F}$, if there exists a sequence of $\mathbb{F}$ -stopping times $(T_n)_{n \geq 0}$ converging to infinity, i.e. $T_n \rightarrow T$ (or ∞ if one is on an infinite time horizon), such that the stopped process X^{T_n} is a (uniformly integrable) martingale with respect to the filtration $\mathbb{F}$.

Any martingale is also a local martingale. It is noteworthy that the class of (continuous) local martingales is essentially larger than the class of (continuous) martingales. Given a continuous local martingale X , it is always possible to stop the local martingale X so that X is bounded and hence square integrable. For example, one can take $\tau_n = \inf\{s : |X_s| > n\}$ and $|X_{\tau_n}| \leq n$.

Remark 5.1.1. Assume that a stochastic process γ belongs to $\mathcal{L}(W)$. The Itô stochastic integral $I(\gamma)$ follows a *continuous local martingale* with respect to the filtration $\mathbb{F}$.

Proposition 5.1.1. Any non-negative local martingale M with respect to $\mathbb{F}$ is also an $\mathbb{F}$ -supermartingale. If, in addition, the expected value of M is constant then M is an $\mathbb{F}$ -martingale.

Proof. Recall that if M is a local martingale with respect to $\mathbb{F}$ then there exists an increasing sequence $(\tau_n)_{n \in \mathbb{N}}$ of $\mathbb{F}$ -stopping times such that

$$\mathbb{P}\left\{\lim_{n \rightarrow \infty} \tau_n = T\right\} = 1$$

and for any $n \in \mathbb{N}$ the stopped process M^{τ_n} is an $\mathbb{F}$ -martingale. This means that we have, for any fixed $n \in \mathbb{N}$ and arbitrary dates $0 \leq u \leq t \leq T$,

$$M_u^{\tau_n} = \mathbb{E}(M_t^{\tau_n} | \mathcal{F}_u).$$

Since the process M is non-negative, using the conditional form of Fatou's lemma (see Appendix), we obtain, for $0 \leq u \leq t \leq T$,

$$\begin{aligned}\mathbb{E}(M_t|\mathcal{F}_u) &= \mathbb{E}(\liminf_{n \rightarrow \infty} M_t^{\tau_n}|\mathcal{F}_u) \leq \liminf_{n \rightarrow \infty} \mathbb{E}(M_t^{\tau_n}|\mathcal{F}_u) \\ &= \liminf_{n \rightarrow \infty} M_u^{\tau_n} = M_u.\end{aligned}$$

We conclude that M is an $\mathbb{F}$ -supermartingale. In particular, M is an integrable process, that is, $\mathbb{E}[M_t] \leq \mathbb{E}[M_0] < \infty$ for $t \in [0, T]$. If in addition the expected value is constant, that is $\mathbb{E}(M_u) = \mathbb{E}(M_t)$ then from 2. of Proposition 2.2.1, the process M is a martingale. $\square$

The assumption that M is non-negative can be replaced by the assumption that $M_u \geq \eta$ for some random variable η with $\mathbb{E}(\eta) > -\infty$.

Corollary 5.1.1. *Let M be a non-negative local martingale M with respect to $\mathbb{F}$. If $M_0 = 0$ then $M_t = 0$ for every $t \in [0, T]$.*

Proof. Assume that $M_{t_0} \neq 0$ for some $t_0 \in [0, T]$. Then $\mathbb{E}(M_{t_0}) > 0 = \mathbb{E}(M_0)$, which contradicts the property that M is a supermartingale. $\square$

The following definition extends the concept of quadratic variation to the class of all continuous local martingales.

Definition 5.1.2. Let M be a continuous local martingale with respect to $\mathbb{F}$. We denote by $\langle M \rangle$ the unique continuous, increasing and $\mathbb{F}$ -adapted process with $\langle M \rangle_0 = 0$ such that the process $M^2 - \langle M \rangle$ is a continuous local martingale with respect to $\mathbb{F}$. The process $\langle M \rangle$ is termed the *quadratic variation* of M .

It is worth noting that the quadratic variation is invariant with respect to an $\mathcal{F}_0$ -measurable shift of M ; specifically, if M is a continuous local martingale and a continuous local martingale N satisfies $N = \psi + M$ for some $\mathcal{F}_0$ -measurable random variable ψ then $\langle N \rangle = \langle M \rangle$. In particular, the quadratic variations of M and $M - M_0$ are identical, that is, $\langle M \rangle = \langle M - M_0 \rangle$.

The existence and uniqueness of the process $\langle M \rangle$ satisfying conditions of Definition 5.1.2 is a non-trivial issue, but it can be deduced from the Doob-Meyer decomposition theorem. We state below without proof a suitable version of this classic result. Let us stress that the result given below is by no means the most general version of the Doob-Meyer decomposition theorem (see, for instance, Karatzas and Shreve (1998)).

Remark 5.1.2. It is known that the quadratic variation $\langle M \rangle$ of a continuous local martingale M can also be interpreted as the following limit (cf. (3.3))

$$\langle M \rangle_t = \lim_{n \rightarrow \infty} \sum_{k=1}^{m_n} (M_{t_k^n} - M_{t_{k-1}^n})^2, \quad (5.1)$$

where the convergence in probability holds for any sequence $(\pi_n)_{n \in \mathbb{N}}$ of finite partitions $\pi_n = \{0 = t_0^n < t_1^n < \dots < t_{m_n}^n = t\}$ of the interval $[0, t]$ satisfying $\lim_{n \rightarrow \infty} |\pi_n| = 0$. Hence the name *quadratic variation* attributed to the process $\langle M \rangle$ is indeed justified.

Lemma 5.1.1. *Let M be a continuous local martingale such that $\langle M \rangle_t = 0$ for $t \in [0, T]$. Then $M_t = M_0$ for every $t \in [0, T]$.*

Proof. We may and do assume, without loss of generality, that M is a continuous bounded martingale and $M_0 = 0$. This is because one can always take $M' = M - M_0$ and stopped the continuous local martingale M' such that it is bounded. Then $\mathbb{E}[M_t^2] = \mathbb{E}[\langle M \rangle_t] = 0$ for any $t \in [0, T]$ and thus $M_t = 0$ for any $t \in [0, T]$. $\square$

Definition 5.1.3. An $\mathbb{F}$ -adapted stochastic process $X = (X_t)_{t \in [0, T]}$ is said to be of *finite variation* if almost all sample paths of X are functions of finite variation on $[0, T]$.

More explicitly, recall that a process X is of finite variation whenever (cf. (3.4) in section 4.)

$$V_{[0, T]}(X) := \sup_{\pi \in \Pi_{[0, T]}} \sum_{k=1}^{n(\pi)} |X_{t_k} - X_{t_{k-1}}| < \infty, \quad \mathbb{P}a.s., \quad (5.2)$$

where $\Pi_{[0, T]}$ is the class of all partitions $\pi = \{0 = t_0 < t_1 < \cdots < t_{n(\pi)} = T\}$ of $[0, T]$.

Proposition 5.1.2. *Let M be a continuous local martingale of finite variation. Then $M_t = M_0$ for every $t \in [0, T]$.*

Proof. It suffices to apply Lemma 5.1.1 and to observe that the assumption that sample paths of M are continuous functions of finite variation implies that the quadratic variation of M vanishes. This follows from the observation that sample path of $\langle M \rangle$ can be obtained as the almost sure limit in (5.1) for some sequence of partitions. The proof is thus based on similar arguments as those already employed in the proof of Corollary 3.1.1. $\square$

The last result shows that continuous local martingales are never processes of finite variation. In particular,

Corollary 5.1.2. *If $M_0 = 0$ and M is a continuous local martingales of finite variation then M vanishes (more precisely, it is indistinguishable from the null process).*

We shall not discuss in detail integration against local martingales. We only mention that as in the special case of a Brownian motion, the stochastic integral of the form

$$\int_0^t \gamma_u dM_u$$

cannot be interpreted as the pathwise Lebesgue-Stieltjes integral. The theory of Itô integration with respect to continuous local martingales parallels the theory developed for the Brownian motion case. The only major modification in the one-dimensional case is that in the definition of spaces $\mathcal{L}^2(M)$ and $\mathcal{L}(M)$ the norm $\|\gamma\|_W^2$ should be replaced by its generalization:

$$\|\gamma\|_M^2 := \mathbb{E} \left(\int_0^T \gamma_u^2 d\langle M \rangle_u \right).$$

Remark 5.1.3. In practice, to check if the Itô integral of some progressively measurable process $\gamma = (\gamma_u)_{u \in [0, T]}$ is well defined, it is suffice to check that a.s

$$\int_0^T |\gamma_u|^2 d\langle M \rangle_u < \infty$$

or the sufficient condition, (which is easier to check)

$$\mathbb{E}[\int_0^T |\gamma_u|^2 d\langle M \rangle_u] < \infty$$

where M is a continuous martingale.

5.1.1 Continuous Semimartingales

The concept of a continuous semimartingale is a natural extension of the notion of a continuous local martingale.

Definition 5.1.4. A real-valued, continuous, $\mathbb{F}$ -adapted process X is called a (*real-valued*) *continuous semimartingale* if it admits a decomposition

$$X_t = X_0 + M_t + A_t, \quad \forall t \in [0, T], \quad (5.3)$$

where X_0 , M and A satisfy:

- (i) X_0 is an $\mathcal{F}_0$ -measurable random variable,
- (ii) M is a continuous local martingale with $M_0 = 0$,
- (iii) A is a continuous process whose almost all sample paths are of finite variation on the interval $[0, T]$ with $A_0 = 0$.

We denote by $\mathcal{S}^c(\mathbb{P})$ the class of all real-valued continuous semimartingales on the probability space $(\Omega, \mathbb{F}, \mathbb{P})$. It appears that the decomposition of X with the properties given in Definition 5.1.4 is unique, up to indistinguishability of stochastic processes, as the following result shows.

Proposition 5.1.3. *Let X be a continuous semimartingale with the decomposition $X = X_0 + M + A$ so that, in particular, $M_0 = A_0 = 0$. If X admits also a decomposition $X = X_0 + \widetilde{M} + \widetilde{A}$ for some continuous local martingale $\widetilde{M}$ with $\widetilde{M}_0 = 0$ and some continuous process $\widetilde{A}$ of finite variation on $[0, T]$ with $\widetilde{A}_0 = 0$ then $M_t = \widetilde{M}_t$ and $A_t = \widetilde{A}_t$ for $t \in [0, T]$.*

Proof. It is enough to observe that a difference of two continuous local martingales is also a continuous local martingales, and a difference of two continuous processes of finite variation also follows a continuous process of finite variation. In our case, we have $M - \widetilde{M} = \widetilde{A} - A$ and $M_0 - \widetilde{M}_0 = \widetilde{A}_0 - A_0 = 0$. Consequently, in view of Corollary 5.1.2, any continuous local martingales of finite variation starting at zero at time zero are zero, we conclude that the equalities $M_t = \widetilde{M}_t$ and $A_t = \widetilde{A}_t$ hold for every $t \in [0, T]$. $\square$

In view of the uniqueness of the decomposition $X = X_0 + M + A$ established in Proposition 5.1.3, it is justified to call it the *canonical decomposition* of a continuous semimartingale X under $\mathbb{P}$. Since the continuous local martingale M in the canonical decomposition of a continuous semimartingale X , it is customary to denote it by X^c .

Definition 5.1.5. The *quadratic variation* $\langle X \rangle$ of a continuous semimartingale X is then defined as the quadratic variation of the continuous martingale part of X given by $X^c = M$, that is, $\langle X \rangle = \langle X^c \rangle$, where

$$\langle X \rangle_t = \lim_{n \rightarrow \infty} \sum_{k=1}^{n(\pi)} (X_{t_k^n} - X_{t_{k-1}^n})^2 = \lim_{n \rightarrow \infty} \sum_{k=1}^{n(\pi)} (X_{t_k^n}^c - X_{t_{k-1}^n}^c)^2$$

where the convergence in probability holds for any sequence $(\pi_n)_{n \in \mathbb{N}}$ of finite partitions $\pi_n = \{0 = t_0^n < t_1^n < \dots < t_{m_n}^n = t\}$ of the interval $[0, t]$ satisfying $\lim_{n \rightarrow \infty} |\pi_n| = 0$.

Corollary 5.1.3. A continuous semimartingale is a continuous local martingale if and only if the process A in its canonical decomposition $X = X_0 + M + A$ vanishes, that is, is indistinguishable from the null process.

The extension of the notion of a continuous semimartingale to a multidimensional setting is rather obvious. Specifically, we say that an $\mathbb{R}^d$ -valued process $X = (X^1, \dots, X^d)$ is a *d-dimensional continuous semimartingale* if each component $X^1, \dots, X^d$ follows a real-valued continuous semimartingale. A *d-dimensional Brownian motion* $W = (W^1, \dots, W^d)$ is a simple example of a *d-dimensional continuous semimartingale*.

Proposition 5.1.4 (The Itô formula). *If $X = X_0 + M + A$ is a real-valued continuous semimartingale, and g is a function of class $C^{2,1}(\mathbb{R} \times [0, T], \mathbb{R})$ then the process $Y_t = g(X_t, t)$ follows a continuous semimartingale with the following canonical decomposition*

$$dY_t = g_t(X_t, t) dt + g_x(X_t, t) dX_t + \frac{1}{2} g_{xx}(X_t, t) d\langle M \rangle_t. \quad (5.4)$$

The *Itô integration by parts formula* is obtained by taking $g(x_1, x_2) = x_1 x_2$.

Proof. By localisation, we can suppose that the processes $X = M + A$, A and $\langle M \rangle$ are bounded. Given a discretized grid $\pi := 0 = t_0 < t_1, \dots, t_{n-1} < t_n = t$, consider the second order Taylor expansion of $g(t, X_t)$ given by

$$\begin{aligned} & g(t_{i+1}, X_{t_{i+1}}) - g(t_i, X_{t_i}) \\ &= \partial_t g(t, X_t) \Delta t_i + \partial_x g(t_i, X_{t_i}) \Delta X_{t_i} + \frac{1}{2} \partial_{xx} g(t_i, \eta_i) (\Delta X_{t_i})^2 \\ & \quad + \frac{1}{2} \partial_{xt} g(t_i, \eta_i) (\Delta X_{t_i}) (\Delta t_i) + \frac{1}{2} \partial_{tt} g(t_i, \eta_i) (\Delta t_i)^2 \end{aligned}$$

where by mean-value theorem there exists θ_i such that $\eta_i = X_{t_i} + \theta_i \Delta X_{t_i}$. By summing

over i and letting $|\pi| \rightarrow 0$, we have

$$\begin{aligned} \sum_{i=0}^{n-1} \frac{1}{2} \partial_{xt} g(t_i, \eta_i) (\Delta X_{t_i}) (\Delta t_i) &\longrightarrow 0 \\ \sum_{i=0}^{n-1} \frac{1}{2} \partial_{xt} g(t_i, \eta_i) (\Delta t_i)^2 &\longrightarrow 0, \end{aligned}$$

Here we have used the fact that continuous function on compact interval is bounded. For example

$$\left| \sum_{i=0}^{n-1} \frac{1}{2} \partial_{xt} g(t_i, \eta_i) (\Delta t_i)^2 \right| \leq Ct \|\partial_{xt} g\|_{\infty} |\pi| \rightarrow 0$$

We now study the other terms given by

$$\begin{aligned} &\sum_{i=0}^{n-1} \partial_t g(t_i, X_{t_i}) \Delta t_i + \sum_{i=0}^{n-1} \partial_t g(t_i, X_{t_i}) \Delta M_{t_i} + \sum_{i=0}^{n-1} \partial_t g(t_i, X_{t_i}) \Delta A_{t_i} + \sum_{i=0}^{n-1} \frac{1}{2} \partial_{xx} g(t_i, \eta_i) (\Delta X_{t_i})^2 \\ &= I_1 + I_2 + I_3 + I_4 \end{aligned}$$

We study each term separately. By the dominated convergence theorem the terms I_1 and I_3 converges almost surely and

$$\begin{aligned} \sum_{i=0}^{n-1} \partial_t g(t_i, X_{t_i}) \Delta t_i &\longrightarrow \int_0^t \partial_s g(s, X_s) ds \\ \sum_{i=0}^{n-1} \partial_t g(t_i, X_{t_i}) \Delta A_{t_i} &\longrightarrow \int_0^t \partial_x g(s, X_s) dA_s \end{aligned}$$

The martingale term I_2 converges in L^2 to an stochastic integral against a martingale. More specifically, from the Itô isometry

$$\mathbb{E} \left[\left| \int_0^t (g_n(s, X_s) - g(s, X_s)) dM_s \right|^2 \right] = \mathbb{E} \left[\int_0^t |g_n(s, X_s) - g(s, X_s)|^2 d\langle M \rangle_s \right]$$

and the right hand side above converges to zero again by dominated convergence theorem as $|\pi| \rightarrow 0$, since $g_n \rightarrow g$.

For the quadratic term I_4 , we have as $|\pi| \rightarrow 0$,

$$\sum_{i=0}^{n-1} \frac{1}{2} \partial_{xx} g(t_i, \eta_i) (\Delta X_{t_i})^2 \longrightarrow \frac{1}{2} \int_0^t \partial_{xx} g(s, X_s) d\langle X \rangle_s$$

We first show that show that

$$\left| \sum_{i=0}^{n-1} \frac{1}{2} \partial_{xx} g(t_i, \eta_i) (\Delta X_{t_i})^2 - \sum_{i=0}^{n-1} \frac{1}{2} \partial_{xx} g(t_i, X_{t_i}) (\Delta X_{t_i})^2 \right| \longrightarrow 0$$

One can write

$$\begin{aligned} & \sum_{i=0}^{n-1} \left| \frac{1}{2} \partial_{xx} g(t_i, \eta_i) (\Delta X_{t_i})^2 - \frac{1}{2} \partial_{xx} g(t_i, X_{t_i}) (\Delta X_{t_i})^2 \right| \\ & \leq C \max_i |\partial_{xx} g(t_i, \eta_i) - \partial_{xx} g(t_i, X_{t_i})| \sum_{i=1} (\Delta X_{t_i})^2 \end{aligned}$$

From the Cauchy-Schwartz inequality and the fact that $\sqrt{\mathbb{E}[|V_n^{(2)}(X)|^2]} < K$ for some $K > 0$, where $V_n^{(2)}(X) := \sum_{i=1} (\Delta X_{t_i})^2$. We then deduce that

$$\begin{aligned} & \mathbb{E} \left[\sum_{i=0}^{n-1} \left| \frac{1}{2} \partial_{xx} g(t_i, \eta_i) (\Delta X_{t_i})^2 - \frac{1}{2} \partial_{xx} g(t_i, X_{t_i}) (\Delta X_{t_i})^2 \right| \right] \\ & \leq C \sqrt{\mathbb{E}[|V_n^{(2)}(X)|^2]} \sqrt{\mathbb{E}[\max_i |\partial_{xx} g(t_i, \eta_i) - \partial_{xx} g(t_i, X_{t_i})|^2]} \longrightarrow 0 \end{aligned}$$

Once we show that (the hard part which we skip)

$$\sum_{i=0}^{n-1} \frac{1}{2} \partial_{xx} g(t_i, X_{t_i}) (\Delta X_{t_i})^2 \rightarrow \sum_{i=0}^{n-1} \frac{1}{2} \partial_{xx} g(t_i, X_{t_i}) \Delta \langle X \rangle_{t_i}$$

then by dominated convergence we have

$$\sum_{i=0}^{n-1} \frac{1}{2} \partial_{xx} g(t_i, X_{t_i}) \Delta \langle X \rangle_{t_i} \longrightarrow \int_0^t \frac{1}{2} \partial_{xx} g(s, X_s) d\langle X \rangle_s$$

Finally, by letting the localising sequence go to infinity, we obtain the Itô formula on the whole time horizon. $\square$

One extension of the Itô formula is to continuous convex functions. In particular, we have the Tanaka-Meyer formula

Proposition 5.1.5 (The Tanaka-Meyer formula). *Let X be a continuous semimartingale, $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous convex function with left derivative f' . Then $f(X)$ is a semimartingale and*

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) dX_s + C_t$$

where C_t is an adapted continuous increasing process with zero initial value.

Corollary 5.1.4. *Let X^1 and X^2 belong to $\mathcal{S}^c(\mathbb{P})$. Then*

$$X_t^1 X_t^2 = X_0^1 X_0^2 + \int_0^t X_u^1 dX_u^2 + \int_0^t X_u^2 dX_u^1 + \langle X^1, X^2 \rangle_t. \quad (5.5)$$

In particular, for any $X \in \mathcal{S}^c(\mathbb{P})$

$$X_t^2 = X_0^2 + 2 \int_0^t X_u dX_u + \langle X \rangle_t. \quad (5.6)$$

Corollary 5.1.5 (The Tanaka Formula). *The process $|X_t|$ is a continuous semimartingale and*

$$|X_t| = \int_0^t \text{sign}(X_s) dX_s + L_t^0(X).$$

where $\text{sign}(x) = -\mathbb{1}_{\{x \leq 0\}} + \mathbb{1}_{\{x > 0\}}$ and the increasing process

$$L^0(X) = \lim_{\epsilon \downarrow 0} \int_0^t \mathbb{1}_{[0, \epsilon)}(X_s) d\langle X \rangle_s$$

is the local time of X at zero.

Exercise: Show that $X_t = |B_t|^\alpha$ for $\alpha \in (0, 1/2)$ is not a semimartingale.

Hint: Use the fact that $L_s(B) = \int_0^t \mathbb{1}_{\{B_s=0\}} dL_s(B)$ and $\{X_t = 0\} = \{B_t = 0\}$.

5.2 Lévy Characterisation

In some instances, it would be convenient to have the possibility of checking whether a given process is a standard Brownian motion by establishing its martingale property and by computing its quadratic variation. We may use the martingale characterization of a Brownian motion, due to Paul Lévy.

5.2.1 One-Dimensional Case

Recall that if W is a one-dimensional Brownian motion on a probability space $(\Omega, \mathbb{F}, \mathbb{P})$ then W is a continuous $\mathbb{F}$ -martingale with $W_0 = 0$ and the process $W_t^2 - t$ is also a continuous $\mathbb{F}$ -martingale.

Theorem 5.2.1. *Let M be a continuous local martingale on a probability space $(\Omega, \mathbb{F}, \mathbb{P})$ such that $M_0 = 0$ and the process $M_t^2 - t$ is a continuous local martingale. Then M is a standard Brownian motion with respect to $\mathbb{F}$.*

The assumption that $M_t^2 - t$ is a continuous local martingale means that the quadratic variation of M satisfies $\langle M \rangle_t = t$ for every $t \in [0, T]$. Under the stronger assumption that M is a square-integrable continuous martingale, this property can be represented as follows

$$\mathbb{E}(M_t^2 - M_u^2 | \mathcal{F}_t) = t - u, \quad u \leq t \leq T, \quad (5.7)$$

or equivalently (in view of the martingale property of M)

$$\mathbb{E}((M_t - M_u)^2 | \mathcal{F}_u) = t - u, \quad u \leq t \leq T.$$

Remark 5.2.1. It should be stressed that the assumption of continuity of sample paths of M is essential in Theorem 5.2.1 and thus it cannot be dropped. Indeed, if we take as M the compensated Poisson process, that is, the process $M_t = N_t - t$, where N is the Poisson process with unit intensity, then it can be easily verified that the process $M_t^2 - t$ is a martingale as well.

Lemma 5.2.1. *A real-valued continuous $\mathbb{F}$ -adapted process X defined on $(\Omega, \mathbb{F}, \mathbb{P})$ is a standard Brownian motion with respect to $\mathbb{F}$ if and only if for any $\lambda \in \mathbb{R}$ the process*

$$M_t^\lambda = \exp\left(\lambda X_t - \frac{1}{2}\lambda^2 t\right), \quad \forall t \in [0, T], \quad (5.8)$$

is a (local) martingale with respect to $\mathbb{F}$ and $M_0^\lambda = 1$.

The process M^λ is a special case of the stochastic exponential examined in some detail later. It is easily seen that when W is a one-dimensional Brownian motion with respect to $\mathbb{F}$ then we have

$$\mathbb{E}_{\mathbb{P}}\left(e^{\lambda(W_t - W_u)} \middle| \mathcal{F}_u\right) = e^{\frac{1}{2}\lambda^2(t-u)}, \quad u \leq t \leq T.$$

This implies that the process M^λ associated with W is an $\mathbb{F}$ -martingale.

Proof of Theorem 1.1 above. *If W is a Brownian motion then its quadratic variation is given by t . Assume now that the process M satisfies the assumptions of Theorem 1.1 and we show that M is a Brownian motion.*

We apply the Itô formula theorem and let, for every $a \in \mathbb{R}$,

$$F(a, x) = e^{i\lambda x}$$

be a function defined on $\mathbb{R}$. Then

$$\frac{\partial F}{\partial x}(\lambda, x) = i\lambda F(\lambda, x), \quad \frac{\partial^2 F}{\partial x^2}(\lambda, x) = -\lambda^2 F(\lambda, x).$$

By the Ito formula

$$F(\lambda, M_t) = e^{i\lambda M_s} + i\lambda \int_s^t e^{i\lambda M_u} dM_u - \frac{1}{2}\lambda^2 \int_s^t e^{i\lambda M_u} du.$$

Hence

$$e^{i\lambda(M_t - M_s)} = 1 + i\lambda \int_s^t e^{ia(M_u - M_s)} dM_u - \frac{1}{2}\lambda^2 \int_s^t e^{i\lambda(M_u - M_s)} du. \quad (5.9)$$

By the properties of stochastic integrals

$$\mathbb{E}\left(\int_s^t e^{i\lambda M_u} dM_u \middle| \mathcal{F}_s\right) = 0.$$

Let $A \in \mathcal{F}_s$. Then

$$\mathbb{E}\left(e^{i\lambda(M_t - M_s)} \mathbf{1}_A\right) = \mathbb{E}[\mathbf{1}_A] - \frac{1}{2}|\lambda|^2 \int_s^t \mathbb{E}\left(e^{i\lambda(M_u - M_s)} \mathbf{1}_A\right) du.$$

If we denote

$$f(t) = \mathbb{E}\left(e^{i\lambda(M_t - M_s)} \mathbf{1}_A\right)$$

then we obtain an equation

$$f(t) = \mathbb{P}(A) - \frac{|\lambda|^2}{2} \int_s^t f(u) du$$

or

$$f'(t) = -\frac{|\lambda|^2}{2} f(t)$$

with the initial condition $f(s) = \mathbb{P}(A)$. It is easy to show that the unique solution to this differential equation is given by the formula

$$f(t) = \mathbb{P}(A) \exp \left(-\frac{|\lambda|^2}{2} (t - s) \right).$$

Finally, taking into account the definition of conditional expectation we proved that

$$\mathbb{E} \left(e^{i\lambda(M_t - M_s)} \mathbb{1}_A \right) = \mathbb{E} \left(\mathbb{1}_A \mathbb{E} \left(e^{i\lambda(M_t - M_s)} \middle| \mathcal{F}_s \right) \right) = \mathbb{E} \left(\mathbb{1}_A \exp \left(-\frac{|\lambda|^2}{2} (t - s) \right) \right)$$

and therefore

$$\mathbb{E} \left(e^{i\lambda(M_t - M_s)} \middle| \mathcal{F}_s \right) = e^{-|\lambda|^2(t-s)/2}$$

and therefore the random variable $(M_t - M_s)$ is independent of $\mathcal{F}_s$ and has the normal $N(0, (t - s))$ distribution.

Theorem 5.2.2. *If $(M_t)_{t \in [0, \infty)}$ is a continuous local martingale such that $\langle M \rangle$ is strictly increasing and $\langle M \rangle_\infty = \infty$, then M is a time changed Brownian motion.*

Proof. The process $\langle M \rangle$ is continuous and strictly increasing therefore the inverse of $\langle M \rangle$ is given by $C_t := \inf \{s : \langle M \rangle_s > t\}$.

We consider the process $B_t := M_{C_t}$ for $t \geq 0$, which is adapted to the filtration $\mathbb{G} = (\mathcal{F}_{C_t})_{t \geq 0}$ (this is not obvious, but we take this for granted). To see that the process $(B_t)_{t \in [0, \infty)}$ is again a $\mathbb{G}$ local martingale, without loss of generality, we suppose that M is bounded and hence a martingale, then from the Doob Optional Sampling Theorem and the fact that for all t , C_t is $\mathbb{F}$ -stopping time we obtain

$$\mathbb{E}[B_t | \mathcal{G}_s] = \mathbb{E}[M_{C_t} | \mathcal{F}_{C_s}] = M_{C_s} = B_s.$$

We note that since C is the (right) inverse of M then $\langle M \rangle_{C_t} = t$ and the process

$$B_t^2 - \langle B \rangle_t = M_{C_t}^2 - \langle M \rangle_{C_t}$$

is a $\mathbb{G}$ local martingale. Therefore by Lévy characterisation theorem we see that B is a Brownian motion in the filtration $\mathbb{G}$.

Finally, since $\langle M \rangle$ is strictly increasing, C is the (left) inverse of $\langle M \rangle$, that is $C_{\langle M \rangle_t} = t$. This shows that

$$M_t = B_{\langle M \rangle_t}$$

where B is a Brownian motion in the filtration $\mathbb{G}$. □

Remark 5.2.2. In general, we do not need to assume $\langle M \rangle$ is strictly increasing and $\langle M \rangle_\infty = \infty$. These assumptions simplify the proof. In general, see the Dubins and Schwarz Theorem.

5.2.2 Multidimensional Case

Theorem 5.2.1 can be extended to a multidimensional setting. Let $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ otherwise.

Theorem 5.2.3. *Let $M = (M^1, \dots, M^d)$ be a continuous $\mathbb{F}$ -adapted process on a filtered probability space $(\Omega, \mathbb{F}, \mathbb{P})$ with the initial value $M_0 = (0, \dots, 0)$. Then M is a d -dimensional standard Brownian motion if and only if the following conditions are satisfied:*

- (i) *for any $i = 1, \dots, d$ the process M^i is an $\mathbb{F}$ -martingale,*
- (ii) *for any $i, j = 1, \dots, d$ the process $M_t^i M_t^j - \delta_{ij}t$ is an $\mathbb{F}$ -martingale.*

Theorem 5.2.4. *Assume that $M = (M^1, \dots, M^d)$ is an $\mathbb{F}$ -adapted process on a probability space $(\Omega, \mathbb{F}, \mathbb{P})$ such that $M_0 = (0, \dots, 0)$ and the components $M^1, \dots, M^d$ are continuous local martingales with $\langle M^i, M^j \rangle_t = \delta_{ij}t$. Then M is a d -dimensional standard Brownian motion.*

5.3 Martingale Representation Property

In this section we shall assume that the filtration $\mathbb{F} = \mathbb{F}^W$ is the standard augmentation of the natural filtration $\sigma\{W_u \mid u \leq t\}$ of a d -dimensional Brownian motion W . In other words, we assume here that the underlying probability space is $(\Omega, \mathbb{F}^W, \mathbb{P})$. We will now state two closely related versions of the so-called predictable representation property of the Brownian filtration.

The first theorem is known as the Itô representation theorem. It states that any square-integrable and $\mathcal{F}_T^W$ -measurable random variables admits a representation as the Itô integral of some stochastic process.

Theorem 5.3.1. *For any random variable $X \in L^2(\Omega, \mathcal{F}_T^W, \mathbb{P})$, there exists a unique $\mathbb{F}$ -predictable process γ from the class $\mathcal{L}_{\mathbb{P}}^2(W)$ such that the following equality is valid*

$$X = \mathbb{E}X + \int_0^T \gamma_u \cdot dW_u. \quad (5.10)$$

The condition that γ belongs to $\mathcal{L}_{\mathbb{P}}^2(W)$ implies that the Itô integral $I(\gamma)$ is a square-integrable $\mathbb{F}$ -martingale and thus we also have that, for every $t \in [0, T]$,

$$\mathbb{E}_{\mathbb{P}}(X \mid \mathcal{F}_t^W) = \mathbb{E}_{\mathbb{P}}X + \int_0^t \gamma_u \cdot dW_u. \quad (5.11)$$

Remark 5.3.1. In Theorem 5.3.1 we assert that γ is an $\mathbb{F}$ -predictable process (for the definition see, for instance, Chapter 5 in Revuz and Yor (1999)). Let us only note that any left-continuous and $\mathbb{F}$ -adapted process is $\mathbb{F}$ -predictable. Also, if a process is $\mathbb{F}$ -predictable then it is $\mathbb{F}$ -progressively measurable, but the converse does not hold, in general.

The second result is commonly known as the martingale representation theorem for the Brownian filtration. Note that it is a rather straightforward consequence of Theorem 5.3.1 (see, in particular, formula (5.11)).

Theorem 5.3.2. *Let a process $M = (M_t)_{t \in [0, T]}$ be an $\mathbb{F}^W$ -martingale such that $\mathbb{E}_{\mathbb{P}} M_T^2 < \infty$. Then there exists a unique $\mathbb{F}$ -predictable process γ from the class $\mathcal{L}_{\mathbb{P}}^2(W)$ such that, for any $t \in [0, T]$,*

$$M_t = M_0 + \int_0^t \gamma_u \cdot dW_u. \quad (5.12)$$

Theorem 5.3.2 can be extended to local martingales with respect to a Brownian filtration. As a consequence, we obtain the following notable result.

Corollary 5.3.1. *Any local martingale with respect to the Brownian filtration $\mathbb{F}^W$ is necessarily a continuous process.*

Example 5.3.1. Let $d = 1$ and let us first take $X = W_T^2$. Using the Itô formula with $g(x, t) = x^2$ and $X = W$, we obtain

$$W_t^2 = \int_0^t 2W_u dW_u + t, \quad \forall t \in [0, T].$$

Since $\mathbb{E}_{\mathbb{P}}[W_T^2] = T$, the equality (5.10) becomes

$$W_T^2 = \mathbb{E}_{\mathbb{P}}[W_T^2] + \int_0^T 2W_u dW_u,$$

so that the process $\gamma_t = 2W_t$, $t \in [0, T]$, is independent of T .

Remark 5.3.2. In general, it is very difficult to find the process γ . Practically speaking, the existence of γ is in some sense means that the market is complete and for any European claim, one can always find a replicating portfolio.

Chapter 6

Stochastic Differential Equations

The theory of stochastic differential equations is covered by numerous textbooks and monographs. In this short section, we mainly deal with the special case of linear stochastic differential equations. Let $\mu : \mathbb{R} \times [0, T] \rightarrow \mathbb{R}$ and $\sigma : \mathbb{R} \times [0, T] \rightarrow \mathbb{R}$ be some given functions and let W be a d -dimensional standard Brownian motion defined on a filtered probability space $(\Omega, \mathbb{F}, \mathbb{P})$.

Intuitively, one can think of stochastic differential equations (SDEs) as ordinary differential equations (ODEs) plus random noise.

Definition 6.0.1. By a *solution* of the stochastic differential equation (SDE, for short)

$$dX_t = \mu(X_t, t) dt + \sigma(X_t, t) \cdot dW_t \quad (6.1)$$

with the initial condition X_0 given as an $\mathcal{F}_0$ -measurable random variable, we mean an $\mathbb{R}^k$ -valued, $\mathbb{F}$ -adapted stochastic process X defined on the probability space $(\Omega, \mathbb{F}, \mathbb{P})$ and such that, for every $t \in [0, T]$,

$$X_t = X_0 + \int_0^t \mu(X_u, u) du + \int_0^t \sigma(X_u, u) \cdot dW_u. \quad (6.2)$$

It is clear from the definition above that a stochastic differential equation, usually represented in its differential form (6.1), is nothing else than the integral equation (6.2), in which the second integral is defined as the Itô stochastic integral. Definition 6.0.1 implicitly assumes that both integrals in the right-hand side of (6.2) are well defined, that is,

$$\mathbb{P}\left(\int_0^T |\mu(X_u, u)| du < \infty\right) = 1$$

and the process γ given as $\gamma_t = \sigma(X_t, t)$, $t \in [0, T]$, belongs to the class $\mathcal{L}_{\mathbb{P}}(W)$. Note that any solution X to the SDE (6.1) is an Itô process.

The concept of uniqueness of solutions to an SDE can be defined in several alternative ways. For our purposes, the most convenient form is the so-called pathwise uniqueness of solutions. If X is a solution to (6.1), we shall informally say that X is driven by W .

Definition 6.0.2. We say that the *pathwise uniqueness* of solutions to (6.1) holds if for any filtered probability space $(\Omega, \mathbb{F}, \mathbb{P})$, any d -dimensional standard Brownian motions W and $\widetilde{W}$ defined on $(\Omega, \mathbb{F}, \mathbb{P})$, and any two solutions X and $\widetilde{X}$ driven by W and $\widetilde{W}$ respectively, the following implication is true

$$\mathbb{P}\{W_t = \widetilde{W}_t \mid \forall t \in [0, T]\} = 1 \implies \mathbb{P}\{X_t = \widetilde{X}_t \mid \forall t \in [0, T]\} = 1.$$

The *pathwise uniqueness of solutions* (6.1) is sometimes referred to as the *strong uniqueness*. This is due to the fact that under pathwise uniqueness any solution to (6.1) is strong, meaning that it is adapted to the filtration generated by the driving Brownian motion W .

6.1 Linear Stochastic Differential Equation

Let us first consider a time-homogeneous linear SDE of the form

$$dX_t = (aX_t + c) dt + (bX_t + f) dW_t, \quad (6.3)$$

where W is a one-dimensional Brownian motion and a, b, c and f are real numbers. It is clear that $\mu(x, t) = ax + c$ and $\sigma(x, t) = bx + f$ so that μ and σ are linear functions.

By setting $c = f = 0$, we obtain the SDE governing the stock price process in the Black-Scholes model of security market. In the present notation, it becomes

$$dX_t = aX_t dt + bX_t dW_t = X_t(a dt + b dW_t). \quad (6.4)$$

Proposition 6.1.1. The unique solution of the SDE (6.4) is given by the expression

$$X_t = X_0 \exp \left(bW_t + \left(a - \frac{1}{2}b^2 \right) t \right). \quad (6.5)$$

Proof. Exercise. □

The following result is useful in the case of Vasicek's model of the short-term interest rates.

Proposition 6.1.2. The unique solution of the SDE

$$dX_t = (aX_t + c) dt + f dW_t \quad (6.6)$$

is given by the formula

$$X_t = X_0 e^{at} + \int_0^t c e^{a(t-u)} du + \int_0^t f e^{a(t-u)} dW_u. \quad (6.7)$$

Proof. Exercise. □

The definition of an SDE can also be extended to the case where the coefficients μ and σ explicitly depend on the time parameter t and ω . For instance, we may assume that the parameters (6.3) follow stochastic processes, so that it becomes

$$dX_t = (a_t X_t + c_t) dt + (b_t X_t + f_t) \cdot dW_t. \quad (6.8)$$

where W is a d -dimensional Brownian motion and the $\mathbb{F}$ -adapted stochastic processes a, c are real-valued, whereas the processes b, f are $\mathbb{R}^d$ -valued. Then we have the following extension of Proposition 6.1.1.

Proposition 6.1.3. (The stochastic exponential) *Let a and b be $\mathbb{F}$ -adapted bounded processes. Then the unique solution of the SDE*

$$dX_t = X_t(a_t dt + b_t \cdot dW_t) \quad (6.9)$$

is given by the following generalization of formula (6.5)

$$X_t = X_0 \exp \left(\int_0^t b_u \cdot dW_u + \int_0^t \left(a_u - \frac{1}{2} |b_u|^2 \right) du \right). \quad (6.10)$$

Proof. To check that the process X given by (6.10) is a solution to (6.9), it suffices to differentiate the right-hand side in (6.10) using the Itô formula. The uniqueness of solutions to (6.9) can be proved directly or deduced from Theorem 6.2.1. $\square$

The following result extends Proposition 6.1.2.

Proposition 6.1.4. *Let a, c and f be $\mathbb{F}$ -adapted bounded processes. Then the unique solution of the linear SDE*

$$dX_t = (a_t X_t + c_t) dt + f_t \cdot dW_t \quad (6.11)$$

is given by the formula

$$X_t = \Phi_t \left(X_0 + \int_0^t \Phi_u^{-1} c_u du + \int_0^t \Phi_u^{-1} f_u \cdot dW_u \right) \quad (6.12)$$

where

$$\Phi_u = \exp \left(\int_0^u a_s ds \right).$$

Proof. To find the form of the solution X , it is suffice to first suppose X satisfies the SDE and then apply Itô's formula (or the integration by parts formula) to $e^{-\int_0^t a_s ds} X_t$ to see that

$$\begin{aligned} X_t e^{-\int_0^t a_s ds} &= X_0 + \int_0^t e^{-\int_0^u a_s ds} dX_u - \int_0^t X_u e^{-\int_0^u a_s ds} a_u du \\ &= X_0 + \int_0^t e^{-\int_0^u a_s ds} ((a_u X_u + c_u) du + f_u \cdot dW_u) - \int_0^t X_u e^{-\int_0^u a_s ds} a_u du \\ &= X_0 + \int_0^t e^{-\int_0^u a_s ds} (c_u du + f_u \cdot dW_u) \end{aligned}$$

The form of the solution X is then obtained by dividing both hand side above by $e^{-\int_0^t a_s ds}$. $\square$

Exercise: Solve the linear SDE

$$dX_t = (aX_t + c)dt + (bX_t + f)dW_t$$

where a , c , b and f are constants.

Exercise: Solve the SDE (square root process or CIR process)

$$dX_t = \left(\frac{1}{4} - bX_t\right)dt + \sqrt{X_t}dW_t, \quad X_0 = 1$$

by considering the $Y_t = \sqrt{X_t}$.

6.2 Itô's Existence and Uniqueness Theorem

We end this section by stating the classic Itô's theorem, which provides sufficient conditions for the existence and uniqueness of solutions to the SDE (6.1).

Theorem 6.2.1. Let $\mu : \mathbb{R} \times [0, T] \rightarrow \mathbb{R}$ and $\sigma : \mathbb{R} \times [0, T] \rightarrow \mathbb{R}$ satisfy the following conditions:

(i) the functions μ and σ are Lipschitz continuous with respect to the variable x , that is, there exists a constant $K_1 > 0$ such that, for any $x, y \in \mathbb{R}$ and $t \in \mathbb{R}_+$,

$$|\mu(t, x) - \mu(t, y)|^2 + |\sigma(t, x) - \sigma(t, y)|^2 \leq K_1|x - y|^2, \quad (6.13)$$

(ii) the functions μ and σ satisfy the linear growth condition: there exists a constant K_2 such that, for any $x \in \mathbb{R}$ and $t \in \mathbb{R}_+$,

$$|\mu(t, x)|^2 + |\sigma(t, x)|^2 \leq K_2(1 + |x|^2). \quad (6.14)$$

Then the SDE has the unique solution X .

Proof. The proof of existence relies on Picard iteration (applying the contraction mapping theorem). We shown only uniqueness below.

Suppose X and Y are two solutions to the SDE and we have appropriately localised X and Y so that $|X|$ and $|Y|$ are bounded. Then by Itô's formula, we have

$$\begin{aligned} (X_t - Y_t)^2 &= \int_0^t 2(X_s - Y_s)d(X_s - Y_s) + \langle X - Y \rangle_t \\ &= \int_0^t 2(X_s - Y_s)[\mu(s, X_s) - \mu(s, Y_s)]ds + 2(X_s - Y_s)[\sigma(s, X_s) - \sigma(s, Y_s)]dW_s \\ &\quad + \frac{1}{2}[\sigma(s, X_s) - \sigma(s, Y_s)]^2 ds \end{aligned}$$

By taking the expected value, then the absolute value and by using (8.3), we obtain for some $C > 0$

$$\mathbb{E}[(X_t - Y_t)^2] = C(1 + K_1) \int_0^t \mathbb{E}[|X_s - Y_s|^2]ds$$

and by Gronwell's lemma (inequality), we obtain $\mathbb{E}[(X_t - Y_t)^2] \leq 0$, which implies for $t \geq 0$, $X_t = Y_t$ a.s. Hence, by continuity of X and Y , the two processes X and Y are indistinguishable. Finally to remove dependence on the localisation sequence, we apply the Fatou Lemma. $\square$

Remark 6.2.1. The square root process or CIR process is a simple example where conditions (i) in Theorem 6.2.1 is not satisfied, but there is still a unique strong solution to the SDE.

For notational simplicity, in Theorem 6.2.1 the symbol $|\cdot|$ is used to denote the Euclidean norms in both $\mathbb{R}$ and $\mathbb{R}^d$, but this slight abuse of notation should not be confusing.

Let us finally mention that an analogous result is valid in the multidimensional case, that is, when the k -dimensional process $X = (X^1, \dots, X^k)$ is given as

$$dX_t = \mu(X_t, t) dt + \sigma(X_t, t) dW_t, \quad X_0 = x, \quad (6.15)$$

with the coefficients $\mu : \mathbb{R}^k \times [0, T] \rightarrow \mathbb{R}^k$ and $\sigma : \mathbb{R}^k \times [0, T] \rightarrow \mathbb{R}^{k \times d}$ satisfying the natural extensions of Itô's conditions (8.3) and (8.4).

6.2.1 Stochastic Exponential

Let W be a d -dimensional standard Brownian motion defined on a filtered probability space $(\Omega, \mathbb{F}, \mathbb{P})$. For an $\mathbb{R}^d$ -valued process γ belonging to $\mathcal{L}(W)$, we define the real-valued $\mathbb{F}$ -adapted process U by setting

$$U_t = I_t(\gamma) = \int_0^t \gamma_u \cdot dW_u, \quad t \in [0, T]. \quad (6.1)$$

The process U defined in this way is, of course, a continuous local martingale under $\mathbb{P}$ with respect to $\mathbb{F}$.

Definition 6.2.1. The *stochastic exponential* (also known as the *Doléans exponential*) of U is given by the formula, for $t \in [0, T]$,

$$\mathcal{E}_t(U) = \mathcal{E}_t\left(\int_0^t \gamma_u \cdot dW_u\right) = \exp\left(\int_0^t \gamma_u \cdot dW_u - \frac{1}{2} \int_0^t |\gamma_u|^2 du\right),$$

that is, $\mathcal{E}_t(U) = \exp(U_t - \langle U \rangle_t/2)$. More generally, for any continuous local martingale M we set $\mathcal{E}_t(M) = \exp(M_t - \langle M \rangle_t/2)$ for $t \in [0, T]$.

Lemma 6.2.1. *The stochastic exponential of U is the unique solution Z of the stochastic differential equation*

$$dZ_t = Z_t \gamma_t \cdot dW_t = Z_t dU_t \quad (6.2)$$

with the initial condition $Z_0 = 1$.

Remark 6.2.2. (i) It follows immediately from Lemma 6.2.1 that

$$d\mathcal{E}_t(U) = \mathcal{E}_t(U) \gamma_t \cdot dW_t = \mathcal{E}_t(U) dU_t, \quad (6.3)$$

Note that $\mathcal{E}(U)$ is a strictly positive continuous local martingale under $\mathbb{P}$ and thus, it follows a supermartingale with respect to $\mathbb{F}$. If, in addition,

$$\mathbb{E}_{\mathbb{P}}(\mathcal{E}_T(U)) = \mathbb{E}_{\mathbb{P}}(\mathcal{E}_0(U)) = 1,$$

so that the expected value $\mathbb{E}_{\mathbb{P}}\mathcal{E}_t(U)$, $t \in [0, T]$ is constant, then the process $\mathcal{E}(U)$ is a strictly positive continuous $\mathbb{F}$ -martingale.

(ii) For any continuous local martingale M , the stochastic exponential $\mathcal{E}(M)$ is the unique solution Z of the stochastic differential equation

$$dZ_t = Z_t dM_t \quad (6.4)$$

with the initial condition $Z_0 = 1$.

Chapter 7

Radon-Nikodým Density and Girsanov Theorem

Two probability measures $\tilde{\mathbb{P}}$ and $\mathbb{P}$ on $(\Omega, \mathcal{F}_T)$ are said to be equivalent if, for any event $A \in \mathcal{F}_T$, the equality $\mathbb{P}(A) = 0$ holds if and only if $\tilde{\mathbb{P}}(A) = 0$. In other words, $\tilde{\mathbb{P}}$ and $\mathbb{P}$ are equivalent on $(\Omega, \mathcal{F}_T)$ if they have the same set of null events in the σ -field $\mathcal{F}_T$. It is easily seen that if $\tilde{\mathbb{P}}$ and $\mathbb{P}$ are equivalent on $\mathcal{F}_T$ then they also enjoy this property on any σ -field $\mathcal{G} \subseteq \mathcal{F}_T$; in particular, $\tilde{\mathbb{P}}$ and $\mathbb{P}$ are equivalent on the σ -field $\mathcal{F}_t$, for any $t \in [0, T]$.

Definition 7.0.1. The Radon-Nikodým density of $\tilde{\mathbb{P}}$ with respect to $\mathbb{P}$ is defined as the unique $\mathcal{F}_T$ -measurable random variable η_T such that we have, for any event $A \in \mathcal{F}_T$,

$$\tilde{\mathbb{P}}(A) = \int_A \eta_T d\mathbb{P} = \mathbb{E}_{\mathbb{P}}(\eta_T \mathbf{1}_A). \quad (7.1)$$

The existence and uniqueness of the random variable η_T satisfying (7.1) is a consequence of the classic Radon-Nikodým theorem. Formula (7.1) implies that for any $\tilde{\mathbb{P}}$ -integrable random variable ψ we have that $\mathbb{E}_{\tilde{\mathbb{P}}} \psi = \mathbb{E}_{\mathbb{P}}(\psi \eta_T)$. Note also that ψ is $\tilde{\mathbb{P}}$ -integrable if and only if $\psi \eta_T$ is $\mathbb{P}$ -integrable. Finally, it is easy to check that $\mathbb{P}\{\eta_T > 0\} = 1$ and $\mathbb{E}_{\mathbb{P}} \eta_T = \tilde{\mathbb{P}}(\Omega) = 1$. Conversely, assume that η_T is any $\mathcal{F}_T$ -measurable random variable such that the last two conditions hold. Then formula (7.1) defines a probability measure $\tilde{\mathbb{P}}$ on $(\Omega, \mathcal{F}_T)$, which is then equivalent to $\mathbb{P}$. To emphasize the role of η_T as the link between the expectations with respect to $\tilde{\mathbb{P}}$ and $\mathbb{P}$, it is customary to use the short-hand notation

$$\eta_T = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}.$$

Definition 7.0.2. The Radon-Nikodým density process $\eta = (\eta_t)_{t \in [0, T]}$ of $\tilde{\mathbb{P}}$ with respect to $\mathbb{P}$ and a given filtration $\mathbb{F}$ is defined by setting, for $t \in [0, T]$,

$$\eta_t = \mathbb{E}_{\mathbb{P}}(\eta_T | \mathcal{F}_t) = \mathbb{E}_{\mathbb{P}}\left(\frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} \mid \mathcal{F}_t\right), \forall t \in [0, T] \quad (7.2)$$

It is clear that Radon-Nikodým density process η is a strictly positive martingale under $\mathbb{P}$. It is also uniformly integrable since clearly $\eta_t = \mathbb{E}_{\mathbb{P}}(\eta_T | \mathcal{F}_t)$. Finally, the random variable η_t is the Radon-Nikodým density of $\tilde{\mathbb{P}}$ with respect to $\mathbb{P}$ on $(\Omega, \mathcal{F}_t)$. The latter property is denoted as

$$\eta_t = \frac{d\tilde{\mathbb{P}}|_{\mathcal{F}_t}}{d\mathbb{P}|_{\mathcal{F}_t}} = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}|_{\mathcal{F}_t}. \quad (7.3)$$

7.1 Abstract Bayes Formula

Suppose that the Radon-Nikodým density of $\mathbb{Q}$ with respect to $\mathbb{P}$ equals

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \eta, \quad \mathbb{P}a.s$$

Note that the random variable η is strictly positive $\mathbb{P}$ -a.s., moreover η is $\mathbb{P}$ -integrable with $\mathbb{E}[\eta] = \mathbb{Q}(\Omega) = 1$. Finally, we have equality $\mathbb{E}_{\mathbb{Q}}[\psi] = \mathbb{E}[\psi\eta]$ holds for any $\mathbb{Q}$ -integrable random variable ψ . We are in the position prove the so-called abstract Bayes formula.

Lemma 7.1.1. *Let $\mathcal{G}$ be a sub- σ -algebra of the σ -algebra $\mathcal{G}$, and let ψ be a random variable integrable with respect to $\mathbb{Q}$. Then the following version of Bayes's formula holds*

$$\mathbb{E}_{\mathbb{Q}}(\psi | \mathcal{G}) = \frac{\mathbb{E}(\psi\eta | \mathcal{G})}{\mathbb{E}(\eta | \mathcal{G})}. \quad (7.4)$$

Proof. It can be easily checked that $\mathbb{E}(\eta | \mathcal{G})$ is strictly positive $\mathbb{P}$ -a.s. so that the right-hand side of (7.4) is well-defined. By our assumption, the random variable $\xi = \psi\eta$ is $\mathbb{P}$ -integrable, it is therefore enough to show that

$$\mathbb{E}(\xi | \mathcal{G}) = \mathbb{E}_{\mathbb{Q}}(\psi | \mathcal{G}) \mathbb{E}(\eta | \mathcal{G}).$$

Since the right-hand side of the last formula defines a $\mathcal{G}$ -measurable random variable, we need to verify that for any set $A \in \mathcal{G}$, we have

$$\int_A \psi\eta \, d\mathbb{P} = \int_A \mathbb{E}_{\mathbb{Q}}(\psi | \mathcal{G}) \mathbb{E}(\eta | \mathcal{G}) \, d\mathbb{P}.$$

But for every $A \in \mathcal{G}$, we get

$$\begin{aligned} \int_A \psi\eta \, d\mathbb{P} &= \int_A \psi \, d\mathbb{Q} = \int_A \mathbb{E}_{\mathbb{Q}}(\psi | \mathcal{G}) \, d\mathbb{Q} = \int_A \mathbb{E}_{\mathbb{Q}}(\psi | \mathcal{G}) \eta \, d\mathbb{P} \\ &= \int_A \mathbb{E}(\mathbb{E}_{\mathbb{Q}}(\psi | \mathcal{G}) \eta | \mathcal{G}) \, d\mathbb{P} = \int_A \mathbb{E}_{\mathbb{Q}}(\psi | \mathcal{G}) \mathbb{E}(\eta | \mathcal{G}) \, d\mathbb{P}. \end{aligned}$$

□

Lemma 7.1.2. *A stochastic process X is an $\mathbb{F}$ -martingale under $\tilde{\mathbb{P}}$ if and only if the product $X\eta$ is an $\mathbb{F}$ -martingale under $\mathbb{P}$.*

Proof. The proof relies on the application of the abstract Bayes formula (see Lemma 7.1.1). Assume first that $X\eta$ is an $\mathbb{F}$ -martingale under $\mathbb{P}$ so that equality $\mathbb{E}_{\mathbb{P}}(X_u\eta_u|\mathcal{F}_t) = X_t\eta_t$ holds for $0 \leq t \leq u \leq T$. Then the Bayes formula yields, for $0 \leq t \leq u \leq T$,

$$\mathbb{E}_{\tilde{\mathbb{P}}}(X_u|\mathcal{F}_t) = \frac{\mathbb{E}_{\mathbb{P}}(X_u\eta_T|\mathcal{F}_t)}{\mathbb{E}_{\mathbb{P}}(\eta_T|\mathcal{F}_t)} = \frac{\mathbb{E}_{\mathbb{P}}(X_u\mathbb{E}_{\mathbb{P}}(\eta_T|\mathcal{F}_u)|\mathcal{F}_t)}{\mathbb{E}_{\mathbb{P}}(\eta_T|\mathcal{F}_t)} = \frac{\mathbb{E}_{\mathbb{P}}(X_u\eta_u|\mathcal{F}_t)}{\eta_t} = \frac{X_t\eta_t}{\eta_t} = X_t$$

We conclude that X is an $\mathbb{F}$ -martingale under $\mathbb{P}$. The proof of the converse implication goes along the same lines. $\square$

7.2 Girsanov's Theorem

We are in the position to state the classic Girsanov theorem. It states that, under mild technical conditions, a Brownian motion with an absolutely continuous drift becomes a standard Brownian motion under an equivalent probability measure. Before proceeding to the general result, let us first consider a special case of a linear drift.

Proposition 7.2.1. *Let W be a one-dimensional standard Brownian motion on a probability space $(\Omega, \mathbb{F}, \mathbb{P})$. For a real number $\gamma \in \mathbb{R}$, we define the process X by setting $\widetilde{W}_t = W_t - \gamma t$ for $t \in [0, T]$. Let the probability measure $\tilde{\mathbb{P}}$, equivalent to $\mathbb{P}$ on $(\Omega, \mathcal{F}_T)$, be defined through the formula*

$$\frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} = \exp\left(\gamma W_T - \frac{1}{2}\gamma^2 T\right) = \eta_T, \quad \mathbb{P}$$

Then X is a standard Brownian motion on the probability space $(\Omega, \mathbb{F}, \tilde{\mathbb{P}})$.

Proof. To establish the proposition, it is enough to make use of the abstract Bayes formula and Lévy's characterisation theorem. In view of the later, it suffices to show that for any $\lambda \in \mathbb{R}$ the process

$$M_t^\lambda = \exp\left(\lambda \widetilde{W}_t - \frac{1}{2}\lambda^2 t\right), \quad \forall t \in [0, T],$$

is an $\mathbb{F}$ -martingale under $\tilde{\mathbb{P}}$. Using the the abstract Bayes formula, we see that it is enough to check that $M^\lambda \eta$ is an $\mathbb{F}$ -martingale under $\mathbb{P}$. But clearly

$$M_t^\lambda \eta_t = \exp\left(\lambda(W_t - \gamma t) - \frac{1}{2}\lambda^2 t\right) \exp\left(\gamma W_t - \frac{1}{2}\gamma^2 t\right) = \exp\left(\alpha W_t - \frac{1}{2}\alpha^2 t\right)$$

where $\alpha = \lambda + \gamma$, and thus this process follows an $\mathbb{F}$ -martingale under $\mathbb{P}$. From the Lévy characterisation theorem we conclude that $\widetilde{W}$ is a standard Brownian motion on $(\Omega, \mathbb{F}, \tilde{\mathbb{P}})$. $\square$

Proposition 7.2.1 can be extended to the case of a stochastic drift term, as the following result shows. Recall that given a continuous semimartingale X , the stochastic exponential $Y_t = \mathcal{E}_t(X)$ is the unique solution to the linear SDE

$$dY_t = Y_t dX_t \quad \text{and} \quad Y_0 = 1.$$

If $X_t = \int_0^t \gamma_u dW_u$ then $Y_t = \exp(\int_0^t \gamma_u dW_u - \frac{1}{2} \int_0^t \gamma_u^2 du)$.

Theorem 7.2.1. *Let W be a standard d -dimensional Brownian motion on a filtered probability space $(\Omega, \mathbb{F}, \mathbb{P})$. Suppose that γ is an $\mathbb{R}^d$ -valued $\mathbb{F}$ -progressively measurable process such that*

$$\mathbb{E}_{\mathbb{P}} \left\{ \mathcal{E}_T \left(\int_0^\cdot \gamma_u \cdot dW_u \right) \right\} = 1. \quad (7.1)$$

Define a probability measure $\tilde{\mathbb{P}}$ on $(\Omega, \mathcal{F}_T)$ equivalent to $\mathbb{P}$ by means of the Radon-Nikodým derivative

$$\frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} = \mathcal{E}_T \left(\int_0^\cdot \gamma_u \cdot dW_u \right), \quad \mathbb{P} \quad (7.2)$$

Then the process $\tilde{W}$ given by the formula

$$\tilde{W}_t = W_t - \int_0^t \gamma_u du, \quad \forall t \in [0, T] \quad (7.3)$$

follows a standard d -dimensional Brownian motion on the space $(\Omega, \mathbb{F}, \tilde{\mathbb{P}})$.

Proof. Again by multidimensional version of the Lévy characterisation theorem and the abstract Bayes formula, it suffices to check that the processes $\tilde{W}^i \eta$ and $(\tilde{W}_t^i \tilde{W}_t^j - \delta_{ij} t) \eta_t$ are continuous local martingales under $\tilde{\mathbb{P}}$. For the first process, an application of the Itô integration by parts formula gives

$$\begin{aligned} d(\tilde{W}_t^i \eta_t) &= \tilde{W}_t^i d\eta_t + \eta_t d\tilde{W}_t^i + d\langle \tilde{W}^i, \eta \rangle_t \\ &= \tilde{W}_t^i \eta_t \gamma_t \cdot dW_t + \eta_t dW_t^i - \eta_t \gamma_t^i dt + d\langle \tilde{W}^i, \eta \rangle_t \\ &= \tilde{W}_t^i \eta_t \gamma_t \cdot dW_t + \eta_t dW_t^i \end{aligned}$$

since $d\eta_t = \eta_t \gamma_t \cdot dW_t$ and thus $d\langle \tilde{W}^i, \eta \rangle_t = \gamma_t^i \eta_t dt$. We thus see that $\tilde{W}^i \eta$ is a continuous local martingale under $\tilde{\mathbb{P}}$.

Let us denote $\tilde{M}_t^{ij} = \tilde{W}_t^i \tilde{W}_t^j - \delta_{ij} t$. Then we have that (recall that $\langle W^i, W^j \rangle_t = 0$ for $i \neq j$ and $\langle W^i, W^i \rangle_t = t$ for $t \in [0, T]$)

$$d\tilde{M}_t^{ij} = \tilde{W}_t^i d\tilde{W}_t^j + \tilde{W}_t^j d\tilde{W}_t^i,$$

and this in turn implies that

$$d\langle \tilde{M}^{ij}, \eta \rangle_t = \eta_t (\tilde{W}_t^i \gamma_t^j + \tilde{W}_t^j \gamma_t^i) dt.$$

By applying once again the Itô integration by parts formula, we obtain

$$\begin{aligned} d(\tilde{M}_t^{ij} \eta_t) &= \tilde{M}_t^{ij} d\eta_t + \eta_t d\tilde{M}_t^{ij} + d\langle \tilde{M}^{ij}, \eta \rangle_t \\ &= \tilde{M}_t^{ij} d\eta_t + \eta_t (\tilde{W}_t^i d\tilde{W}_t^j + \tilde{W}_t^j d\tilde{W}_t^i) + \eta_t (\tilde{W}_t^i \gamma_t^j + \tilde{W}_t^j \gamma_t^i) dt \\ &= \tilde{M}_t^{ij} \eta_t \gamma_t \cdot dW_t + \eta_t (\tilde{W}_t^i dW_t^j + \tilde{W}_t^j dW_t^i) \end{aligned}$$

since $\tilde{W}_t^i = W_t^i - \gamma_t^i t$. We conclude that the process $(\tilde{W}_t^i \tilde{W}_t^j - \delta_{ij} t) \eta_t$ follows a continuous local martingale under $\tilde{\mathbb{P}}$. $\square$

Theorem 7.2.2. *Let W be a standard d -dimensional Brownian motion on a filtered probability space $(\Omega, \mathbb{F}, \mathbb{P})$. Suppose that γ is an $\mathbb{R}^d$ -valued $\mathbb{F}$ -progressively measurable process such that*

$$\mathbb{E}_{\mathbb{P}}\left\{\mathcal{E}_T\left(\int_0^\cdot \gamma_u \cdot dW_u\right)\right\} = 1. \quad (7.4)$$

Define a probability measure $\widetilde{\mathbb{P}}$ on $(\Omega, \mathcal{F}_T)$ equivalent to $\mathbb{P}$ by means of the Radon-Nikodým derivative

$$\frac{d\widetilde{\mathbb{P}}}{d\mathbb{P}} = \mathcal{E}_T\left(\int_0^\cdot \gamma_u \cdot dW_u\right), \quad \mathbb{P} \text{ a.s.} \quad (7.5)$$

Then the process $\widetilde{W}$ given by the formula

$$\widetilde{W}_t = W_t - \int_0^t \gamma_u du, \quad \forall t \in [0, T] \quad (7.6)$$

follows a standard d -dimensional Brownian motion on the space $(\Omega, \mathbb{F}, \widetilde{\mathbb{P}})$.

In the following, suppose $\gamma \in \mathcal{L}_{\mathbb{P}}(W)$ and we set

$$U_t := \int_0^t \gamma_s dW_s, \quad t \in [0, T]. \quad (7.7)$$

In order to make use of the Girsanov theorem, we need to make sure that the the Radon Nikodým density prescribed in (7.5) or

$$\eta_T = \mathcal{E}_T(U)$$

satisfies the condition (7.4) or in other words

$$\mathbb{E}_{\mathbb{P}}(\mathcal{E}_T(U)) = 1. \quad (7.8)$$

The next result furnishes a sufficient condition or the equality (7.8) to hold. The inequality (7.9) is commonly referred to as Novikov's condition.

Proposition 7.2.2 (Novikov's condition). *Assume that*

$$\mathbb{E}_{\mathbb{P}}\left\{\exp\left(\frac{1}{2}\int_0^T |\gamma_u|^2 du\right)\right\} < \infty. \quad (7.9)$$

Then $\mathbb{E}_{\mathbb{P}}(\mathcal{E}_T(U)) = 1$, where U is given by (7.7). Consequently, the process $\mathcal{E}(U)$ is a strictly positive continuous $\mathbb{F}$ -martingale.

In particular, if the process γ is uniformly bounded (that is, there exists a constant K such that $|\gamma_t| \leq K$ for $t \in [0, T]$) then Novikov's condition is satisfied and thus $\mathbb{E}_{\mathbb{P}}(\mathcal{E}_T(U)) = 1$.

A weaker, but also sufficient for the validity of (7.8), is the Kazamaki condition

$$\mathbb{E}_{\mathbb{P}}\left\{\exp\left(\frac{1}{2}\int_0^t \gamma_u \cdot dW_u\right)\right\} < \infty, \quad \forall t \in [0, T]. \quad (7.10)$$

Let us comment briefly on relevant filtrations. Obviously, we always have that $\mathcal{F}_t^{\widetilde{W}} \subseteq \mathcal{F}_t$ for any $t \in [0, T]$. The filtrations generated by W and $\widetilde{W}$ do not coincide, in general. In the special case where the underlying filtration $\mathbb{F}$ is the $\mathbb{P}$ -augmentation of the natural filtration of W , the inclusion $\mathcal{F}_t^{\widetilde{W}} \subseteq \mathcal{F}_t^W$ holds for any $t \in [0, T]$, but the filtration generated by $\widetilde{W}$ can be essentially smaller than $\mathbb{F}^W$.

The next well-known result shows that, if the underlying filtration is generated by a d -dimensional Brownian motion, the Radon-Nikodým density process of any probability measure equivalent to $\mathbb{P}$ has necessarily the form of the stochastic exponential (7.5) for some process γ .

Proposition 7.2.3. *Assume that the filtration $\mathbb{F}$ is the usual augmentation of the natural filtration of W , that is, $\mathbb{F} = \mathbb{F}^W$. Then for any probability measure $\widetilde{\mathbb{P}}$ on $(\Omega, \mathcal{F}_T)$ equivalent to $\mathbb{P}$ there exists an $\mathbb{F}^W$ -progressively measurable, $\mathbb{R}^d$ -valued process γ such that the Radon-Nikodým density of $\widetilde{\mathbb{P}}$ with respect to $\mathbb{P}$ equals*

$$\frac{d\widetilde{\mathbb{P}}}{d\mathbb{P}} = \mathcal{E}_T \left(\int_0^\cdot \gamma_u \cdot dW_u \right), \quad \mathbb{P} \text{ a.s.} \quad (7.11)$$

Proof. Let η_T be the Radon-Nikodým density of $\widetilde{\mathbb{P}}$ with respect to $\mathbb{P}$ on $(\Omega, \mathcal{F}_T)$ (The existence is guaranteed by the Radon-Nikodým theorem). It is clear that the process $\eta_t = \mathbb{E}_{\mathbb{P}}(\eta_T | \mathcal{F}_t)$ is an $\mathbb{F}$ -martingale and $\eta_0 = 1$. Since the underlying filtration is a Brownian filtration, from the martingale representation theorem, Theorem 5.3.2, we can deduce the existence of a process $\widetilde{\gamma} \in \mathcal{L}_{\mathbb{P}}(W)$ such that

$$\eta_t = 1 + \int_0^t \widetilde{\gamma}_u \cdot dW_u, \quad \forall t \in [0, T]. \quad (7.12)$$

Since $\mathbb{P}\{\eta_T > 0\} = 1$, we also have that $\mathbb{P}\{\eta_t > 0\} = 1$ for any $t \in [0, T]$, and thus, in view of continuity of η , which is apparent from representations (7.12), we obtain $\mathbb{P}\{\eta_t > 0, \forall t \in [0, T]\} = 1$. Therefore, the process $\gamma_t = \widetilde{\gamma}_t \eta_t^{-1}$ is well defined and we manifestly have

$$\eta_t = 1 + \int_0^t \widetilde{\gamma}_u \cdot dW_u = 1 + \int_0^t \eta_u \gamma_u \cdot dW_u.$$

We conclude that the Radon-Nikodým density process is the unique solution to the SDE

$$d\eta_t = \eta_t \gamma_t \cdot dW_t, \quad (7.13)$$

and thus, in view of the form of the stochastic exponential, it satisfies, for any $t \in [0, T]$,

$$\eta_t = \mathcal{E}_t \left(\int_0^\cdot \gamma_u \cdot dW_u \right).$$

This completes the proof of the proposition. $\square$

Case of Continuous Semimartingales

We end this section by stating a generalization of Girsanov's theorem to the case of continuous semimartingales.

Let $\tilde{\mathbb{P}}$ and $\mathbb{P}$ be two equivalent probability measures on a common filtered probability space $(\Omega, \mathcal{F}_T)$ with Radon-Nikodým density η_T . We assume, in addition, that the Radon-Nikodým density process η given by

$$\eta_t = \mathbb{E}_{\mathbb{P}}(\eta_T | \mathcal{F}_t) = \mathbb{E}_{\mathbb{P}}\left(\frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} \mid \mathcal{F}_t\right), \forall t \in [0, T] \quad (7.14)$$

is continuous (we know already that this holds if $\mathbb{F} = \mathbb{F}^W$ for some Brownian motion W). Let us set, for any $t \in [0, T]$,

$$U_t = \int_0^t \eta_u^{-1} d\eta_u = \mathcal{L}_t(\eta)$$

which is called the *Stochastic Logarithm* of η so that U is a continuous local martingale and $\eta_t = \mathcal{E}_t(U)$ for $t \in [0, T]$.

The stochastic exponential and stochastic logarithm are reverse operations of each other. We show here that $\mathcal{L}_t(\mathcal{E}(\eta)) = \eta_t$. By setting $X_t = \mathcal{E}_t(\eta)$ and applying Itô's formula

$$d\mathcal{L}_t(X) = \frac{1}{X_t} dX_t = \frac{1}{X_t} X_t d\eta_t = d\eta_t$$

which shows that $\mathcal{L}_t(\mathcal{E}(\eta)) = \eta_t$.

Exercise: Let $U_t = \mathcal{L}_t(\eta)$ and show that $\mathcal{E}_t(U) = \eta_t$. Hint: Show that $\log(\eta_t) = U_t - \frac{1}{2}\langle U \rangle_t$.

Theorem 7.2.3. Let $\tilde{\mathbb{P}}$ be a probability measure equivalent to $\mathbb{P}$ on $(\Omega, \mathcal{F}_T)$ and such that the Radon-Nikodým density process η given by (7.14) is continuous. Then any continuous real-valued $\mathbb{P}$ -semimartingale X is a continuous $\tilde{\mathbb{P}}$ -semimartingale. If the canonical decomposition of X under $\mathbb{P}$ is $X = X_0 + M + A$ then its canonical decomposition under $\tilde{\mathbb{P}}$ is $X = X_0 + \tilde{M} + \tilde{A}$ where

$$\tilde{M}_t = M_t - \int_0^t \eta_u^{-1} d\langle \eta, M \rangle_u = M_t - \langle U, M \rangle_t$$

and

$$\tilde{A}_t = A_t + \int_0^t \eta_u^{-1} d\langle \eta, M \rangle_u = A_t + \langle U, M \rangle_t.$$

In particular, X follows a local martingale under $\tilde{\mathbb{P}}$ if and only if the process $A + \langle U, M \rangle$ vanishes identically, that is, $A_t + \langle U, M \rangle_t = 0$ for every $t \in [0, T]$.

Proof. We proof the case where X is a local martingale that is $X = M$. Recall that $\tilde{M}$ is a martingale under $\tilde{\mathbb{P}}$ if and only if $\eta \tilde{M}$ is a martingale under $\mathbb{P}$. Therefore we compute

$$\mathbb{E}_{\mathbb{P}}[\eta_t \tilde{M}_t | \mathcal{F}_s] = \mathbb{E}_{\mathbb{P}}[\eta_t (M_t - \langle U, M \rangle_t) | \mathcal{F}_s]$$

By applying the integration by parts formula we obtain

$$\begin{aligned} & \eta_t(M_t - \langle U, M \rangle_t) - \eta_s(M_s - \langle U, M \rangle_s) \\ &= \int_s^t \eta_u dM_u + \int_s^t M_u d\eta_u + \int_s^t d\langle \eta, M \rangle_u - \int_s^t \eta_u d\langle U, M \rangle_u - \int_s^t \langle U, M \rangle_u d\eta_u. \end{aligned}$$

To proceed, we note that

$$\int_s^t \eta_u dM_u + \int_s^t M_u d\eta_u - \int_s^t \langle U, M \rangle_u d\eta_u$$

is a $\mathbb{P}$ martingale for $t \in [s, T]$, and

$$\int_s^t d\langle \eta, M \rangle_u - \int_s^t \eta_u d\langle U, M \rangle_u = \int_s^t d\langle \eta, M \rangle_u - \int_s^t \eta_u \eta_u^{-1} d\langle \eta, M \rangle_u = 0.$$

This shows that

$$\mathbb{E}_{\mathbb{P}}[\eta_t \widetilde{M}_t | \mathcal{F}_s] = \mathbb{E}_{\mathbb{P}}[\eta_t (M_t - \langle U, M \rangle_t) | \mathcal{F}_s] = \eta_s (M_s - \langle U, M \rangle_s) = \eta_s \widetilde{M}_s$$

which conclude the proof. $\square$

Remark 7.2.1. More generally, if $\eta = \mathcal{E}(U)$, where U is a local martingale under $\mathbb{P}$ (not necessarily with continuous sample paths), the last theorem remains valid under the assumption that the cross-variation $\langle U, M \rangle$ exists.

7.3 Applications

In the following, we give some applications of the Girsanov's transform.

7.3.1 Martingale Measures

One of standard applications of Girsanov's theorem is related to the issue of existence and uniqueness of a martingale measure for a given process.

Definition 7.3.1. Let $X = (X_t)_{t \in [0, T]}$ be an $\mathbb{F}$ -adapted process on a filtered probability space $(\Omega, \mathbb{F}, \mathbb{P})$. We say that a probability measure $\widetilde{\mathbb{P}}$ equivalent to $\mathbb{P}$ on $(\Omega, \mathcal{F}_T)$ is a *martingale measure* for X whenever X is a local martingale under $\widetilde{\mathbb{P}}$ with respect to $\mathbb{F}$.

Some authors prefer to make a clear terminological distinction between the concepts of a martingale measure and a local martingale measure. We decided to use only one term, and thus our martingale measure corresponds to what is sometimes called a local martingale measure.

Let us examine in detail the existence and uniqueness of a martingale measure for the unique solution X of the SDE (The Black Scholes Model)

$$dX_t = X_t(a dt + b dW_t) \tag{7.1}$$

where a and $b \neq 0$ are real numbers and W is a one-dimensional standard Brownian motion. Note that X is an $\mathbb{F}$ -martingale under $\mathbb{P}$ whenever $a = 0$.

From Theorem 7.2.2, we know that for any real number γ the process $\widetilde{W}_t = W_t - \gamma t$ is a one-dimensional standard Brownian motion on a probability space $(\Omega, \mathbb{F}, \widetilde{\mathbb{P}})$, where $\widetilde{\mathbb{P}}$ satisfies

$$\frac{d\widetilde{\mathbb{P}}}{d\mathbb{P}} = \exp\left(\gamma W_T - \frac{1}{2}\gamma^2 T\right), \quad \mathbb{P}\text{a.s.} \quad (7.2)$$

By inserting the differential $dW_t = d\widetilde{W}_t + \gamma dt$ into (7.1), we see that X satisfies under $\widetilde{\mathbb{P}}$ the following SDE

$$dX_t = X_t((a + b\gamma) dt + b d\widetilde{W}_t). \quad (7.3)$$

Hence X is an $\mathbb{F}$ -martingale under $\widetilde{\mathbb{P}}$ provided that γ is chosen in such a way that the drift term in under (7.3) vanishes, that is, when $a + b\gamma = 0$. Of course, this means that $\gamma = -a/b$. This lead to the following result.

Proposition 7.3.1. *Let W be a one-dimensional standard Brownian motion and let $\mathbb{F} = \mathbb{F}^W$ be generated by W . Then the unique martingale measure $\widetilde{\mathbb{P}}$ for the solution X to the SDE (7.1) is given by (7.2) with $\gamma = -a/b$.*

Proof. We have already checked that $\widetilde{\mathbb{P}}$ is a martingale measure for X . The uniqueness can be easily deduced from Proposition 7.2.3. $\square$

The uniqueness in Proposition 7.3.1 is no longer true when W is a d -dimensional standard Brownian motion for some $d > 1$, so that $b \in \mathbb{R}^d$. By applying Theorem 7.2.2 and Proposition 7.2.3, we conclude that a martingale measure for X corresponds to any $\mathbb{R}^d$ -valued $\mathbb{F}$ -progressively measurable process γ satisfying the equation $a + b \cdot \gamma_t = 0$, $t \in [0, T]$, and such that (7.4) is satisfied for γ . It is rather clear that there are infinitely many such processes, and thus uniqueness of a martingale measure for X fails to hold.

7.3.2 Distribution of the First Passage Time

Let the process Y follow a Brownian motion with the standard deviation σ and the drift μ , specifically,

$$dY_t = \mu dt + \sigma dW_t, \quad Y_0 = y_0,$$

where W is a standard one-dimensional Brownian motion under $\mathbb{P}$ with respect to $\mathbb{F}$. Put another way

$$Y_t = y_0 + \mu t + \sigma W_t \quad (7.4)$$

for some constants $\mu \in \mathbb{R}$ and $\sigma > 0$. Let us note that Y inherits from a Brownian motion W the strong Markov property with respect to the reference filtration $\mathbb{F}$.

Let τ stand for the first passage time to zero by the process Y , that is, the $\mathbb{F}$ -stopping time τ given by

$$\tau = \inf\{t \geq 0 \mid Y_t = 0\}.$$

It is well known that in an arbitrarily small time interval $[0, t]$ the sample path of the Brownian motion started at 0 passes through origin infinitely many times (see, for

instance, Page 42 in Krylov (1995)). Using Girsanov's theorem and the strong Markov property of the Brownian motion, it is not difficult to show that first passage time by Y to zero coincides with the first crossing time by Y of the level 0, i.e., with probability 1,

$$\tau = \inf\{t \geq 0 \mid Y_t \leq 0\} = \inf\{t \geq 0 \mid Y_t < 0\}.$$

Let us first prove the following well known result.

Lemma 7.3.1. *Let W be a one-dimensional standard Brownian motion under $\mathbb{P}$ and let $X_t = \mu t + \sigma W_t$, $t \in \mathbb{R}_+$, where $\sigma > 0$ and $\mu \in \mathbb{R}$. Then for every $x > 0$ and any $s > 0$ we have*

$$\mathbb{P}\left(\sup_{0 \leq u \leq s} X_u \leq x\right) = N\left(\frac{x - \mu s}{\sigma \sqrt{s}}\right) - e^{2\nu\sigma^{-2}x} N\left(\frac{-x - \mu s}{\sigma \sqrt{s}}\right) \quad (7.5)$$

and for every $x < 0$ and any $s > 0$

$$\mathbb{P}\left(\inf_{0 \leq u \leq s} X_u \geq x\right) = N\left(\frac{-x + \mu s}{\sigma \sqrt{s}}\right) - e^{2\nu\sigma^{-2}x} N\left(\frac{x + \mu s}{\sigma \sqrt{s}}\right). \quad (7.6)$$

Proof. To derive the first equality, we will use Girsanov's theorem and the reflection principle for a Brownian motion. Assume first that $\sigma = 1$. Let $\mathbb{P}^*$ be the probability measure on $(\Omega, \mathcal{F}_T)$ given by

$$\frac{d\mathbb{P}^*}{d\mathbb{P}} = e^{-\mu W_T - \frac{\mu^2}{2}T}, \quad \mathbb{P}\text{-a.s.},$$

so that the process $W_t^* = X_t = W_t + \mu t$, $t \in [0, T]$, follows a standard Brownian motion under $\mathbb{P}^*$ and

$$\frac{d\mathbb{P}}{d\mathbb{P}^*} = e^{\mu W_T^* - \frac{\mu^2}{2}T}, \quad \mathbb{P}^*\text{-a.s.}$$

Moreover

$$\mathbb{P}\left(\sup_{0 \leq u \leq s} X_u > x, X_s \leq x\right) = \mathbb{E}_{\mathbb{P}^*}\left(e^{\mu W_s^* - \frac{\mu^2}{2}s} \mathbb{1}_{\{\sup_{0 \leq u \leq s} W_u^* > x, W_s^* \leq x\}}\right).$$

Let us define an $\mathbb{F}$ -stopping time T_x by setting

$$T_x = \inf\{t \geq 0 \mid W_t^* = x\},$$

and let us introduce an auxiliary process $\widetilde{W}_t$, $t \in [0, T]$, defined by the formula

$$\widetilde{W}_t = W_t^* \mathbb{1}_{\{\tau_x \geq t\}} + (2x - W_t^*) \mathbb{1}_{\{\tau_x < t\}}.$$

By virtue of the reflection principle, the process $\widetilde{W}$ also follows a standard Brownian motion under $\mathbb{P}^*$. Moreover, we have

$$\left\{\sup_{0 \leq u \leq T} \widetilde{W}_u > x, \widetilde{W}_T \leq x\right\} = \{W_T^* \geq x\} \subset \{T_x \leq T\}.$$

Let us denote

$$J = \mathbb{P}\left(\sup_{0 \leq u \leq T} (W_u + \mu u) \leq x\right).$$

Then we obtain

$$\begin{aligned}
J &= \mathbb{P}(X_T \leq x) - \mathbb{P}\left(\sup_{0 \leq u \leq T} X_u > x, X_T \leq x\right) \\
&= \mathbb{P}(X_T \leq x) - \mathbb{E}_{\mathbb{P}^*}\left(e^{\mu W_T^* - \frac{\mu^2}{2}T} \mathbf{1}_{\{\sup_{0 \leq u \leq T} W_u^* > x, W_T^* \leq x\}}\right) \\
&= \mathbb{P}(W_T + \mu T \leq x) - \mathbb{E}_{\mathbb{P}^*}\left(e^{\mu \widetilde{W}_T - \frac{\mu^2}{2}T} \mathbf{1}_{\{\sup_{0 \leq u \leq T} \widetilde{W}_u > x, \widetilde{W}_T \leq x\}}\right) \\
&= \mathbb{P}(X_T \leq x) - \mathbb{E}_{\mathbb{P}^*}\left(e^{\mu(2x - W_T^*) - \frac{\mu^2}{2}T} \mathbf{1}_{\{W_T^* \geq x\}}\right) \\
&= \mathbb{P}(X_T \leq x) - e^{2\mu x} \mathbb{E}_{\mathbb{P}^*}\left(e^{\mu W_T^* - \frac{\mu^2}{2}T} \mathbf{1}_{\{W_T^* \leq -x\}}\right) \\
&= \mathbb{P}(X_T \leq x) - e^{2\mu x} \mathbb{P}\{W_T + \mu T \leq -x\} \\
&= N\left(\frac{x - \mu T}{\sqrt{T}}\right) - e^{2\mu x} N\left(\frac{-x - \mu T}{\sqrt{T}}\right).
\end{aligned}$$

This ends the proof of the first equality for $\sigma = 1$. We have, for any $\sigma > 0$,

$$\mathbb{P}\left(\sup_{0 \leq u \leq s} (\sigma W_u + \mu u) \leq x\right) = \mathbb{P}\left(\sup_{0 \leq u \leq s} (W_u + \mu \sigma^{-1} u) \leq x \sigma^{-1}\right),$$

and this implies (7.5). Since $-W$ follows a standard Brownian motion under $\mathbb{P}$ we have, for any $x < 0$,

$$\mathbb{P}\left(\inf_{0 \leq u \leq s} (\sigma W_u + \mu u) \geq x\right) = \mathbb{P}\left(\sup_{0 \leq u \leq s} (\sigma W_u - \mu u) \leq -x\right),$$

and thus the second equality is a simple consequence of the first. $\square$

Proposition 7.3.2. *Let Y be given by (7.4), where $\mu \in \mathbb{R}$, $\sigma > 0$ and W is a standard Brownian motion under $\mathbb{P}$. Then the random variable τ has an inverse Gaussian probability distribution under $\mathbb{P}$, that is, for $0 < s < \infty$,*

$$\mathbb{P}(\tau \leq s) = \mathbb{P}(\tau < s) = N(h_1(s)) + e^{-2\nu\sigma^{-2}y_0} N(h_2(s)), \quad (7.7)$$

where N is the standard Gaussian cumulative distribution function and

$$h_1(s) = \frac{-y_0 - \mu s}{\sigma\sqrt{s}}, \quad h_2(s) = \frac{-y_0 + \mu s}{\sigma\sqrt{s}}.$$

Proof. Let us first observe that

$$\mathbb{P}(\tau \geq s) = \mathbb{P}\left(\inf_{0 \leq u \leq s} Y_u \geq 0\right) = \mathbb{P}\left(\inf_{0 \leq u \leq s} X_u \geq -y_0\right), \quad (7.8)$$

where $X_u = \sigma W_u + \mu u$. We know from Lemma 7.3.1 that for every $x < 0$ we have

$$\mathbb{P}\left(\inf_{0 \leq u \leq s} X_u \geq x\right) = N\left(\frac{-x + \mu s}{\sigma\sqrt{s}}\right) - e^{2\nu\sigma^{-2}x} N\left(\frac{x + \mu s}{\sigma\sqrt{s}}\right).$$

When combined with (7.8), this yields (7.7). $\square$

The following corollary is a consequence of Proposition 7.3.2 and the strong Markov property of the process Y with respect to the filtration $\mathbb{F}$.

Corollary 7.3.1. *Under the assumptions of Proposition 7.3.2 we have, for any $t < s$ on the event $\{\tau > t\}$,*

$$\mathbb{P}(\tau \leq s | \mathcal{F}_t) = N\left(\frac{-Y_t - \mu(s-t)}{\sigma\sqrt{s-t}}\right) + e^{-2\nu\sigma^{-2}Y_t} N\left(\frac{-Y_t + \mu(s-t)}{\sigma\sqrt{s-t}}\right).$$

7.3.3 Brownian motion with drift and its running supremum.

In the next step, we will examine the joint distribution of the process Y and its first passage time τ . For the process $X_t = \mu t + \sigma W_t$, we write

$$M_T^X = \sup_{u \in [0, T]} X_u, \quad m_T^X = \inf_{u \in [0, T]} X_u.$$

By the Girsanov theorem, the process $\sigma^{-1}X$ is a Brownian motion under an equivalent probability measure and thus we can deduce that, for any $s > 0$

$$\mathbb{P}(M_T^X > 0) = 1, \quad \mathbb{P}(m_T^X < 0) = 1.$$

Lemma 7.3.2. *For any $s > 0$, the joint distribution of X_s and M_s^X is given by the formula*

$$\mathbb{P}(X_s \leq x, M_s^X \geq y) = e^{2\nu y \sigma^{-2}} \mathbb{P}(X_s \geq 2y - x + 2\nu s), \quad (7.9)$$

for every $x, y \in \mathbb{R}$ such that $y \geq 0$ and $x \leq y$.

Proof. Since

$$I = \mathbb{P}\{X_s \leq x, M_s^X \geq y\} = \mathbb{P}\{X_s^\sigma \leq x\sigma^{-1}, M_s^{X^\sigma} \geq y\sigma^{-1}\},$$

where $X_t^\sigma = W_t + \mu t \sigma^{-1}$, it is clear that we may and do assume, without loss of generality, that $\sigma = 1$.

We shall use similar arguments as in the proof of Lemma 7.3.1. It follows from Girsanov's theorem that X is a standard Brownian motion under the probability measure $\mathbb{P}^*$ given on $(\Omega, \mathcal{F}_T)$ by the formula

$$\frac{d\mathbb{P}^*}{d\mathbb{P}} = e^{-\mu W_T - \frac{\mu^2}{2}T}, \quad \mathbb{P}\text{-a.s.}$$

Therefore

$$\frac{d\mathbb{P}}{d\mathbb{P}^*} = e^{\mu W_T^* - \frac{\mu^2}{2}T}, \quad \mathbb{P}\text{-a.s.},$$

where $W_s^* = X_s = W_s + \mu s$, $s \in [0, T]$, is a standard Brownian motion under $\mathbb{P}^*$. It is thus clear that

$$I = \mathbb{E}_{\mathbb{P}^*} \left(e^{\mu W_s^* - \frac{\mu^2}{2}s} \mathbb{1}_{\{X_s \leq x, M_s^X \geq y\}} \right) = \mathbb{E}_{\mathbb{P}^*} \left(e^{\mu W_s^* - \frac{\mu^2}{2}s} \mathbb{1}_{\{W_s^* \leq x, M_s^{W^*} \geq y\}} \right).$$

Since W^* is a standard Brownian motion under $\mathbb{P}^*$, an application of the reflection principle yields

$$\begin{aligned} I &= \mathbb{E}_{\mathbb{P}^*} \left(e^{\mu(2y-W_s^*) - \frac{\mu^2}{2}s} \mathbf{1}_{\{2y-W_s^* \leq x, M_s^{W^*} \geq y\}} \right) \\ &= \mathbb{E}_{\mathbb{P}^*} \left(e^{\mu(2y-W_s^*) - \frac{\mu^2}{2}s} \mathbf{1}_{\{W_s^* \geq 2y-x\}} \right) \\ &= e^{2\mu y} \mathbb{E}_{\mathbb{P}^*} \left(e^{-\mu W_s^* - \frac{\mu^2}{2}s} \mathbf{1}_{\{W_s^* \geq 2y-x\}} \right), \end{aligned}$$

since under the present assumptions we have $2y - x \geq y$. Let us define an equivalent probability measure $\tilde{\mathbb{P}}$ on $(\Omega, \mathcal{F}_T)$ by setting

$$\frac{d\tilde{\mathbb{P}}}{d\mathbb{P}^*} = e^{-\mu W_T^* - \frac{\mu^2}{2}T}, \quad \mathbb{P}^*\text{-a.s.}$$

Then we have that

$$I = e^{2\mu y} \mathbb{E}_{\mathbb{P}^*} \left(e^{-\mu W_s^* - \frac{\mu^2}{2}s} \mathbf{1}_{\{W_s^* \geq 2y-x\}} \right) = e^{2\mu y} \tilde{\mathbb{P}}\{W_s^* \geq 2y-x\}.$$

The process $\widetilde{W}_t = W_s^* + \mu s$, $s \in [0, T]$, is a standard Brownian motion under $\tilde{\mathbb{P}}$ and we have

$$I = e^{2\mu y} \tilde{\mathbb{P}}(\widetilde{W}_s + \mu s \geq 2y - x + 2\nu s).$$

The asserted formula (7.9) follows easily from the last equality. $\square$

Let us observe that (a similar remark applies to any formula given below)

$$\mathbb{P}(X_s \leq x, M_s^X \geq y) = \mathbb{P}(X_s < x, M_s^X > y).$$

The following result is a straightforward consequence of Lemma 7.3.2.

Proposition 7.3.3. *We have, for any $x, y \in \mathbb{R}$ satisfying $y \geq 0$ and $x \leq y$,*

$$\mathbb{P}(X_s \leq x, M_s^X \geq y) = e^{2\nu y \sigma^{-2}} N\left(\frac{x - 2y - \mu s}{\sigma \sqrt{s}}\right). \quad (7.10)$$

Hence for every $x, y \in \mathbb{R}$ such that $x \leq y$ and $y \geq 0$

$$\mathbb{P}(X_s \leq x, M_s^X \leq y) = N\left(\frac{x - \mu s}{\sigma \sqrt{s}}\right) - e^{2\nu y \sigma^{-2}} N\left(\frac{x - 2y - \mu s}{\sigma \sqrt{s}}\right). \quad (7.11)$$

Proof. For the first equality, note that

$$\mathbb{P}(X_s \geq 2y - x + 2\nu s) = \mathbb{P}(-\sigma W_s \leq x - 2y - \mu s) = N\left(\frac{x - 2y - \mu s}{\sigma \sqrt{s}}\right),$$

since $-\sigma W_t$ has Gaussian distribution with zero mean and variance $\sigma^2 t$. To establish (7.11), it is enough to observe that

$$\mathbb{P}(X_s \leq x, M_s^X \leq y) + \mathbb{P}(X_s \leq x, M_s^X > y) = \mathbb{P}(X_s \leq x)$$

and to apply (7.10). $\square$

It is clear that, for every $y \geq 0$,

$$\mathbb{P}(M_s^X \geq y) = \mathbb{P}(X_s \geq y) + \mathbb{P}(X_s \leq y, M_s^X \geq y)$$

and thus

$$\mathbb{P}(M_s^X \geq y) = \mathbb{P}(X_s \geq y) + e^{2\nu y \sigma^{-2}} \mathbb{P}(X_s \geq y + 2\nu s). \quad (7.12)$$

Hence

$$\mathbb{P}(M_s^X \leq y) = 1 - \mathbb{P}(M_s^X \geq y) = \mathbb{P}(X_s \leq y) - e^{2\nu y \sigma^{-2}} \mathbb{P}(X_s \geq y + 2\nu s).$$

This leads to the following corollary.

Corollary 7.3.2. *The following formula holds for every $y \geq 0$*

$$\mathbb{P}(M_s^X \leq y) = N\left(\frac{y - \mu s}{\sigma \sqrt{s}}\right) - e^{2\nu y \sigma^{-2}} N\left(\frac{-y - \mu s}{\sigma \sqrt{s}}\right). \quad (7.13)$$

Let us now focus on the distribution of the infimum of X . Observe that we have, for any $y \leq 0$,

$$\mathbb{P}\left(\sup_{0 \leq u \leq s} (\sigma W_u - \mu u) \geq -y\right) = \mathbb{P}\left(\inf_{0 \leq u \leq s} (-\sigma W_u + \mu u) \leq y\right) = \mathbb{P}\left(\inf_{0 \leq u \leq s} X_u \leq y\right),$$

where the last equality follows from the symmetry of the Brownian motion. Consequently we have, for every $y \leq 0$,

$$\mathbb{P}(m_s^X \leq y) = \mathbb{P}(M_s^{\tilde{X}} \geq -y),$$

where the process $\tilde{X}$ equals $\tilde{X}_t = \sigma W_t - \mu t$. The following result is thus not difficult to prove.

Proposition 7.3.4. *For every $s > 0$, the joint distribution of (X_s, m_s^X) satisfies, for any $x, y \in \mathbb{R}$ such that $y \leq 0$ and $y \leq x$,*

$$\mathbb{P}(X_s \geq x, m_s^X \geq y) = N\left(\frac{-x + \mu s}{\sigma \sqrt{s}}\right) - e^{2\nu y \sigma^{-2}} N\left(\frac{2y - x + \mu s}{\sigma \sqrt{s}}\right).$$

Corollary 7.3.3. *The following formula is valid for every $y \leq 0$*

$$\mathbb{P}(m_s^X \geq y) = N\left(\frac{-y + \mu s}{\sigma \sqrt{s}}\right) - e^{2\nu y \sigma^{-2}} N\left(\frac{y + \mu s}{\sigma \sqrt{s}}\right).$$

Recall that $Y_t = y_0 + X_t$, where $X_t = \mu t + \sigma W_t$. We write

$$m_T^X = \inf_{u \in [0, T]} X_u, \quad m_T^Y = \inf_{u \in [0, T]} Y_u.$$

Corollary 7.3.4. *For any $s > 0$ and $y \geq 0$ we have*

$$\mathbb{P}(Y_s \geq y, \tau \geq s) = N\left(\frac{-y + y_0 + \mu s}{\sigma \sqrt{s}}\right) - e^{-2\nu \sigma^{-2} y_0} N\left(\frac{-y - y_0 + \mu s}{\sigma \sqrt{s}}\right).$$

Proof. Since

$$\mathbb{P}(Y_s \geq y, \tau \geq s) = \mathbb{P}(Y_s \geq y, m_s^Y \geq 0) = \mathbb{P}(X_s \geq y - y_0, m_s^X \geq -y_0),$$

the formula is rather obvious. $\square$

More generally, by applying the Markov property and time-homogeneity of Y we obtain the following result.

Lemma 7.3.3. *Under the assumptions of Lemma 7.3.2 we have, for any $t < s$ and $y \geq 0$ on the event $\{\tau > t\}$,*

$$\mathbb{P}(Y_s \geq y, \tau \geq s | \mathcal{F}_t) = N\left(\frac{-y + Y_t + \mu(s-t)}{\sigma\sqrt{s-t}}\right) - e^{-2\nu\sigma^{-2}Y_t} N\left(\frac{-y - Y_t + \mu(s-t)}{\sigma\sqrt{s-t}}\right).$$

Chapter 8

Feynman-Kac Formula

Let us first consider a particular example of a general set-up covered by Proposition 8.1.2. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a bounded function and $g : \mathbb{R} \rightarrow \mathbb{R}$ is a function bounded from below then the unique bounded solution $v(t, x)$ of the partial differential equation (PDE)

$$\frac{\partial v}{\partial t}(t, x) = \frac{1}{2} \frac{\partial^2 v}{\partial x^2}(t, x) - g(x)v(t, x) \quad (8.1)$$

with the initial condition $v(0, x) = f(x)$ can be represented by the following version of the Feynman-Kac formula

$$v(t, x) = \mathbb{E}_{\mathbb{P}} \left(f(x + W_t) \exp \left(- \int_0^t g(x + W_u) du \right) \right). \quad (8.2)$$

It is clear that this correspondence between expected values of certain functionals of a Brownian motion and solutions to partial differential equations allows us to compute some probabilistic quantities using the PDE approach and, conversely, to solve initial value problems for certain PDEs through purely probabilistic techniques (for instance, the Monte Carlo simulation of sample paths of a stochastic process).

8.1 One-Dimensional Case

Let us first examine the one-dimensional version of the Feynman-Kac formula.

Recall that $\mu : \mathbb{R} \times [0, T] \rightarrow \mathbb{R}$ and $\sigma : \mathbb{R} \times [0, T] \rightarrow \mathbb{R}^d$ is said to be Lipschitz continuous with respect to x if there exists a constant $K_1 > 0$ such that, for any $x, y \in \mathbb{R}$ and $t \in \mathbb{R}_+$,

$$|\mu(t, x) - \mu(t, y)|^2 + |\sigma(t, x) - \sigma(t, y)|^2 \leq K_1 |x - y|^2, \quad (8.3)$$

The functions μ and σ satisfy the linear growth condition: there exists a constant K_2 such that, for any $x \in \mathbb{R}$ and $t \in \mathbb{R}_+$,

$$|\mu(t, x)|^2 + |\sigma(t, x)|^2 \leq K_2(1 + |x|^2). \quad (8.4)$$

Proposition 8.1.1. *Let $\mu, \sigma : \mathbb{R} \rightarrow \mathbb{R}$ satisfy the Lipschitz condition (8.3) and the linear growth condition (8.4). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a bounded function and $g : \mathbb{R} \rightarrow \mathbb{R}$ be a function bounded from below. Assume that $v : \mathbb{R} \times [0, T] \rightarrow \mathbb{R}$ is the unique bounded solution to the PDE*

$$\frac{\partial v}{\partial t}(t, x) = \frac{1}{2}\sigma^2(x)\frac{\partial^2 v}{\partial x^2}(t, x) + \mu(x)\frac{\partial v}{\partial x}(t, x) - g(x)v(t, x) \quad (8.5)$$

with the initial condition $v(0, x) = f(x)$. Then v has the following representation

$$v(t, x) = \mathbb{E}_{\mathbb{P}}\left(f(X_t^x) \exp\left(-\int_0^t g(X_u^x) du\right)\right), \quad (8.6)$$

where X^x is a solution to the SDE

$$dX_t^x = \mu(X_t^x) dt + \sigma(X_t^x) dW_t$$

with the initial condition $X_0^x = x$.

The infinitesimal generator $\mathcal{A}$ of X is given by the formula

$$\mathcal{A}f(x) = \frac{1}{2}\sigma^2(x)\frac{\partial^2 f}{\partial x^2}(x) + \mu(x)\frac{\partial f}{\partial x}(x).$$

Then equation (8.5) can be rewritten as follows

$$\frac{\partial v}{\partial t} = \mathcal{A}v - gv.$$

In the proof of Proposition 8.1.1 we will use the following lemma.

Lemma 8.1.1. *Assume that the σ -field $\mathcal{F}_0$ is trivial. Let $M = (M_t)_{t \in [0, T]}$ be a local martingale with respect to $\mathbb{F}$ such that the random variable $M_T^* = \sup_{t \leq T} |M_t|$ is integrable. Then M is an $\mathbb{F}$ -martingale.*

Proof. Let $(\tau_n)_{n \in \mathbb{N}}$ be any reducing sequence of stopping times for M . For any $n \in \mathbb{N}$ and $t \leq T$ the martingale property of the stopped process M^{τ_n} yields

$$\mathbb{E}_{\mathbb{P}}(M_{T \wedge \tau_n} | \mathcal{F}_t) = M_{t \wedge \tau_n}. \quad (8.7)$$

Since the random variable M_T^ dominates $M_{T \wedge \tau_n}$ and $X_{t \wedge \tau_n}$, using the dominated convergence theorem, we may pass to the limit in (8.7) and thus we obtain, for any $t \in [0, T]$, $\mathbb{E}_{\mathbb{P}}(M_T | \mathcal{F}_t) = M_t$, which is, of course, the desired martingale property of M . $\square$*

Proof of Proposition 8.1.1. The proof relies on finding, for a fixed $t \in [0, T]$, a martingale $M = (M_s)_{s \in [0, t]}$ such that $\mathbb{E}_{\mathbb{P}}[M_0] = v(t, x)$ and

$$\mathbb{E}_{\mathbb{P}}[M_t] = \mathbb{E}_{\mathbb{P}}\left(f(X_t^x) \exp\left(-\int_0^t g(X_u^x) du\right)\right).$$

Since for any martingale M we have that $\mathbb{E}_{\mathbb{P}}[M_0] = \mathbb{E}_{\mathbb{P}}[M_t]$, this will imply (8.6).

Let us fix $t > 0$ and let us define the process $M = (M_s)_{s \in [0, t]}$ by the formula

$$M_s = v(t - s, X_s^x) \exp \left(- \int_0^s g(X_u^x) du \right). \quad (8.8)$$

The Itô formula, when combined with the assumption that the function v solves (8.5), yield

$$dM_s = \exp \left(- \int_0^s g(X_u^x) du \right) \sigma(X_s^x) \frac{\partial v}{\partial x}(t - s, X_s^x) dW_s$$

and thus M is manifestly a local martingale. We assumed that v, g are bounded functions, so that $v(x) \leq L_1$ and $g(x) \geq -L_2$ for some constants L_1 and L_2 . From (8.8), we thus get

$$\sup_{u \leq t} |M_u| \leq L_1 \exp(L_2 t) < \infty,$$

so that $\mathbb{E}_{\mathbb{P}} M_t^* < \infty$. It follows from Lemma 8.1.1 that M is a martingale. Hence

$$\begin{aligned} v(t, x) &= \mathbb{E}_{\mathbb{P}}[M_0] = \mathbb{E}_{\mathbb{P}}[M_t] = \mathbb{E}_{\mathbb{P}} \left(v(0, X_t^x) \exp \left(- \int_0^t g(X_u^x) du \right) \right) \\ &= \mathbb{E}_{\mathbb{P}} \left(f(X_t^x) \exp \left(- \int_0^t g(X_u^x) du \right) \right), \end{aligned}$$

since obviously $v(0, X_0^x) = v(0, x) = f(x)$. We conclude that (8.6) holds. $\square$

8.1.1 Multidimensional Case

Let W be a d -dimensional standard Brownian motion on the underlying probability space $(\Omega, \mathbb{F}, \mathbb{P})$. We consider the k -dimensional diffusion process X given as

$$dX_t = \mu(X_t) dt + \sigma(X_t) dW_t, \quad X_0 = x, \quad (8.9)$$

where the coefficients $\mu : \mathbb{R}^k \rightarrow \mathbb{R}^k$ and $\sigma : \mathbb{R}^k \rightarrow \mathbb{R}^{k \times d}$ satisfy the Lipschitz condition (8.3) and the linear growth condition (8.4). By virtue of the multidimensional version of Theorem on existence and uniqueness of solution to SDEs, we conclude that the SDE (8.9) admits a unique solution. Then X is a continuous process and it enjoys the Markov property with respect to $\mathbb{F}$.

Let the d -dimensional matrix $a(x) = [a_{ij}(x)]$ be equal to $\sigma(x)\sigma^t(x)$, where $\sigma^t(x)$ is the transpose of the matrix $\sigma(x)$. We associate with the Markov process X its infinitesimal generator $\mathcal{A}$, which is given by the formula

$$\mathcal{A}f(x) = \sum_{i=1}^k \mu_i(x) \frac{\partial f}{\partial x_i}(x) + \frac{1}{2} \sum_{i,j=1}^k a_{ij}(x) \frac{\partial^2 f}{\partial x_i \partial x_j}(x),$$

for any function $f : \mathbb{R}^k \rightarrow \mathbb{R}$ belonging to a suitably defined class of functions, referred to as the domain $D(\mathcal{A})$ of the differential operator $\mathcal{A}$.

Typically, it is rather difficult to determine explicitly the domain $D(\mathcal{A})$ of the infinitesimal generator $\mathcal{A}$ of a diffusion process X . However, $\mathcal{A}$ is always well defined

on the class of all functions with compact support (i.e., vanishing outside a bounded interval) belonging to the class $C^2(\mathbb{R}^k)$. Let us denote by $C_c^2(\mathbb{R}^k)$ the class of all such functions. Then we have $D(\mathcal{A}) \subseteq C_c^2(\mathbb{R}^k)$.

Proposition 8.1.2. *Let $g \in C(\mathbb{R}^k)$ be a function bounded from below and let f belong to the class $C_c^2(\mathbb{R}^k)$. Define the function $v : \mathbb{R}_+ \times \mathbb{R}^k \rightarrow \mathbb{R}$ by the formula*

$$v(t, x) = \mathbb{E}_{\mathbb{P}} \left(f(X_t^x) \exp \left(- \int_0^t g(X_s^x) ds \right) \right).$$

Then v satisfies the PDE

$$\frac{\partial v}{\partial t} = \mathcal{A}v - gv \quad (8.10)$$

with the initial condition $v(0, x) = f(x)$ for $x \in \mathbb{R}^k$.

Example: Consider the case $g(x) = 0$. Then we show that

$$v(t, x) = \mathbb{E}(f(x + W_t))$$

satisfies the PDE, that is $v(0, x) = f(x)$ and

$$\frac{\partial v}{\partial t}(t, x) = \frac{1}{2} \frac{\partial^2 v}{\partial x^2}(t, x).$$

Since $W_0 = 0$, it is clear that $v(0, x) = f(x)$. We need only to compute the derivatives

$$v(t, x) = \mathbb{E}(f(x + W_t)) = \int_{\mathbb{R}} f(x + u)g(t, u)du = \int_{\mathbb{R}} f(z)g(t, z - x)dz$$

where $g(t, u) = \frac{1}{\sqrt{2\pi t}}e^{-u^2/2t}$. Then by using the fact that (Using Dominated convergence theorem)

$$\begin{aligned} \partial_t \int_{\mathbb{R}} f(z)g(t, z - x)dz &= \int_{\mathbb{R}} f(z)\partial_t g(t, z - x)dz \\ \partial_{xx}^2 \int_{\mathbb{R}} f(z)g(t, z - x)dz &= \int_{\mathbb{R}} f(z)\partial_{xx}^2 g(t, z - x)dz \end{aligned}$$

and also the fact that

$$\partial_t g(t, x) = \frac{1}{2} \partial_{xx}^2 g(t, x)$$

we see that $v(t, x)$ satisfies the PDE.

Under assumptions of Proposition 8.1.2 we also have the following converse result.

Proposition 8.1.3. *If $w \in C^{1,2}(\mathbb{R} \times \mathbb{R}^k)$ is bounded on $[-K, K] \times \mathbb{R}^d$ for any $K > 0$ (equivalently, the function w is bounded on $A \times \mathbb{R}^d$ for any compact subset $A \subset \mathbb{R}$) and w is a solution to the PDE (8.10) with the initial condition $w(0, x) = f(x)$ for $x \in \mathbb{R}^k$ then $w(t, x) = v(t, x)$.*

Chapter 9

Exercise

In the following are some exercises, some of them are taken from the books

- Oksendal B. Stochastic Differential Equations

- Karatzas I. and Shreve S. E.: Stochastic Calculus and Brownian Motion

In the following, unless otherwise stated we assume that the probability space is $(\Omega, \mathcal{A}, \mathbb{P})$ and the image space is $(\Psi, \mathcal{G})$ or $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

1. Suppose $\mathbb{P}(A) = \frac{3}{4}$ and $\mathbb{P}(B) = \frac{1}{3}$. Show that

$$\frac{1}{12} \leq \mathbb{P}(A \cap B) \leq \frac{1}{3}$$

2. Let $(\mathcal{G}_a)_{a \in A}$ be a family of σ -algebra defined on some abstract space Ω . Show that $\mathcal{H} := \cap_{a \in A} \mathcal{G}_a$ is also a σ -algebra.

3. Consider $X : (\Omega, \mathcal{A}) \rightarrow (\Psi, \mathcal{G})$, show that $\sigma(X) = X^{-1}(\mathcal{G})$. Hint: Show that the family of set $X^{-1}(\mathcal{G}) = \{X^{-1}(A) : A \in \mathcal{G}\}$ is a σ -algebra.

4. Let $\Omega = \{-1, 0, 1\}$ and consider the random variable $X(\omega) = |\omega|$, find $\sigma(X)$ and repeat for $X(\omega) = \omega^2$.

5. If $X = Y$ a.s then the distribution of X and Y are the same. Hint: Check that X and Y have the same CDF.

6. Given an example of an event B such that $\mathbb{P}(B) = 1$, but $B \neq \Omega$.

7. Given random variables X and Y , show that

(a) For $a \in \mathbb{R}$, aX is a random variable

(b) $X + Y$ is a random variable.

(c) XY is a random variable.

(d) $\max(X, Y)$ is a random variable.

8. Let $(X_n)_n$ be a sequence of $\mathcal{A}$ -measurable functions such that $X := \lim_{n \rightarrow \infty} X_n$ exists. Show that X is also $\mathcal{A}$ -measurable.

9. Let B_t be a 1-dimensional Brownian motion and let $c > 0$ be a constant. Show that

$$\widehat{B}_t := \frac{1}{c} B_{c^2 t}$$

is also a Brownian motion in its natural filtration.

10. Check whether the following processes X_t are martingales wrt to $\mathbb{F}$.

- (a) $X_t = B_t + 4t$
- (b) $X_t = B_t^2$
- (c) $X_t = t^2 B_t - 2 \int_0^t s B_s ds$

11. Let X be a random variable such that $\mathbb{E}(X^2) < \infty$ and let $\mathcal{H} \subset \mathcal{A}$. Show that

$$\mathbb{E}[(\mathbb{E}(X|\mathcal{H}))^2] \leq \mathbb{E}[X^2]$$

12. Let W be a Brownian motion starting at $x \geq 0$, suppose $x \leq b$ and let $T_b = \inf\{s : W_s = b\}$. Show that for $\lambda > 0$

$$\begin{aligned} \mathbb{E}[e^{-\lambda T_0} \mathbf{1}_{\{T_0 < T_b\}}] &= \frac{\sinh((b-x)\sqrt{2\lambda})}{\sinh(b\sqrt{2\lambda})} \\ \mathbb{E}[e^{-\lambda T_b} \mathbf{1}_{\{T_b < T_0\}}] &= \frac{\sinh(x\sqrt{2\lambda})}{\sinh(b\sqrt{2\lambda})}. \end{aligned}$$

Hint: Use optional sampling theorem.

- (a) Show that $\mathbb{E}[T_0 \wedge T_b] = x(b-x)$ for $0 \leq x \leq b$.
- (b) Show that for $0 \leq x \leq b$

$$\mathbb{P}(T_0 < T_b) = \frac{b-x}{b} \quad \text{and} \quad \mathbb{P}(T_b < T_0) = \frac{x}{b}.$$

13. Let B_t be an one dimensional Brownian motion with $B_0 = 0$. Define

$$\beta_k(t) = \mathbb{E}(B_t^k); \quad k = 0, 1, 2, \dots \quad t \geq 0$$

- (a) Use the Itô formula to prove that

$$\beta_k(t) = \frac{1}{2} k(k-1) \int_0^t \beta_{k-2}(s) ds$$

- (b) Deduce that $\mathbb{E}(B_t^4) = 3t^2$ and find $\mathbb{E}(B_t^6)$.
- (c) Show that $\mathbb{E}(B_t^{2k+1}) = 0$.

14. Let Y is a $\mathcal{F}_T$ -measurable random variable such that $\mathbb{E}(|Y|^2) < \infty$. Let

$$M_t = \mathbb{E}[Y|\mathcal{F}_t] \quad t \in [0, T]$$

- (a) Check that $\mathbb{E}(M_t^2) < \infty$ for all $t \in [0, T]$
 (b) Check that $(M_t)_{t \in [0, T]}$ is a martingale with respect to $\mathbb{F}$.
 (c) According to martingale representation theorem, there exists a $g \in \mathcal{L}(W)$ such that

$$M_t = \mathbb{E}(M_0) + \int_0^t g_s dB_s$$

Find g in the following cases: $Y = B_T^2$ and $Y = \exp(\sigma B_T)$.

15. The Hermite polynomial of order n is given by

$$h_n(x) = (-1)^n \exp(x^2/2) \frac{d^n}{dx^n} \{ \exp(-x^2/2) \}$$

and let $H_n(x, t) = t^{n/2} h_n(x/\sqrt{t})$.

- (a) Prove the following recursive relation for $n \geq 0$,

$$h_{n+1}(x) = xh_n(x) - \partial_x h_n(x).$$

- (b) Show that $H_n(B_t, t)$, $t \in [0, T]$ is a martingale for $n = 0, 1, 2$. Here B is Brownian motion. (In fact $H_n(B_t, t)$ is a martingale for all $n \in \mathbb{N}$).

16. (Hitting time of a symmetric random walk) For $n = 1, 2, \dots$, let $X_n = X_{n-1} + Y_n$, $X_0 = 0$ and Y_n be independent random variables which takes values 1 or -1 with equal probability. Show that $\tau = \inf\{n : X_n = 2\}$ is a stopping time with respect to the natural filtration generated by $X = (X_n)_{n \in \mathbb{N}}$.

17. Assume that we are given a filtered probability space $(\Omega, \mathbb{F}, \mathbb{P})$. For any $s < t$ and any event $A \in \mathcal{F}_s$, we define the random variable $\tau : \Omega \rightarrow \mathbb{R}_+$ by setting

$$\tau = s\mathbb{1}_A + t\mathbb{1}_{A^c}$$

- (a) Show that the random variable τ given above is an $\mathbb{F}$ -stopping time.
 (b) Let X be an $\mathbb{F}$ -adapted, integrable process. Show that the following two conditions (i) and (ii) are equivalent:
 (i) The process X is an $\mathbb{F}$ -martingale.
 (ii) Equality $\mathbb{E}(X_\tau) = \mathbb{E}(X_0)$ holds for any stopping time τ of the form given above.

18. Let $(\Omega, \mathbb{P}) = ([0, 1], \lambda)$ where $\mathbb{P}([a, b]) = \lambda([a, b]) = b - a$. In other words, $\mathbb{P}(d\omega) = \lambda(d\omega) = d\omega$.

- (a) Consider the random variables

$$X(\omega) = \begin{cases} \omega, & \omega \in [0, \frac{1}{3}] \\ \frac{1}{3}, & \omega \in (\frac{1}{3}, 1] \end{cases}$$

and

$$Y(\omega) = \begin{cases} 1, & \omega \in \{X = \frac{1}{3}\} \\ 2, & \omega \in \{X < \frac{1}{3}\} \end{cases}$$

(b) Compute the CDF of X and hence $\mathbb{E}[X]$ and $\text{Var}(X)$.

(c) What is $\sigma(Y)$ and $\mathbb{E}[X^2|\sigma(Y)]$

19. Let $t \in [0, 1]$, argue that $\sigma(\mathcal{F}_t^W, W_1) = \sigma(\mathcal{F}_t^W, W_1 - W_t)$ and if X is a process whose future increments are independent from the past filtration generated by the Brownian motion, show that $\mathbb{E}[X_t - X_s | \mathcal{F}_s^W, W_1 - W_s] = \mathbb{E}[X_t - X_s | W_1 - W_s]$ for $s \leq t \leq 1$.

20. The brownian bridge is the solution to

$$dY_t = \frac{b - Y_t}{1 - t} dt + dB_t \quad Y_0 = a$$

Show that $Y_t = a(1 - t) + bt + (1 - t) \int_0^t \frac{1}{1-s} dB_s$, $t \in [0, 1]$.

21. Integrating factor can be applied to more general non-linear stochastic differential equations of the form

$$dX_t = f(t, X_t)dt + c(t)X_t dB_t$$

where f and c are continuous functions.

(a) Use the integrating factor

$$F_t = \exp\left(-\int_0^t c(s)dB_s + \frac{1}{2}\int_0^t c^2(s)ds\right)$$

Show that

$$d(F_t X_t) = F_t f(t, X_t)dt$$

(b) Let $Y_t = F_t X_t$ so that $X_t = F_t^{-1} Y_t$. Deduce that

$$\frac{dY_t}{dt} = F_t f(t, F_t^{-1} Y_t)$$

which is just a random ODE.

(c) Apply this method to solve the stochastic differential equation

$$dX_t = \frac{1}{X_t} dt + \alpha X_t dB_t \quad X_0 = x > 0$$

where α is a constant.

22. (Black-Scholes PDE). Consider the following PDE

$$\begin{cases} \partial_t u = -\rho u + \alpha x \partial_x u + \frac{1}{2} \beta^2 x^2 \partial_{xx}^2 u & x \in \mathbb{R} \\ u(0, x) = (x - K)^+ & x \in \mathbb{R} \end{cases}$$

where $\rho > 0$, α, β and $K > 0$ are constants and $(x - K)^+ = \max(x - K, 0)$. Use the Feynman-Kac formula to prove that the solution u of this equation is given by

$$u(t, x) = \frac{e^{-\rho t}}{\sqrt{2\pi t}} \int_{\mathbb{R}} (x e^{(\alpha - \frac{1}{2}\beta^2)t + \beta y} - K)^+ e^{-\frac{y^2}{2t}} dy$$

23. Consider two real number δ, μ and an one dimensional Brownian motion W , let $W_t^\mu = W_t + \mu t$. Show that the process

$$X_t = \int_0^t \exp \left(\delta(W_t^\mu - W_s^\mu) - \frac{1}{2} \delta^2(t-s) \right) ds$$

satisfies the Shriyaev-Roberts equation

$$dX_t = (1 + \delta\mu X_t)dt + \delta X_t dW_t.$$

24. Let W be a Wiener process and let the process ρ be defined by

$$\rho_t = \exp \left(-b \int_0^t W_s dW_s - \frac{b^2}{2} \int_0^t W_s^2 ds \right), \quad t \in [0, T].$$

Let for every $A \in \mathcal{F}_T$

$$\mathbb{P}^b(A) = \mathbb{E}_{\mathbb{P}}(\rho_T \mathbf{1}_A).$$

(a) Show that the process W is a $\mathbb{P}^b$ Ornstein-Uhlenbeck process.

(b) Show that

$$\mathbb{E}_{\mathbb{P}} \left(\exp \left(-\alpha W_t^2 - \frac{b^2}{2} \int_0^t W_s^2 ds \right) \right) = \mathbb{E}_{\mathbb{P}}^b \left(\exp \left(-\alpha W_t^2 + \frac{b}{2} (W_t^2 - t) \right) \right).$$

(c) Deduce from (b) the Lévy area formula**:

$$\mathbb{E}_{\mathbb{P}} \left(\exp \left(-\alpha W_t^2 - \frac{b^2}{2} \int_0^t W_s^2 ds \right) \right) = \frac{1}{\sqrt{\cosh bt + 2 \frac{\alpha}{b} \sinh bt}}.$$

25. Let $(W_t)_{t \geq 0}$ be a Brownian motion and let X be the unique solution to the following stochastic differential equation, under $\mathbb{P}$:

$$dX_t = X_t(\alpha_t dt + \beta_t dW_t)$$

where α and β bounded and adapted processes, and $\sigma > 0$ almost surely.

(a) Show that $X_t \exp(-\int_0^t \alpha_s ds)$ is a martingale.

(b) Find a probability $\mathbb{Q}$ equivalent to $\mathbb{P}$ under which X is a (local) martingale.

26. Consider the filtered probability space $(\Omega, \mathcal{F}_T, \mathbb{F}, \mathbb{P})$ with a $\mathbb{F}$ -Brownian motion B . For $\mu > 0$, we consider the Radon-Nikodym density process on $[0, T]$ given by

$$Z_t = e^{\mu B_t - \frac{\mu^2}{2} t}$$

and define an equivalent probability measure $\mathbb{Q}$ on $\mathcal{F}_T$ by $\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}}(Z_T \mathbf{1}_A)$.

(a) For every $t \geq 0$, show that $\{T_b \leq t\} \in \mathcal{F}_{T_b \wedge t}$. Recall that for any $\mathbb{F}$ stopping time σ .

$$\mathcal{F}_\sigma = \{A \in \mathcal{F}_T : \forall t \in [0, T], A \cap \{\sigma \leq t\} \in \mathcal{F}_t\}$$

- (b) Suppose $b > 0$, consider the first hitting time of b for the Brownian motion with drift $T_b := \inf\{s : W_s + \mu s = b\} = \inf\{s : B_s = b\}$. By computing for $s \leq T$ the quantity

$$\mathbb{Q}(T_b \leq s) = \mathbb{E}_{\mathbb{P}}(\mathbf{1}_{\{T_b \leq s\}} Z_T), \quad (9.1)$$

show that under $\mathbb{Q}$ the probability density function for T_b is given by

$$f_{T_b}^{\mathbb{Q}}(t) = \frac{b}{\sqrt{2\pi t^3}} \exp\left(-\frac{(b - \mu t)^2}{2t}\right), \quad t \geq 0.$$

Hint: The density process Z is a $(\mathbb{P}, \mathbb{F})$ martingale, and condition on $\mathcal{F}_{T_b \wedge s}$ and apply optional sampling theorem.

27. Consider the probability space $(\Omega, \mathcal{F}) = ([0, 1], \mathcal{B}([0, 1]))$ and the probability measure $\mathbb{P} = \lambda$, where λ is the Lebesgue measure. That is $\lambda([a, b]) = b - a$ for any subinterval $[a, b]$ of $[0, 1]$. Consider the functions

$$X(\omega) = 2\omega^2 - 1, \quad \omega \in [0, 1]$$

and $Y = \text{sign}(X)$, that is

$$Y(\omega) = \begin{cases} 1, & \text{if } X > 0 \\ -1, & \text{if } X \leq 0. \end{cases}$$

- (a) Write down the σ -algebra $\sigma(Y)$.
- (b) Compute the expected values $\mathbb{E}(X)$.
- (c) Find $\mathbb{E}(X|Y)$ as function of the variable ω .

Chapter 10

Appendix

10.1 Modes of convergences

1. We write $X_n \rightarrow X$ surely (or everywhere) if $X_n(\omega) \rightarrow X(\omega)$ for all $\omega \in \Omega$.
2. We write $X_n \rightarrow X$ almost surely (or almost everywhere) if $\mathbb{P}(\lim_{n \rightarrow \infty} X_n \neq X) = 0$.

Below, we list different type of convergence one might meet in this course.

- $X_n \rightarrow X$ in L^p for $p \geq 1$ (or L^p convergence) if $\mathbb{E}(|X|^p) < \infty$ for all n and

$$\lim_{n \rightarrow \infty} \mathbb{E}[|X_n - X|^p] = 0$$

- $X_n \rightarrow X$ in probability if for all $\epsilon > 0$

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > \epsilon) = 0$$

- $X_n \rightarrow X$ in distribution if $\mathbb{P}(X_n \leq x) \rightarrow \mathbb{P}(X \leq x)$ for all x at which $\mathbb{P}(X \leq x)$ is continuous.

The following implications also holds.

- Convergence surely $\implies$ Convergence a.s $\implies$ Convergence in probability $\implies$ Convergence in distribution.
- L^p convergence $\implies$ Convergence in probability.
- If a sequence converges in L^2 , then there exists a subsequence converging almost surely to the same limit.
- Convergence in $L^p \implies$ Convergence in L^r for $p \geq r \geq 1$.

Intuitively, one can think that if the variance of $X_n - c$ is going to zero, then the mean of X_n is going to c . That is L^2 convergence implies L^1 convergence.

10.2 Convergence Theorems

Theorem 10.2.1 (Convergence Theorems). *Let $(X_n)_{n \geq 1}$ is a sequence of positive random variables*

- *(Monotone Convergence Theorem) Suppose $X_n \leq X_{n+1}$ for all $n = 1, \dots$, and $\lim_{n \rightarrow \infty} X_n = X$ a.s then*

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \mathbb{E}[\lim_{n \rightarrow \infty} X_n] = \mathbb{E}[X]$$

- *(Fatou lemma) Let $(X_n)_{n \geq 1}$ be a sequence of non-negative random variables. Define X by $\liminf_{n \rightarrow \infty} X_n$ then*

$$\mathbb{E}[X] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[X_n]$$

- *(Dominated Convergence Theorem) Suppose $X_n \in L^1$ for all $n \geq 1$ and is such that (i) $X_n \rightarrow X$, a.s. (ii) there exists a non-negative r.v $Y \in L^1$ such that $|X_n| \leq Y$ for all $n \geq 1$, then $X \in L^1$ and*

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \mathbb{E}[\lim_{n \rightarrow \infty} X_n] = \mathbb{E}[X].$$